



Asset Performance Management

User's Guide

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CONTENTS

Chapter 1 - About This Guide	8
1.1 Purpose	8
1.2 Revision History	8
1.3 References	8
1.4 Audience	8
Chapter 2 - Introduction	9
2.1 About Honeywell Asset Performance Management	9
2.1.1 Functions of Asset Performance Management	9
2.1.2 Benefits of using Asset Performance Management	10
2.2 Reasons for Failures	10
2.3 Getting Started with Asset Performance Management	11
2.3.1 Opening Asset Performance Management on the client computers	11
2.3.2 Configuring Internet Explorer on the Client Computers	12
Chapter 3 - Asset Performance Management Concepts	14
3.1 Assets and Events	14
3.1.1 Asset Categories	14
3.1.2 Asset Class	14
3.1.3 Asset Types	14
3.1.4 Asset	14
3.1.5 Asset Status	15
3.1.6 Asset Rating	15
3.1.7 Asset Attributes	15
3.1.8 Asset Links	16
3.1.9 Asset Hierarchy	16
3.1.10 Attribute Property	17
3.1.11 Fault	17

3.1.12 Symptom	17
3.1.13 Event Status	17
3.1.14 Fault Models	18
3.1.15 Fault and Symptom Management Process	19
3.1.16 Health Index	20
3.1.17 Performance Index	20
3.1.18 Energy Index	21
3.2 Monitoring	21
3.2.1 Process Monitoring	21
3.2.2 Performance Monitoring	21
3.2.3 Energy/Utility Monitoring	21
3.2.4 Condition Monitoring	21
3.2.5 Instrument Monitoring	22
3.2.6 Safety Monitoring	22
3.2.7 Limit Monitoring	22
Chapter 4 - Using Asset Performance Management	23
4.1 Asset Tree	23
4.1.1 Asset Status Icons	24
4.2 Contents Tree	25
4.3 Viewing Custom Hierarchies	26
Chapter 5 - Performance Management	27
5.1 Performance Management	27
5.1.1 Asset Status	27
5.1.2 Asset Health Dashboard	27
5.2 Using the Treemap	27
5.2.1 Reading the Treemap	28
5.2.2 Treemap Settings	29
5.2.3 Treemap Filters	30
5.3 Attribute Overview	30
5.3.1 Opening Attribute Overview	31
5.3.2 Attribute Overview Interface	32
5.3.3 Columns in the Attribute Overview	35
5.3.4 Exporting Attributes to an Excel File	36

5.3.5 Filtering Attributes	36
5.3.6 Grouping Fields in Attribute Overview	38
5.4 Asset Health Dashboard	39
5.4.1 Introduction	39
5.4.2 Objective of Asset Health Dashboard	39
5.4.3 Workflow for Asset Health Dashboard	40
5.4.4 Knowing the Interface of Asset Health Dashboard	40
5.4.5 Identify	43
5.4.6 Investigate	45
5.4.7 Act	45
5.4.8 Configuration	45
5.5 Performance Widget	46
5.5.1 Introduction	46
5.5.2 Interface	48
5.5.3 Default View	49
5.5.4 Performance Widget Configuration	52
5.5.5 Opening Performance Dashboard	53
5.5.6 Using Performance Dashboard	57
5.5.7 Calculation Flow Diagram	74
Chapter 6 - Event Management	75
6.1 Event Management Views	75
6.1.1 Event Monitor	75
6.1.2 Event Investigation	76
6.1.3 Event History	76
6.1.4 Filters in Event Management	77
6.2 Event Management Work-flow	78
6.2.1 Event Monitoring	78
6.2.2 Event Investigation	79
6.3 Event Management Example	79
6.3.1 Compressor Vibration Increased	79
6.4 Using the Event Viewer	80
6.4.1 Opening the Event Viewer	80
6.4.2 Interface	80
6.4.3 Editing Columns in the Event Viewer	83

6.4.4 Filters in Event Viewer	84
6.4.5 Configuring Event Viewer Time to Milliseconds	85
6.4.6 Instance Level UoM in Event Viewer	85
6.4.7 Accept or Decline an Event	86
6.4.8 Enabling Event Closeout and Reject Comment	86
6.4.9 Close Out Event	87
6.4.10 Reject an Event	90
6.4.11 Add a Comment	92
6.4.12 Lock an Event	93
6.4.13 Export Events to Excel	96
6.4.14 Viewing Event Details	96
6.4.15 Insert an Event	97
6.4.16 Event Acknowledge Button	98
6.4.17 Play or Resume Event Updates in the Event Viewer	99
6.4.18 Generate CMMS Notification	99
6.5 Fault Tree	100
6.6 Using the Trend	102
6.6.1 Interface	102
6.6.2 Attributes	104
6.6.3 Events	105
6.6.4 Date Range Picker	105
6.6.5 Trend for the Events	108
6.6.6 Save Trend Configuration and Templatize	109
6.6.7 Threshold Limit View	110
6.6.8 Plot Fault Overlay	111
6.6.9 Trending of Contributing Variables of Faults and Symptoms	112
6.6.10 Trend Attributes from Other Assets	112
6.6.11 Multiple Attributes on Trend	114
6.6.12 Zoom In and Zoom Out	114
6.6.13 Multi-state Digital Data	115
Chapter 7 - System Monitoring	117
7.1 Calculation Status	117
7.1.1 Status Viewer Interface	118
7.1.2 Filtering Calculations	118

7.1.3 Asset Type Level UOM in Calculation Status	119
7.1.4 Failed Calculations	120
7.2 Data Source Status	120
Chapter 8 - Asset Analysis	121
8.1 Asset Views	121
8.1.1 Asset Details	121
8.1.2 Trends	121
8.2 Viewing Asset Details	121
8.2.1 Adding Asset Details and Key Metrics	122
8.2.2 Changing Asset Status	123
8.2.3 Asset Comments	124
8.2.4 Resetting Asset Metrics	125
8.3 Asset Links and Notifications	125
8.3.1 Asset Links	126
8.3.2 Notifications	126
Chapter 9 - Viewing Reports	127
9.1 Reports Overview	127
9.1.1 Accessing the Asset Performance Management Reports	128
9.2 Asset Reports	128
9.2.1 Asset Status Report	129
9.2.2 Important Priority Assets	130
9.2.3 Monitored Assets	130
9.2.4 Normal Priority Assets	130
9.2.5 Out of Service	131
9.2.6 UnMonitored Assets	131
9.2.7 Urgent Priority Assets	132
9.2.8 Viewing KPI Reports	132
9.3 Fault Reports	134
9.3.1 Viewing Fault reports	135
9.4 Key Performance Indicator (KPI)Reports	136
9.5 Changing Report Properties	137

ABOUT THIS GUIDE

1.1 Purpose

This guide describes how to work with the Asset Performance Management application.

1.2 Revision History

This guide describes how to work with the application.

Revision	Date	Description
1	August 2023	Asset Performance Management R540.1 General Availability (GA) release

1.3 References

The following list identifies all documents that are sources of reference for material discussed in this publication.

Document Name	Document ID
Asset Performance Management Installation Guide	APMDOC-X390-en-R540.1
Asset Performance Management Configuration Guide	APMDOC-X391-en-R540.1

For the most up-to-date version of the documentation, see [https:// www.honeywellprocess.com](https://www.honeywellprocess.com) (click the Support tab). User registration and login are required to access the Honeywell Process site.

1.4 Audience

The intended audiences of this guide are the engineers, operators, and managers who can understand the working of the Asset Performance Management application.

This chapter provides an overview of the Asset Performance Management application and describes its functionalities and benefits.

2.1 About Honeywell Asset Performance Management

Asset Performance Management is an application designed and developed by Honeywell to monitor, maintain, and track the reliability of assets in a plant. The Reliability Engineer uses the Asset Performance Management application in a plant to monitor the reliability of equipment such as, pumps, valves, compressors, heat exchangers, and to identify any failure/fault. The monitoring and quick identification of failures helps in taking decisions and planning for maintenance.

In Asset Performance Management, certain fault models are built to analyze the failure of an asset. In all equipment, a failure of an equipment is determined only when the baseline value of an equipment increases. The corresponding progression of the fault is tracked and the subsequent maintenance action is initiated depending on the priority or the age of the asset. For example, if the priority is to save the equipment from any failures, then the maintenance action is initiated at an earlier stage, thus saving the life of the equipment. If the priority is to utilize the life of the component to the fullest, then the maintenance action is initiated at a later stage.

The Asset Performance Management application plays a vital role in synergizing various functional areas of an industry such as maintenance department, operations department, and reliability department. The application provides various operating conditions of an equipment to the operator. It also consists of a calculation engine that helps the operator to calculate efficiency, plant load factor, and failure modes depending on the operating conditions.

Asset Performance Management uses the wireless condition-based monitoring to monitor the reliability of rotating assets (pumps, compressors) using wireless devices. The approach of using the wireless devices effectively manages the maintenance cost of the equipment. Asset Performance Management acquires wireless vibration data from SKF wireless DAQ devices.

The Asset Performance Management application receives data through the OPC A&E data source. The events are generated in the data source based on a complicated scenario. The events are received by the Asset Performance Management application as symptoms. The data from the following devices flows into the Asset Performance Management application.

- Instrumentation data such as notifications coming from HART or FF devices.
- Process information or process data collected by OPC A&E or OPC DA client.
- Manually collected data.
- Plant historian data - the process related or other sources such as lab data.

2.1.1 Functions of Asset Performance Management

- Asset Performance Management provides a single platform for performance monitoring and vibration monitoring for different departments like operations, maintenance, and reliability.

For example, cavitation in a centrifugal pump is caused due to high flow and low pressure head. When the operations department increases the production by increasing the flow rate beyond the best efficiency point (BEP), then the action induces cavitation and leads to a subsequent rise in vibration. If the rise in the vibration is very large, then the operator can view the indication and change the operating conditions of the pump. The reliability department views the severity of the cavitations and the secondary damage caused to the equipment. The maintenance department schedules maintenance actions and orders spare equipment to plan the shutdown of the maintenance activity.

- The assets in a unit are displayed in a tree-view for easier monitoring of the assets and its components.
- The condition of assets are indicated by different colors such as yellow for warning, red for alert, green for normal operating assets.
- Sensor information from multiple sensors are consolidated and presented as a single consolidated health indicator. For example, a centrifugal pump may be monitored for its process performance and mechanical subsystems vibration. The measurements collected by all these sensors are consolidated and presented for easy monitoring and tracking.
- It helps the reliability engineer to analyze the failure modes and the Fast Fourier Transform (FFT) data.
- Process performance monitoring for process equipment like pump, compressor, steam turbine, gas turbine, and shell and tube heat exchanger is supported in Asset Performance Management. Process performance monitoring models are used to calculate the operating performance output, design performance output, and deviation between operating performance output and design performance output.

2.1.2 Benefits of using Asset Performance Management

The main advantages of using the Asset Performance Management application are:

- Reduces the equipment's maintenance.
- Reduces scheduled and unscheduled downtime.
- Limits the production of off-specification product.
- Reduces energy loss.
- Provides timely one-stop access to information, which increases the efficiency of maintenance and reliability engineers.
- Prevents the failure of an equipment by identifying the deteriorating conditions at the component level.
- Early detection of changes in asset performance, allowing more time for planned maintenance or operational changes, to reduce stress on the monitored asset.
- Reduces Mean Time To Repair (MTTR) and increases Mean Time Between Failures (MTBF) by identifying improper operation and process changes for mitigating further equipment damage.
- Improves the communication between specialists (vibration, oil analysts, operations, and maintenance planning).
- Increases the availability through reduced failure rates and MTTR, resulting in more production capability and profit potential.
- Collection of maintenance, process, and reliability data to facilitate Root Cause Failure Analysis (RCFA). The identified causes enable to reduce or eliminate future failures resulting in reduced failure rates and reliability growth.

2.2 Reasons for Failures

In a plant, different techniques are adopted to measure the health of the equipment. Asset Performance Management uses the concept of CBM to assess the health of the equipment. CBM is a predictive maintenance procedure, where the health/condition of a specific area on the equipment is monitored to detect early signs of equipment failure. Failure in equipment can happen due to the following reasons.

- **Improper design:** The design of the asset is the most important reason for its failure. The tolerance limits of any bearing and shaft coupling must be accurate. The design of key and key holes, stress concentration factor of such holes, and so on, play an important role in increasing stress and reducing the life of the asset. Similarly, stress concentration factor of each and every component of the asset also adds stress to the asset.
- **Improper installation:** A lot of caution has to be taken while installing the asset. The tightening of the bolts must be as per the rating. An over-rated tightening of the bolt can cause rigidity to the system while an under-rated tightening of the bolt can induce looseness in the asset. Similarly, the shaft and the alignment of the asset and its structure must be properly aligned during installation.
- **Improper operating conditions:** Rotating equipment like pumps and compressors are designed to operate at certain conditions known as rated conditions. For a pump, these conditions may be rated as “load”, “speed”, “flow”, and “pressure”. The asset operates more efficiently if it is operated near its rated conditions. The operators of these equipment may not follow the ideal conditions due to pressure of meeting production targets. This often results in early and unexpected failure.
- **Improper maintenance:** There are times when the equipment fails due to lack of proper maintenance which can range anywhere from improper alignment during the last repair to lack of oiling. Improper assembly during turn around can also cause equipment failure. Overall improper maintenance can account for a good percentage of equipment failure.
- **Ageing:** Every material wears down due to natural fatigue. Bearings and other moving parts have a rated life and need to be replaced at the end of their rated life. Preventive maintenance is employed to capture the gradual degradation of parts.
- **Secondary damage:** A fault that has occurred on a component if not corrected at the earliest, can progress fast and damage the component. Moreover, there can be secondary damages in other components which are coupled to the damaged component. For example, if a bearing failure, if not corrected at the right time, causes damage to the shaft coupled to it. Similarly, the damaged shaft can further damage the other bearings coupled to it. In some cases, the bearing with wrong specifications is installed on an equipment due to which the bearing is likely to fail faster than expected.

Performance rating applies only at the level of an individual symptom and must be combined with asset criticality to provide an overall assessment of risk for decision-making.

- A serious performance rating indicates a critical fault, where the instrument no longer functions. Failures at this level typically require immediate action from NRA (e.g. restart / cycle power to the device).
- A tolerable performance rating indicates a fault, where the instrument continues to function although with reduced performance or will have the potential to fail completely. Failures at this level typically require maintenance which can be up to 6 weeks.
- A minor performance rating indicates a minor fault, where the instrument can continue to function for a long term. Failures at this level typically require repairs during the annual campaign maintenance.
- Significant and critical performance ratings indicate that the instrument is not in use.

2.3 Getting Started with Asset Performance Management

2.3.1 Opening Asset Performance Management on the client computers

To open Asset Performance Management

1. Open Internet Explorer.

Type the URL in the Address box of the Internet Explorer (using the protocol configured for your server):

http://FullyQualifiedWebServerName:Port No/MES/Default.aspx

https://FullyQualifiedWebServerName:Port No/MES/Default.aspx

The **Connect to AM Server** dialog box appears.

2. In the **Username** box, type the user name for logging on to the application.

NOTE

If you are logging on to the Asset Performance Management application for the first time, use the same logon credentials that you used while installing the application.

3. In the **Password** box, type the password.
The **Honeywell Applications** home page appears.
4. Click **Asset Performance Management**.
The Asset Performance Management page appears.

2.3.2 Configuring Internet Explorer on the Client Computers

To ensure that Asset Performance Management functions as expected, configure your Internet Explorer settings to allow access to the required servers and settings.

To configure Internet Explorer on a client computer

1. Open **Internet Explorer**.
2. Choose **Tools** (Gear icon) > **Internet Options**.
3. On the **General** tab, in the **Browsing history** area, clear the **Delete browsing history on exit** check box.
4. Click the **Security** tab.
5. Do all of the following to configure the security settings:
 - a. Click **Custom Level**.
The Security Settings dialog box appears.
 - b. In the **Settings** list, select the following settings:
 - a. For **Allow Scriptlets**, select **Prompt**.
 - b. For **Automatic prompting for ActiveXcontrols**, select **Enable**.
 - c. For **Download signed activeX controls**, select **Prompt**.
 - d. For **Download unsigned activeX controls**, select **Prompt**.
 - e. For **Automatic prompting for file downloads**, select **Enable**.
 - f. For **Initialize and script activeX controls that not marked as safe for scripting**, select **Prompt**.
 - c. Click **OK**.
6. Do all of the following to add the servers to the Local intranet site:
 - a. On the **Security** tab, select **Local intranet**, and then click **Sites**.
 - b. Click **Advanced**.
 - c. If the Asset Performance Management (both fully-qualified and non-fully qualified domain names) do not appear on the list, for both the fully-qualified and non-fully qualified domain names, enter the address in the **Add this website to the zone** list, and then click **Add**.

NOTE

Ensure the servers are added by using the correct protocol (http or https).

- d. Click **Close** to close the Advanced local intranet settings dialog box.
 - e. Click **OK** to close the Local intranet dialog box.
- 7. Do all of the following to turn off Enhanced Protection Mode:
 - a. Click the **Advanced** tab.
 - b. Scroll to the **Security** section.
 - c. Clear the **Enable Enhanced Protected Mode** check box.
- 8. Click **OK** to close the Internet Options dialog box.

NOTE

When accessing Asset Performance Management, compatibility mode should be off.

ASSET PERFORMANCE MANAGEMENT CONCEPTS

This chapter explain the concepts used in Asset Performance Management

3.1 Assets and Events

3.1.1 Asset Categories

Asset Performance Management categorizes assets by Category, Class, and Type. Asset Categories are used to categorize assets at a very high level and are the first level of asset classification.

For example, all equipment in an industrial plant can be categorized into a single asset category named **Equipment** and all materials (for example, crude, gasoline) in a refinery can be categorized into single asset category named **Material**.

3.1.2 Asset Class

Asset classes are custom grouping of assets based on the common characteristics of different types of equipment. An asset class is a sub-set of an asset category. (In other words, you can define one or more asset classes under an asset category.) All assets below the site level have class and type definitions that allow you to create fault models for similar assets.

An example of an asset class is a **Pump**. Different types of pumps like the centrifugal pump, reciprocating pump, rotary pump, and so on are grouped under the **Pump** class.

3.1.3 Asset Types

Asset Type is the next level of grouping similar assets. An Asset class can have one or more asset types. The asset type represents the primary purpose of the asset, model, and the revision of the device. Asset Performance Management displays the common asset types and provides the ability to create custom asset types.

When you consider an example of a **Turbine** class, wind turbine, steam turbine, hydroelectric turbine, gas turbine, and so on are similar sets of asset types grouped or associated to a turbine class.

3.1.4 Asset

Asset is an entity at the finest level of granularity within the system, on which you want to:

- Perform data manipulations or calculations.
- Define rules (faults and symptoms) for event detection and notifications.
- View the status, collect metrics, or report information.

For example, the asset "A1-SE434437SE01" identifies a particular pressure transmitter in a plant. This equipment is of class "Process Sensor" and of type "Pressure Transmitter".

System assets are assets which are defined by including other asset types and calculations, or rules aggregating the information from sub assets to monitor.

Some examples of system assets: Compressor System (includes bearing, shaft, lubrication sub system, motor, or turbine), FCC Reactor section, and mining truck.

Logical assets are logical entities in a plant to define groups. These assets can also be configured with calculations to aggregate asset metrics.

Some examples of logical assets: Site, Plant, Unit, and Area.

3.1.5 Asset Status

Asset status changes in field due to planned maintenance, unexpected event and trip unit or put in service. In Asset Performance Management, the asset is in one of the following modes or states:

- **Monitor:** Asset calculations and rules are configured to execute. The asset is available for operation when required.
- **UnMonitor:** Asset calculations and rules are disabled. The asset is available for operation when required.
- **OutofService:** Asset calculations and rules are disabled. The asset is NOT available for operation.

Each asset status is mapped to an icon image, which is displayed in the asset hierarchy.

3.1.6 Asset Rating

The Asset Performance Management application uses the following 3 asset concepts to determine the relative importance of an asset and an event.

- Criticality
- Severity
- Priority

Criticality is an attribute (numeric value) associated to an asset. It denotes the relative importance of an asset based on financial and safety considerations. The values of criticality are:

- 1: None
- 2: Negligible
- 3: Minor Process
- 4: Major Process
- 5: Safety

Severity is an attribute (numeric value) associated with faults and symptoms. It indicates the relative performance impact (severity) of the fault and symptom on an asset. The values of severity are:

- 1: None
- 2: Minor
- 3: Required
- 4: Important
- 5: Serious
- 6: Critical

Priority is a parameter of an asset which indicates the health of the asset. The priority value is calculated by multiplying the numeric values of severity and criticality. The priority value determines the asset's icon color which indicates a fault. It enhances sorting capabilities within the asset's view.

3.1.7 Asset Attributes

Asset attributes are dimensions associated to a certain fault model for its identification.

For example, the attributes of a pump model are:

Attribute Name	Attribute Description
Voltage	Voltage

Attribute Name	Attribute Description
DesignPower	Power at current flow from curve
PropertyData	Includes Temperature, Specific Gravity and VapourPressure
Density	Density of the fluid being pumped
DeviationNPSH	Deviation between required and available net positive suction head (NPSH)
VapourPressure	Pressure of the vapour

3.1.8 Asset Links

Assets can be configured with a link to additional information in a document (pdf or word format) or a URL. For example, an asset can be a link to a saved Performance Curve to allow quick access to information about asset performance.

NOTE

Asset links cannot be configured in the UI but only via bulkexcel.

Asset Link Templates are URL defined with or without a standard parameter, such as asset name or display name. Asset Link Templates are associated to the asset type.

`http://UASServer/MES/AM/UI/home/Trend?assetName=[AssetName]`

Asset Links or link instances are configured on an individual asset manually, or created from the links associated to their respective asset template.

`http://UASServer/MES/AM/UI/home/Trend?assetName=Asset1`

3.1.9 Asset Hierarchy

In Asset Performance Management, an asset hierarchy is created to classify an asset into several segments and represent the segments in the tree view. Each segment in the hierarchy consists of an asset class, asset type, and an asset. The hierarchy enable you to view the whole structure of the plant in a graphical tree.

By default, "AMAsset" is the root node displayed in the Asset Performance Managementtree view. You can build various levels/segments and create the hierarchy. You can create multiple levels of the hierarchy in Asset Performance Management. A four level hierarchy is shown as a sample below.

- Area
- Plant
- Unit
- Asset

Site: A site occupies the highest level in the asset hierarchy and contains various plants.

Plant: A plant occupies the level immediately lower to the site in the asset hierarchy. A plant can contain many units.

Unit: A unit occupies the level immediately lower to the plant in the asset hierarchy. You must configure a plant before configuring a unit.

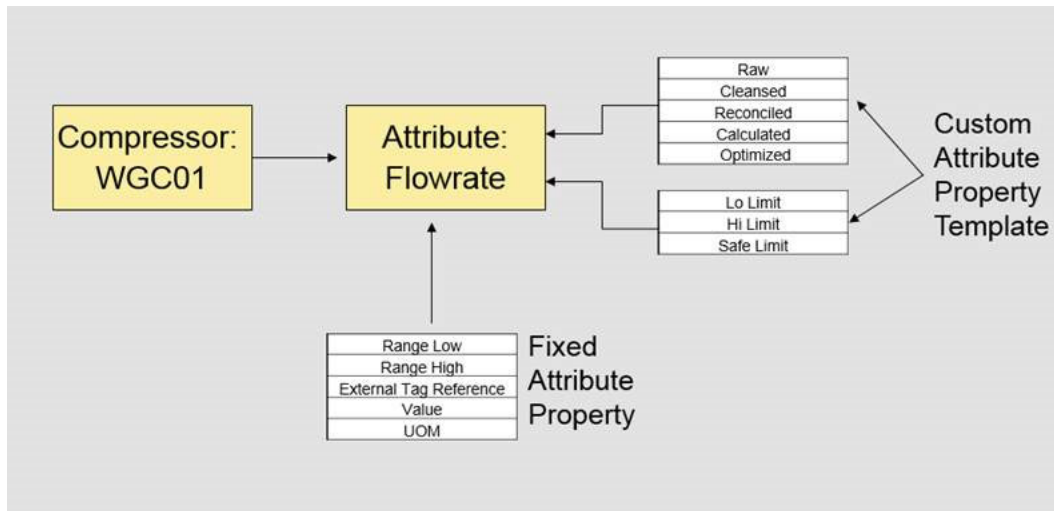
Equipment/Asset: An equipment occupies the lowest level in the asset hierarchy. You must configure a plant or a unit before configuring an equipment.

NOTE

It is not mandatory to create all these levels to build a hierarchy.

3.1.10 Attribute Property

Properties are additional information of attribute to support improved visualization and managing attribute.



Custom Properties can be used in calculation or fault model by specifying "Attribute.Property".

For example: FlowRate.Cleansed > FlowRate.HighLimit.

3.1.11 Fault

A fault is a state or condition which leads to a functional failure of an equipment. Faults are defined using symptoms.

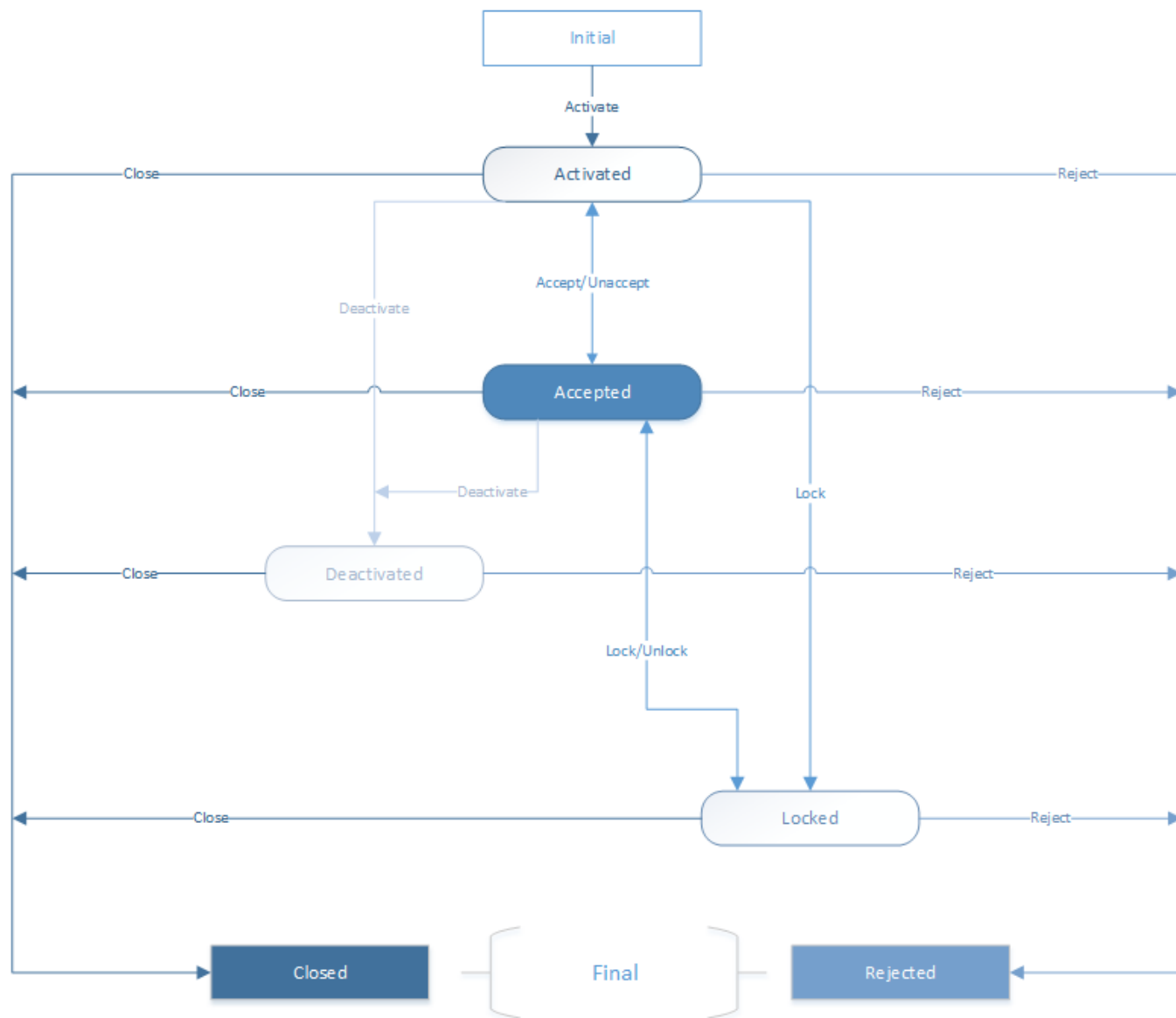
3.1.12 Symptom

A symptom is a state or condition which can potentially lead to a fault. Symptoms are defined using asset attributes.

3.1.13 Event Status

All events travel through transition statuses during their life cycle in Asset Performance Management:

- **Potential:** one or more symptoms is active, but the fault conditions have not been reached.
- **Active:** the expression associated with the event (for example, a fault or symptom) returns value of !=0.
- **Inactive (Deactivated):** the expression associated with the event (for example, a fault or symptom) returns value of 0.
- **Accepted:** the event has been accepted by a user to address.
- **Locked:** event is locked by the user so the state does not change irrespective of the fault or the symptom condition. This is typically used to avoid duplicate events when the event is being addressed in field.
- **Closed:** event has been closed by the user (includes a record of the reason code, comments).
- **Rejected:** event has been falsely identified by the system and rejected by a user (possibly marked as a false positive).



For example, whenever a fault and one symptom are active in the system, the fault status gradually changes to a potential fault. This fault moves to an active fault status only when the condition of the fault is satisfied. The repair actions associated to these faults are activated and the respective groups are notified (based on the notifications configured).

NOTE

All the potential fault will be rejected by the application if it goes to active or inactive state from potential fault state.

The rejected potential fault information will be displayed under the Event History section of Event Monitor page.

3.1.14 Fault Models

In a process plant, FMEA is performed to determine the failure modes and its causes for an equipment. The outcome of the FMEA is converted to a Fault-Symptom model in Asset Performance Management. The failure modes from the FMEA analysis are converted to faults and all the causes from the FMEA analysis are converted to symptoms.

The Fault-symptom models from Allied Reliability is used to build fault -symptom models for process equipment that are widely used.

Other than the fault-symptom model, Asset Performance Management also supports calculations. Calculations are used for process performance calculations and KPI calculations.

The Asset Performance Management library consists of pre-built models for process equipment. The built-in process performance calculation models are available for Pumps, Compressors, Heat Exchangers, Steam Turbines, and Gas Turbines.

A Performance Monitoring model is used to calculate the operating (actual) performance output, design (expected) performance output, and deviation between operating performance and design performance output. Fault models can be configured on process performance outputs like efficiency, capacity, and so on.

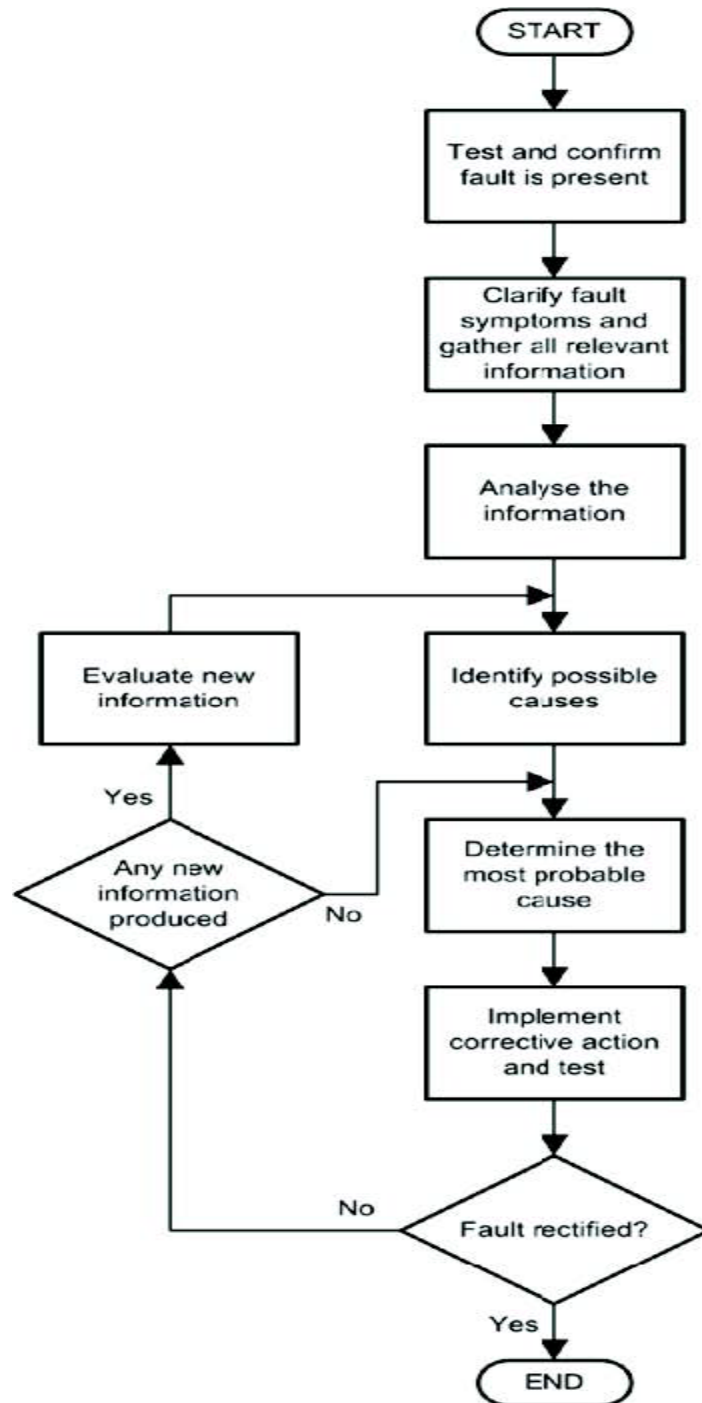
A vibration model is also available for monitoring vibrations of rotating equipment.

3.1.15 Fault and Symptom Management Process

When a fault or a symptom is identified in Asset Performance Management, the information about the state of the affected equipment, the time at which the fault was raised, and the status of its resolution is displayed. You must perform the following steps to act on the fault.

1. Testing: test the equipment and confirm if any faults are present in the system.
2. Gathering information: gather all relevant information pertaining to the fault such as information on how and when the fault occurred in the system.
3. Analyzing the information: involves analyzing the test results.
4. Identifying fault: involves separating effects from their causes and identifying faults based on their criticality. The most critical fault is identified for rectification.
5. Removing fault: involves removing the identified fault using corrective action or the right type of measuring technique. If this fault is not removed from the system, there is a chance that it might reoccur.
6. Rectifying fault: involves identifying the correct rectification procedures. Some of the rectification procedures include:
 - Inspection and assessment
 - Adjustments
 - Replacements
 - Repair

- Substitution



3.1.16 Health Index

Health Index refers to the physical health of an equipment. In Asset Performance Management, for all the rotating equipment, the health indices are computed by custom calculations based on the limit differences between current value and expected operating values for various mechanical/elsectical condition parameters. The value of the health index is 0 to 1 where 0 is healthy and 1 is unhealthy.

3.1.17 Performance Index

Performance Index refers to the process performance of an equipment. In Asset Performance Management, the advanced equipment models for assets such as turbine, compressor and pumps calculates actual and expected performance at current operating conditions. The same can also be configured for other asset type to compare performance of various assets across the fleet. The value of the performance index is calculated based the deviation of actual and expected between 0 to 1, where 0 is bad performance and 1 is good performance.

Example

$$\text{PerformanceIndex} = ((\text{ExpectedHead} - \text{OperatingPolyHead}) / \text{ExpectedHead})$$

3.1.18 Energy Index

Energy Index refers to the energy efficiency of an equipment. The energy term can be shaft power or heat duty and so on which is calculated by advanced models for asset type such as turbine, compressor and pumps calculates actual and expected energy consumption at current operating conditions. The same can also be configured for other asset type to compare performance of various assets across the fleet. The value of the energy index is calculated based the deviation of actual and expected between 0 to 1, where 0 is bad performance and 1 is good performance. Note: Thermo dynamic property packages are available to compute energy consumption through engineering scripting calculations.

Example

```
OperatingShaftPower= abs((ExpectedShaftPower - OperatingShaftPower) /  
OperatingShaftPower)
```

```
Duty= E101Feed.Flow * E101Feed.Cp *( OutletT-inletT)
```

3.2 Monitoring

Asset Performance Management contains pre-built contents to monitor process and assets to improve overall performance and reliability.

3.2.1 Process Monitoring

Process engineers monitor the process in each asset or unit to understand changes and their impact to production targets or equipment performance. Process monitoring involves the following on selected major equipments or units:

- Material balance
- Energy balance
- Thermodynamic parameter tracking
- Catalyst performance

3.2.2 Performance Monitoring

Process and reliability engineers monitor the equipment performance of various equipments to identify actions to improve the reliability and extension of asset life. Examples of performance monitoring involve the following:

- Rotary equipment performance (for example, pumps or compressors)
- Static equipment performance (for example, furnaces or valves)

3.2.3 Energy/Utility Monitoring

Process engineers identify bad actor assets in the plant hierarchy with respect to energy based on actual energy deviation from dynamic energy target (calculated in real time based on current loading conditions). Unit or plant level energy balance based on consumed energy, generated energy, and imported energy is calculated and reported.

3.2.4 Condition Monitoring

Fleet monitors and mechanical or reliability engineers monitor the condition of assets which involves the conditions leading to failure or after failure effects. Typically this involves a combination of logic or rule engines to detect state of the equipments based on vibration, current, and various other conditions around equipments.

3.2.5 Instrument Monitoring

Instruments and associated system such as FIM or C200 are monitored based on the data and events from an Experion system. Smart instruments provide basic field events which Asset Performance Management groups into more actionable events and tracks. In addition to smart events, sensor faults such as bias, drift, and outliers are tracked using the basic statistic function on real time data streams and controller performance calculated, and an interface with Control Performance Monitor can be configured for more details.

3.2.6 Safety Monitoring

Monitoring the healthiness of instruments and process provides data for evaluating the following:

- Safety Instrumented function loop tracking
- SIL level
- Safety risk (increase from design)
- Layer of protection healthiness / status tracking

3.2.7 Limit Monitoring

Limit Monitoring in Asset Performance Management provides basic exception monitoring:

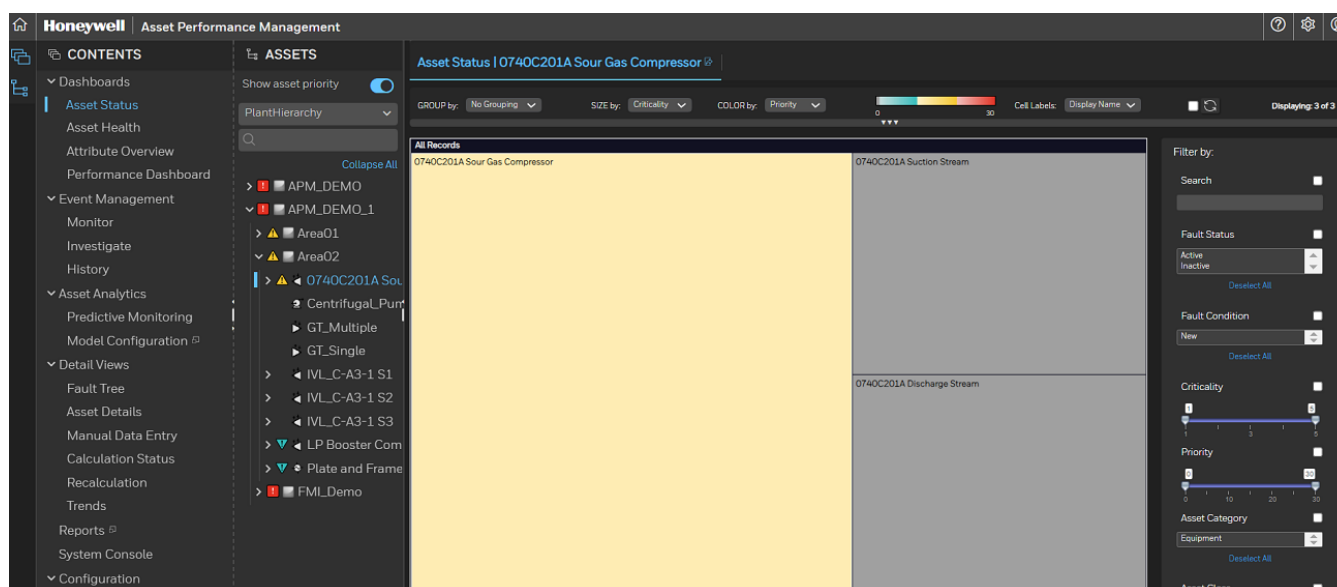
- Monitor variables against fixed limits or dynamic limits or periodic baseline limits. Limit Types supported are: low limit, high limit, and target limit.
- Generate simple events to manage events without requiring configuration of a fault model for the fixed limits, or dynamic limits or statistical limits violation in a proactive technical monitoring application.

The following different types of limits are supported:

- **Fixed Limits:** Fixed limits are limits obtained from design and optionally imported from PHD or Limit Repository. Limitation: Single mode is supported. For example, you can monitor reactor temperature and product yield based on the design limits.
- **Dynamic Limits:** Dynamic limits are limits those are obtained from calculation within Asset Performance Management or from external system. Limitation: Mode of operation is handled within calculation or external system based on source of limits. For example, for Compressor Surge Monitoring, which requires minimum and maximum flow obtained from manufacturer performance curve for specific operating conditions.
- **Statistical Limits:** Statistical limits are control limits derived from specific set of periodic baseline. For example, in batch operation reference or a golden batch is identified, control limits derived from those are used as a reference to control process.

USING ASSET PERFORMANCE MANAGEMENT

The following figure provides information on various elements of the Asset Performance Management user interface.



1. **Asset Tree:** Contains assets that represent assets and its types in a hierarchical view. Show or Hide the tree using the sidebar icon .
2. **Contents Tree:** Contains the configured displays on the systems, including dashboards, event management displays, and asset displays. Show or hide the tree using the sidebar icon .
3. **Sidebar Icons:** Contains the side bar icons that denotes the function of Asset Performance Management. A tool-tip provides the name of the function when you place the mouse pointer on the icon.
4. **Workspace:** Contains the configuration panes, trends, report, and messages based on the task.
5. **Status bar:** Displays the result of the actions performed on the menu commands.

NOTE

For asset type and asset class based grouping in asset status page, you need to set the following settings "showAssetTypeAndClassGroupingInAssetStatus" to true in ILMSettings.config file.

In case of split server, the following configuration has to be done in Webserver.

Navigate to the following install location :

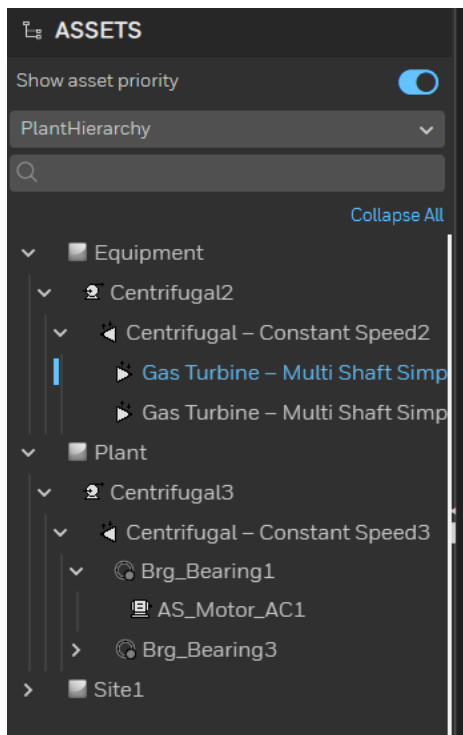
<InstallLocation>\Honeywell\MES\AssetManager\WebServer\WUI

4.1 Asset Tree

The asset tree is a hierarchical display of the assets in your plant hierarchy. Selecting an asset on the tree sets the context for the pages displayed in the workspace. When configured, the assets are shown with icons indicating the asset status and asset type.

Switching on the **Show asset priority** displays the priority level in risk matrix of each asset.

Example of Asset Tree



TIP

Click **Collapse All** to shrink all the nested hierarchy in a at a time.

By default, the asset tree shows all the assets on the system. You can filter the tree using the asset group selector or the search bar.

To filter the asset tree

Do one of the following:

- 1. From the asset group list, select the asset group to be displayed.
 - 2. In the Search box, begin typing the name of the assets to display.
- The tree updates to show only the assets matching the filter selections.

To open an asset link from the asset tree

- 1. Right-click the asset in the asset tree and then select the asset link from the menu.
- The asset link opens in the workspace. (See “To open asset links for PDFs” on page 25 for opening a PDF asset link.)

4.1.1 Asset Status Icons

The asset status icons indicate the current status of the assets. If a branch is collapsed, the icon indicates the highest asset status on any asset in the collapsed branch.

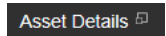
Priority	Active	Inactive
High		

Priority	Active	Inactive
High, Potential		
Medium		
Medium, Potential		
Low		
Low, Potential		
Out of Service		
Unmonitored		

4.2 Contents Tree

The content tree is a list of all displays configured on the system. Selecting an item on the content tree opens the related display. For example, selections under the Event Management category open the event viewer, to the context currently selected in the asset tree.

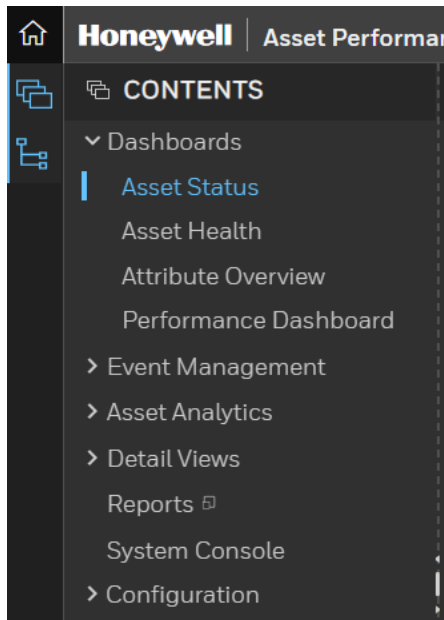
Some of the views in the contents tree can be opened either in the current tab, or in a new tab. If the option exists, the Open in New Tab icon appears when you hover the mouse over the selection in the tree.



Click the name to open the display in the current tab and click the icon to open the display in a new tab.

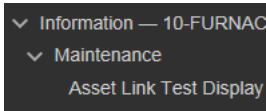
NOTE

Some displays can only open in a new tab. In this case, the Open in New Tab icon is visible at all times.



In addition to the default Contents tree listings, custom links configured for the selected asset are displayed at the bottom of the tree as information links. They are displayed in the following format:

- Information – [AssetName]
 - [Category]
 - [Asset Link Display Name]



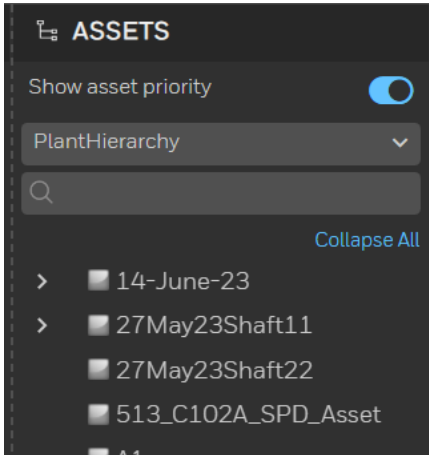
If an asset link opens a PDF, it can be blocked by system settings. Update the Internet Explorer security settings to ensure the PDF links can open as expected.

To open asset links for PDFs

1. In Internet Explorer, on the **Tools** menu, select **Internet Options**.
2. Select the **Privacy** tab.
3. Click **Sites**.
4. Type the server name and then click **Allow**.

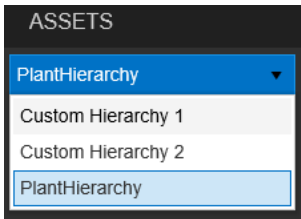
4.3 Viewing Custom Hierarchies

If your system is configured to use custom hierarchies, you can change the asset tree to use one of these alternate hierarchies. Once a custom hierarchy is selected, pages in Asset Performance Management, such as the Asset Status, Attribute Overview, and Event Management views will display the assets and events related to the assets shown in the custom hierarchy (including using the custom hierarchy’s parent and child relationship).



To select a custom hierarchy

On the asset tree, from the hierarchy bar, select the name of the custom hierarchy to use.



PERFORMANCE MANAGEMENT

Performance Management includes the following review tasks:

- Asset Status
- Attribute Overview

5.1 Performance Management

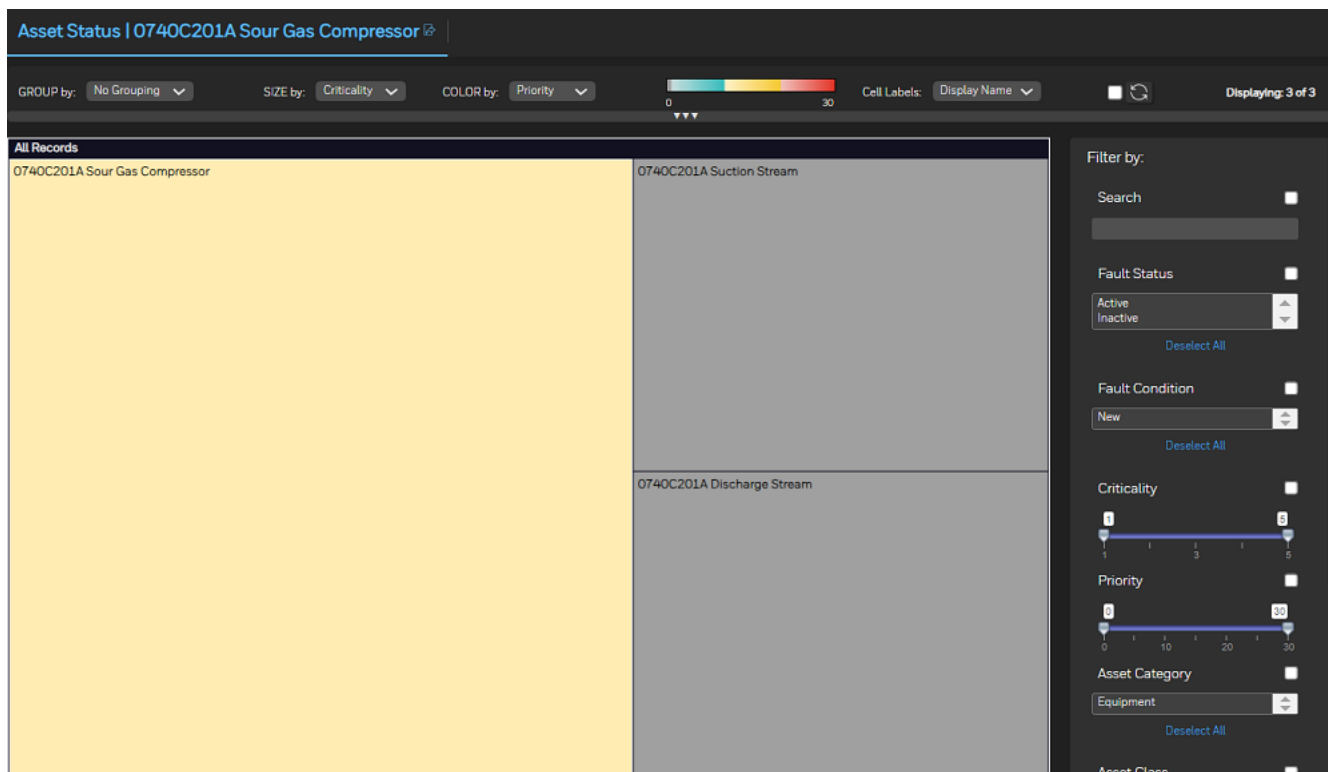
5.1.1 Asset Status

The Asset Status displays a treemap showing the criticality and priority values of the item currently selected on the asset tree. This allows you to view possible issues on performance from a high-level, and then drill-down to view more detailed information.

5.1.2 Asset Health Dashboard

The Asset Health Dashboard provides an overview of the frequently failing assets, the asset status based on priority of the fault, the asset failures, asset conditions, and it also provides a widget for even summary. You can use the Plant Health to get a quick overview of the health for the selected asset. You can also navigate to Attribute Overview and Event Summary pages from here.

5.2 Using the Treemap



To Open the Treemap

1. On the asset tree, select an asset to set the context.
2. On the Contents tree, under s, select **Asset Status**.

To Zoom In or Out on the Treemap

1. Hover the mouse over a rectangle for an asset or group.
The asset details appear in a box.
2. Click the rectangle to open the context menu and the ability to interact with the asset details box.
3. Do the following:
 - a. To zoom in, click **Zoom In**.
The treemap zooms in to the selected group.
 - b. To zoom out, click **Zoom Out**.
If the treemap has not been zoomed in at least once, the Zoom Out option is not listed. Do one of the following:

To Reload Treemap Data

1. Do one of the following:
 - a. To manually reload the treemap data, click the Reload Data  button.
 - b. To have the treemap automatically reload data on a one minute interval, select the check-box beside the Reload Data button.

To View Asset Details or Context Menus on the Treemap

1. Hover the mouse over a rectangle.
The asset details appear in a box.
2. Click the rectangle to open the context menu and the ability to interact with the asset details box.
3. Do any of the following:
 - a. To open the [Asset Status](#) for the child assets, select **Drilldown Asset Status**.
 - b. To open the [Attribute Overview](#) for the selected asset, click **Attribute Overview**.
 - c. To open the [Event Monitor](#) for the selected asset, click **Event Monitor**.
 - d. To open the [Fault Trend](#) for the select Asset, click **Fault Tree**.
 - e. To open the [Trends](#) for the selected Asset, click **Trends**.

5.2.1 Reading the Treemap

A treemap is a visual indication of the priority and criticality of an asset. Every asset on the treemap is represented by a rectangle. The size and colour of the rectangles are controlled by the selected properties.

For example, you can set the criticality of an asset to be indicated by the size of the rectangle and the priority of the asset to be indicated by the color of the rectangle.

In this case, the higher the criticality value, larger the size of the rectangle and lower the criticality value, smaller the size. You can read the criticality values of an asset by viewing the tool tip displayed on the rectangle. The following are the criticality values of an asset:

- 1- None
- 2 - Negligible
- 3 - Minor
- 4 - Major
- 5 - Safety

In the same case, the following would be shown for the priority values:

- **1 to 10** - Indicates that the health of the asset is “Normal” and is displayed in green.
- **11 to 20** - Indicates that the health of the asset is “Warning”, and is displayed in yellow.
- **Above 20** - Indicates that the health of the asset is “Urgent”, and is displayed in red.

Priority value is calculated by multiplying the numeric values of severity of a fault and criticality of an equipment. Priority indicates the relative importance of an asset with another asset. When the asset is selected, its priority value is displayed on the treemap when you hover the mouse pointer on the treemap.

5.2.2 Treemap Settings

The treemap settings control which properties are used to display the rectangles. You can change the property and color settings to update the information shown in the treemap.

NOTE

For asset type and asset class based grouping in asset status page, you need to set the following settings "showAssetTypeAndClassGroupingInAssetStatus" to true in ILMSettings.config file.

In case of split server, the following configuration has to be done in Webserver.

Navigate to the following install location :

InstallLocation\Honeywell\MES\AssetManager\WebServer\WUI

To Change the Treemap Settings

1. To change the property used to group the rectangles, from the **GROUP by** list, select one of the available properties. For example, Asset Class, Asset Category, Asset Type, Criticality, Priority, Location, or No Grouping.
2. To change the property indicated by the size of the rectangles, from the **SIZE by** list, select one of the following:
 - a. Criticality
 - b. Priority
 - c. Max Severity
 - d. Same Size
3. To change the property indicated by the color of the rectangles, from the **COLOR by** list, select one of the following:
 - a. Max Severity
 - b. Status
 - c. Priority
 - d. Criticality
4. To change the property used to label the cells, from the **Cell Labels** list, select one of the following:

- a. Display Name
- b. Description

5. To reload the treemap data, click the Refresh button, or select the Auto Refresh check box to have the treemap data refresh on a regular basis.

5.2.3 Treemap Filters

The assets shown on the treemap can be limited through the use of filters.

To filter the treemap

1. In the Filter panel, use any of the following that apply (multiple filters can be applied at the same time):
 - a. To filter by a string or number, in the Search box, type the string that must be present for the asset to be displayed.
 - b. To filter by the current fault status, in the Fault Status list, select the fault status to display.
Press the CTRL key to select multiple individual statuses, or the SHIFT key to select all items between the first and second click.
 - c. To filter by the current fault condition, in the Fault Condition List, select the fault condition to display.
 - d. To filter by criticality, use the Criticality sliders to select the criticality value range. (See “Reading the Treemap” on page 30 for a list of the criticality associated with each value.)
 - e. To filter by priority, use the Priority sliders to select the priority value range. (See “Reading the Treemap” on page 30 for a list of the priorities associated with each value.)
 - f. To filter by asset category, from the Asset Category, select the asset category to display.
 - g. To filter by the asset class, in the Asset Class list, select the asset class to display. Press the CTRL key to select multiple individual statuses, or the SHIFT key to select all items between the first and second click.
 - h. To filter by the asset type, in the Asset Type list, select the asset type to display. Press the CTRL key to select multiple individual statuses, or the SHIFT key to select all items between the first and second click.
 - i. To filter by a filter label, in the Filter Label box, type the filter label to display.

To clear a filter on the treemap

1. In the Filter panel, do either of the following to clear a filter:
 - a. To clear the filter, but maintain the selections/settings, clear the check box next to the filter name.
 - b. To clear the filter and remove the selections/settings from the lists, under the list, click Deselect All.

5.3 Attribute Overview

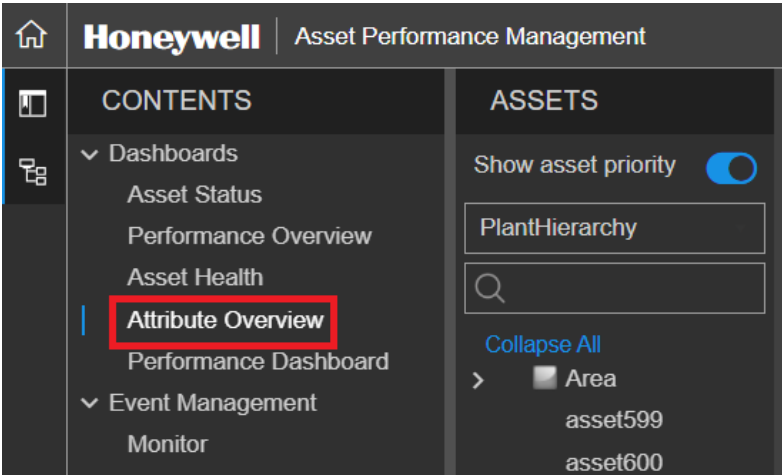
The Attribute Overview page displays the attributes of the selected asset.

Using the Attribute overview page, you can:

- View the attribute values
- Manually provide values to attributes
- View the quality of the input data received for the attribute
- View the historical trend of the attribute

5.3.1 Opening Attribute Overview

- 1. From the Honeywell Application landing page, click **Asset Performance Management**.
The **Asset Performance Management** page appears.
- 2. From the Dashboards drop-down menu in the **CONTENTS** pane, click **Attribute Overview**.

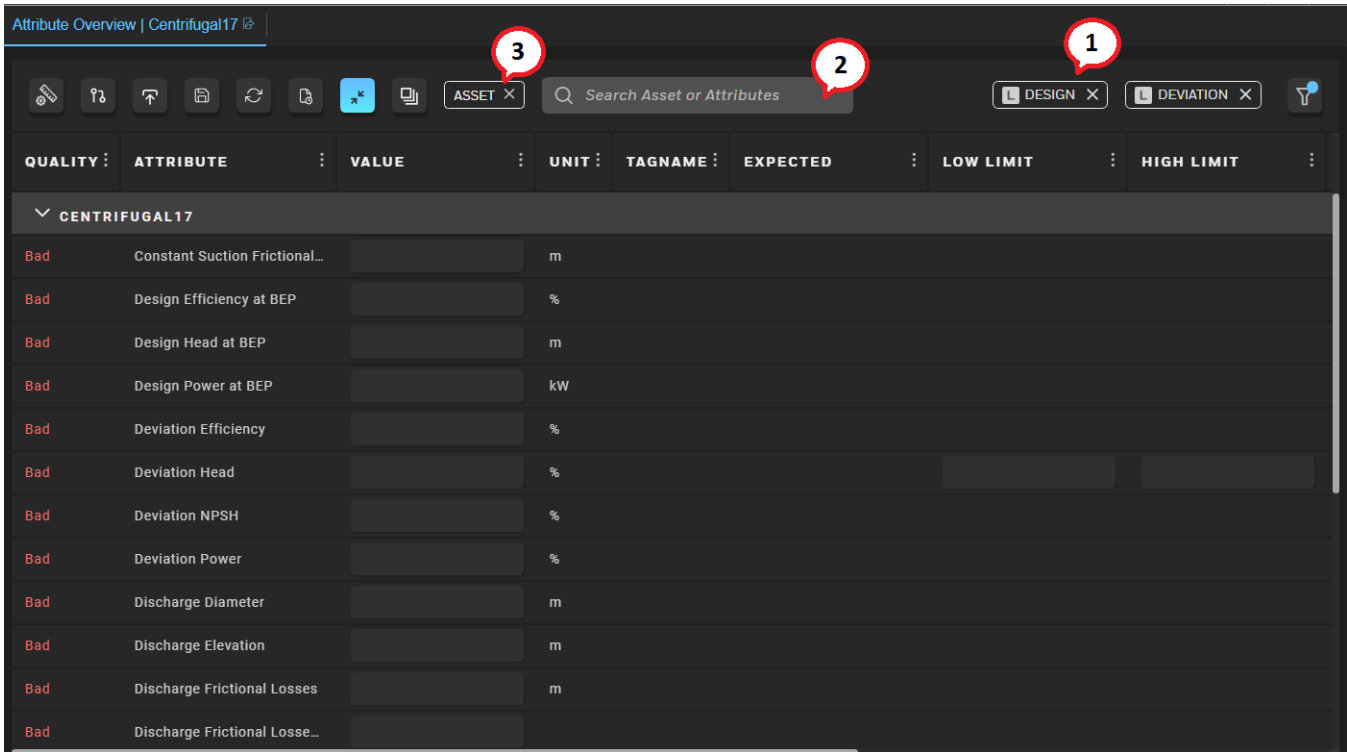


- 3. Select the asset for which you want to view the attribute overview page.
The attribute overview page for the selected asset is displayed in the right pane.

QUALITY	ATTRIBUTE	VALUE	UNIT	TAGNAME	EXPECTED	LOW LIMIT	HIGH LIMIT	DEVIATION	THRESHOLD LOW
Bad	CombustorEff		%					^NaN %	
Bad	CombustorHeatOfCo...		kJ/hr					^NaN %	
Bad	CombustorJacketHea...		kJ/hr					^NaN %	
Bad	CombustorPresLossF...							^NaN %	
Bad	CombustorRatedEff		%					^NaN %	
Bad	CompressorDisPresC...							^NaN %	
Good	CompressorInletArea	90.000	m2					^Infinity %	
Bad	CompressorMechEff		%					^NaN %	
Bad	CompressorOption							^NaN %	
Bad	CompressorRecycleB...							^NaN %	
Bad	DesignCompressorIs...		%					^NaN %	
Bad	DesignCompressorPo...		%					^NaN %	
Bad	DesignCompressorS...		kW					^NaN %	

NOTE
By default, 90 attributes are displayed and on scrolling down, the next set of 90 attributes are displayed. Contact the system administrator if you want to change the default number of attributes displayed in the grid.

5.3.2 Attribute Overview Interface



Tool-bar

Button	Name	Description
1	Filter Label	Displays the filter labels selected from the filter panel.
2	Search the Asset or Attributes	Filters the attributes based on asset or attribute. NOTE If filter labels are selected, then the search is limited within the selected filter labels. To search across all attributes, click Clear All in the filter panel.
3	Group by	Groups the attributes based on category, asset, or filter labels.
	Asset Type Level UOM	Displays the attribute values with the unit of measurements at the asset type level. By default, attribute values are displayed for the unit of measurement at the asset level.
	Include Children	Displays the attributes of the child assets of the selected asset. NOTE The list of filter label selections will vary depending on whether include children is enabled or not. Disabling the include children button automatically removes the filter labels selected for the child assets in the main grid and filter panel.

Button	Name	Description
		TIP If the selected asset is a root node in the asset hierarchy and has many child assets, clicking the Include Children button can result in a Timeout error and and this might result in attributes not being retrieved.
	Export to Excel	Export the current list of asset attributes to an Excel file.
	Save Changes	Saves any changes made to items in the grid. TIP The number on top of the save icon indicates the number of attributes with unsaved changes.
	Refresh Attributes	Refreshes the content displayed in the grid.
	Show Properties	Displays the properties for the selected attribute in the property panel. Alternatively, double-click on the attribute for viewing the property panel. NOTE The property panel is displayed only if the selected attribute has properties associated with it.
	Expand All	Toggle for expanding all groups or collapsing all groups on the grid. This button is enabled only if you have selected the Group by option.
	Group by	Group the results based on Assets or Category or Filter Label. NOTE Group by icon is disabled when the number of attributes for the asset exceed 600. You can, however, view the attributes in a flat list on the grid. If the attribute count for the asset exceeds 600 and you wish to view the attributes in a grouped manner, you can increase the count in the ILM settings file in the webserver: Navigate to <Install Location>:\Honeywell\MES\AssetManager\WebServer\WUIILMSettings.config and update the attributeOverviewGroupSoftLimit value.
	Filter By	Filter and retrieve attributes based on Attribute Category or Attribute Filter Label or both. Refer to Adding a Filter in Attribute Overview for information on using the Filter panel.

Grid

The attributes of the asset are displayed.

Column Name	Description
Quality	The quality of the data for the asset attribute: Good (Green), Bad (Red), or Uncertain (Yellow).

Column Name	Description
Attribute	Attributes are dimensions associated to a certain fault model for its identification. Attribute name column can be sorted based on ascending or descending order. Click on the three dots provided in the Value column and select Sort ascending or Sort descending. NOTE The attributes are displayed in the following order: • Descending order of Deviation percentage • Infinity % • NAN
Value	The current value of the asset attribute. NOTE • The Value is displayed in red color if it does not fall in between the low and high limits. • The corresponding limit the value is falling out of, is displayed with an red background. The color can be customized if needed. Contact your system administrator if you must change the color. • The attributes are displayed in the descending order of deviation percentages.
Unit	The unit of measure for the value. For information on the units of measure settings, see “Units of Measure” of the Asset Performance Management Configuration Guide.
Expected	Asset attribute property: Expected value.
Low Limit	Asset attribute property: Low Limit value.
High Limit	Asset attribute property: High Limit value.
Deviation	The deviation is calculated as the difference between the current value and the expected value. NOTE • The attributes are displayed in the descending order of deviation percentages. • The deviation color can be customized if needed. Contact your system administrator if you must change the color.
Threshold Low Limit	The DCS low limit.
Threshold High Limit	The DCS high limit.
Historical	The short history of asset attribute (displayed as a sparkline) over the last 2 hours by default. Clicking the trend opens the Trends page in a new tab. The attribute value and limits are trended for the last 2 hours by default.

Column Name	Description
	NOTE The default period for trending the attribute values can be changed by contacting the system administrator.
TagName	The Tag Name of the Asset Attribute.
System Attribute	By default, the system attribute column is disabled. To enable the system attribute column, navigate to the below path, open the ILMSettings.config file, and then update the hideSystemAttributes key value to false . <Install Location>:\Program Files (x86)\Honeywell\MES\AssetManager\WebServer\WUI\ILMSettings.config
Min	The minimum recorded value in the last two hours for the asset attribute.
Max	The maximum recorded value in the last two hours for the asset attribute.
Avg	The average recorded value in the last two hours for the asset attribute. NOTE Contact the system administrator if you want to change the default time interval for Min , Max and Avg .
Category	The category to which the asset attribute belongs.
Asset	The asset for which the attribute is displayed. You can sort the attributes based on the Asset name by clicking on the three dots on the Asset name and clicking Sort ascending or Sort descending.
Description	The description of the asset attribute.
Time Stamp	The timestamp when the attribute value or properties was last updated. NOTE The time displayed is local client time.

5.3.3 Columns in the Attribute Overview

Default Columns

Below are the default columns displayed in the grid:

Quality, Attribute, Value, Unit, Expected, Low limit, High limit, Deviation, Threshold Low Limit, Threshold High Limit, Historical, and Tag Name

You can view additional columns on the grid by clicking the dots on any column header and selecting the necessary column.

Sorting of Attributes or Asset Values

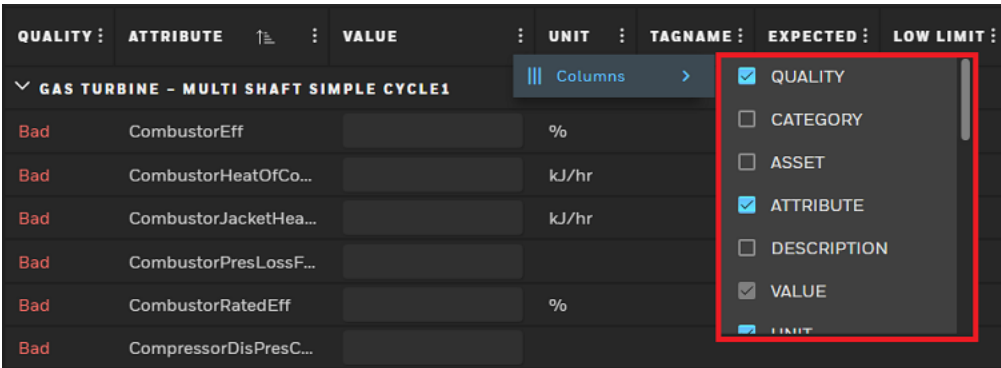
By default, the Attribute column or Asset column is displayed in the following order:

- Descending order of Deviation percentage
- Infinity percentage
- NAN

The first click on Attribute column or Asset column will sort it in Ascending order, the second click will sort the column in Descending order, and the third click will revert it back to default order.

Show or Hide Columns

1. Click the three dots to display the columns pop-up.



2. Click **Columns** and then select or clear the checkboxes to show or hide the columns.

Change the Column Order

Click and drag the column over the column headers, and drop it in the new position.

5.3.4 Exporting Attributes to an Excel File

Open the attribute overview page for an asset and click  icon to export the attributes to an Excel.

NOTE

- The attributes are exported to an excel file with the default columns of the Attribute Overview page,
- Columns with calculated values on the attribute page are not exported to the excel file. For example, Deviation.

5.3.5 Filtering Attributes

Default Filter Labels

Below filter labels are selected by default, and the attributes having these filter labels are displayed in the grid:

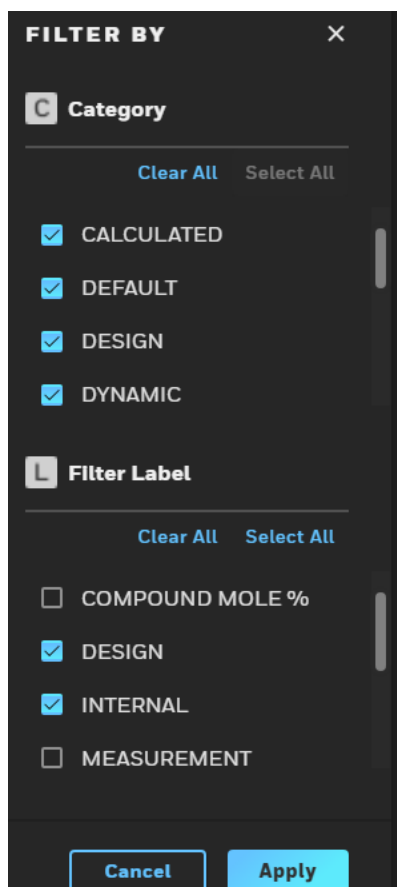
KPI, Input, Output, Key Metrics

NOTE

Contact the system administrator if you want to change the filter labels based on which the attributes must be displayed.

Using Filter

1. Click the  icon to open the filter panel.



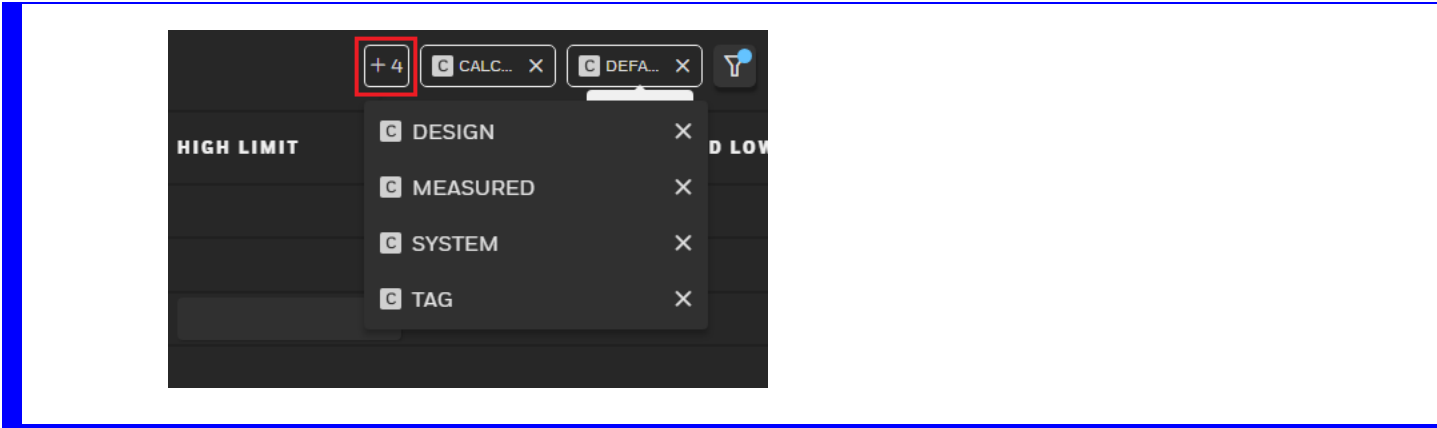
TIP

- Blue mark on the filter icon means a filter has been applied to the displayed attribute overview page.
- Remove the filter to display all the attributes of the selected asset.

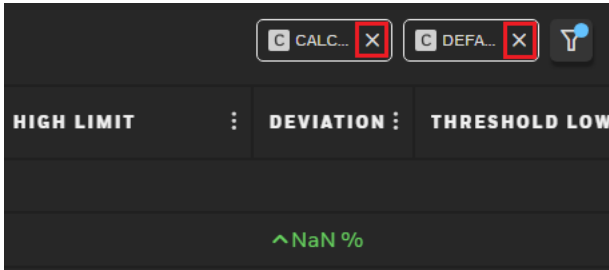
2. From the **Category** list, select the category of attributes that you want to view.
The attributes based on the selected category are displayed.
3. From the **Filter Label** list, select the filter labels of the attributes you want to view..
The attributes in the selected labels are displayed.

NOTE

- Chips on the top right corner of the page indicate the filter labels you have selected.
- The display of these chips is dynamic based on the screen resolution. The labels that cannot be fit in the main page is displayed inside a numbered chip, as shown in the below example.



4. Click **Apply**.
The list of attributes for the selected asset is displayed as per the selected filters.
5. To stop seeing attributes of a filter label, click the cross button on the respective filter label chip.



5.3.6 Grouping Fields in Attribute Overview

You can group the attributes displayed on the page based on Asset, Category or Filter labels.

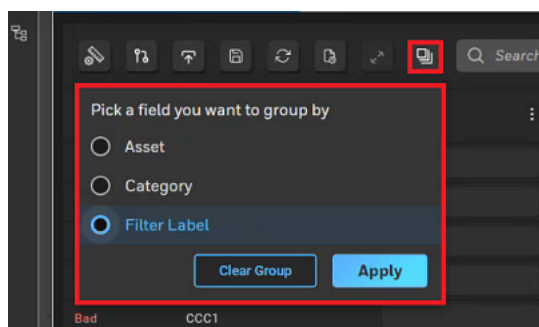
NOTE

Group by icon is disabled when the number of attributes for the asset exceed 600. You can, however, view the attributes in a flat list on the grid.

If the attribute count for the asset exceeds 600 and you wish to view the attributes in a grouped manner, you can increase the count in the ILM settings file in the webserver:

Navigate to <Install Location>:\\Honeywell\\MES\\AssetManager\\WebServer\\WUI\\ILMSettings.config and update the **attributeOverviewGroupSoftLimit** value.

1. Click the  button to open the group by panel.



2. Select the field based on which the results must be grouped and click **Apply**.
 - a. **Asset**: Group the fields based on the asset name.
 - b. **Category**: Group the fields based on the attribute category.
 - c. **Filter Label**: Group the field based on the filter labels.
3. Click **Apply**.
The attributes are displayed as per the selected group. The first group is open by default.
4. Toggle with the expand group  button to expand all the groups or to collapse all groups.

5.4 Asset Health Dashboard

5.4.1 Introduction

The Asset health dashboard is a one stop shop, which gives an accurate representation of all assets that needs immediate attention to bring the assets back to the optimum health and performance. The dashboard enables you to take actionable steps immediately, avoiding disruptions, and downtime.

5.4.2 Objective of Asset Health Dashboard

At a high level the objective of the asset health dashboard can be categorized into 3 stages:

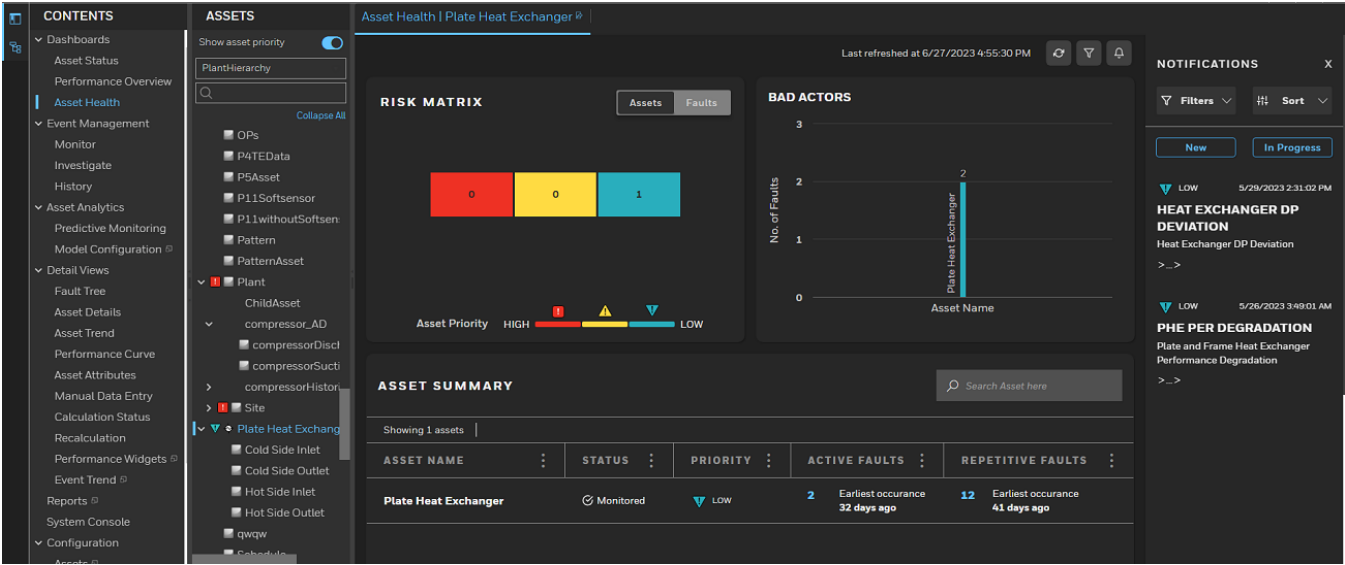
- Identify - identify the top assets/faults that are impacting the plant's health and performance.
- Investigate - drill-down further on the identified asset/fault to identify the root cause by using common tools like Event Detail, Event trend, attribute overview etc.
- Act - take necessary actions to rectify the asset to improve the asset's health and performance, which in turn increases the plant's up-time and productivity.

5.4.3 Workflow for Asset Health Dashboard

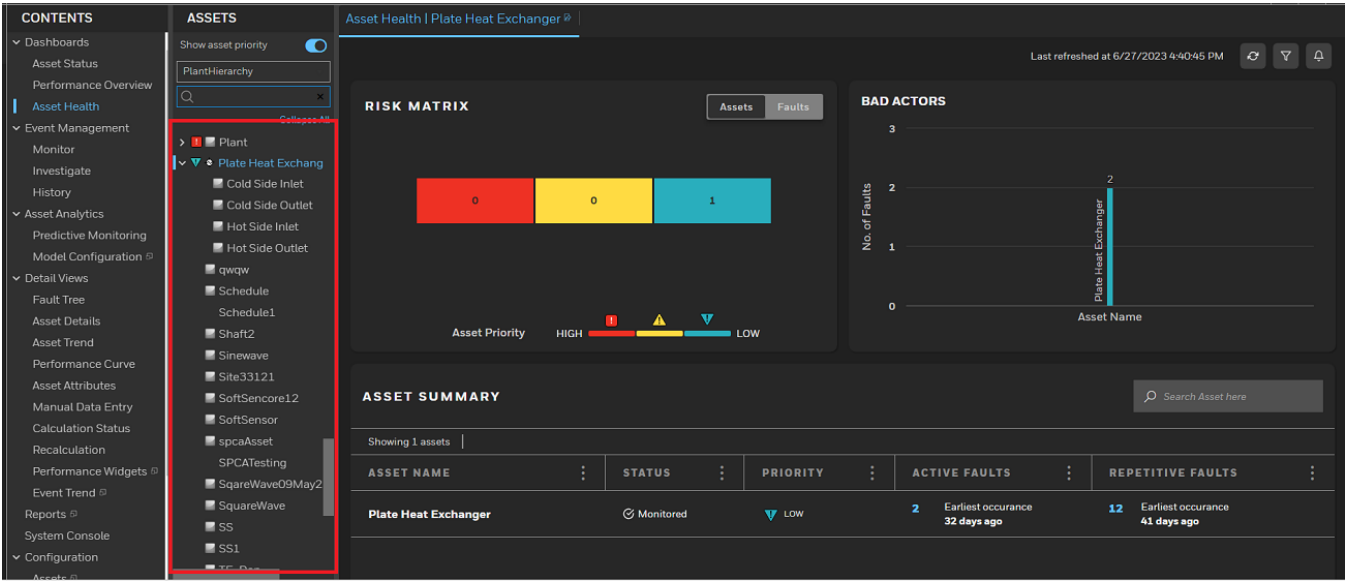


5.4.4 Knowing the Interface of Asset Health Dashboard

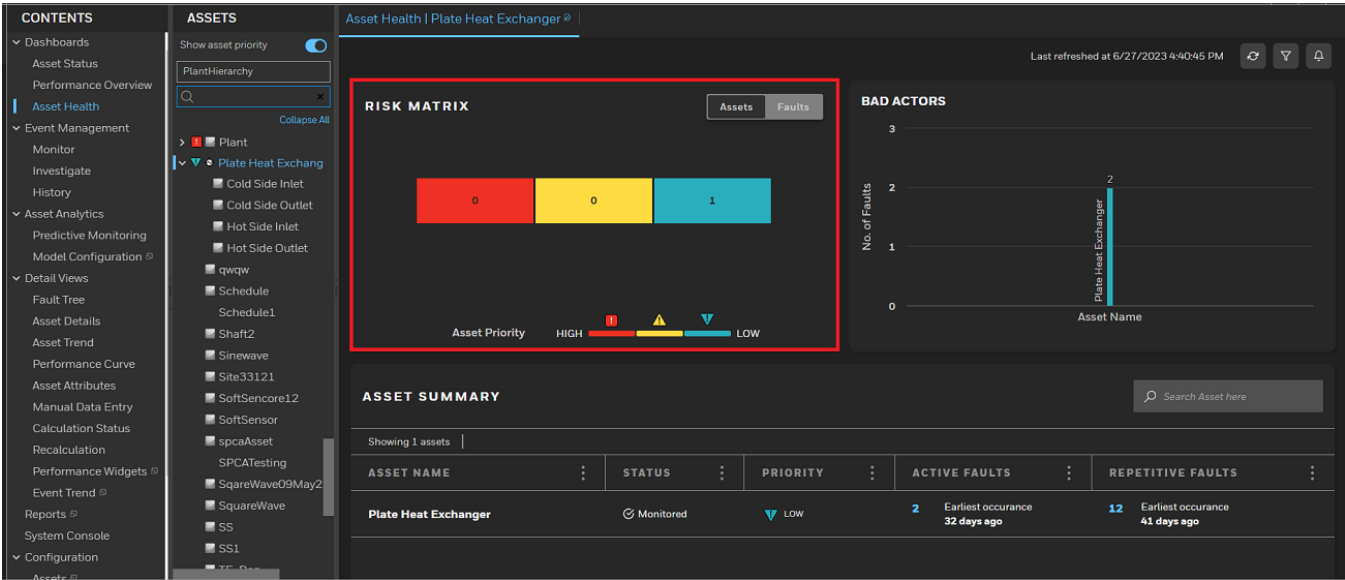
To open the Asset Health Dashboard > Contents > Asset Health.



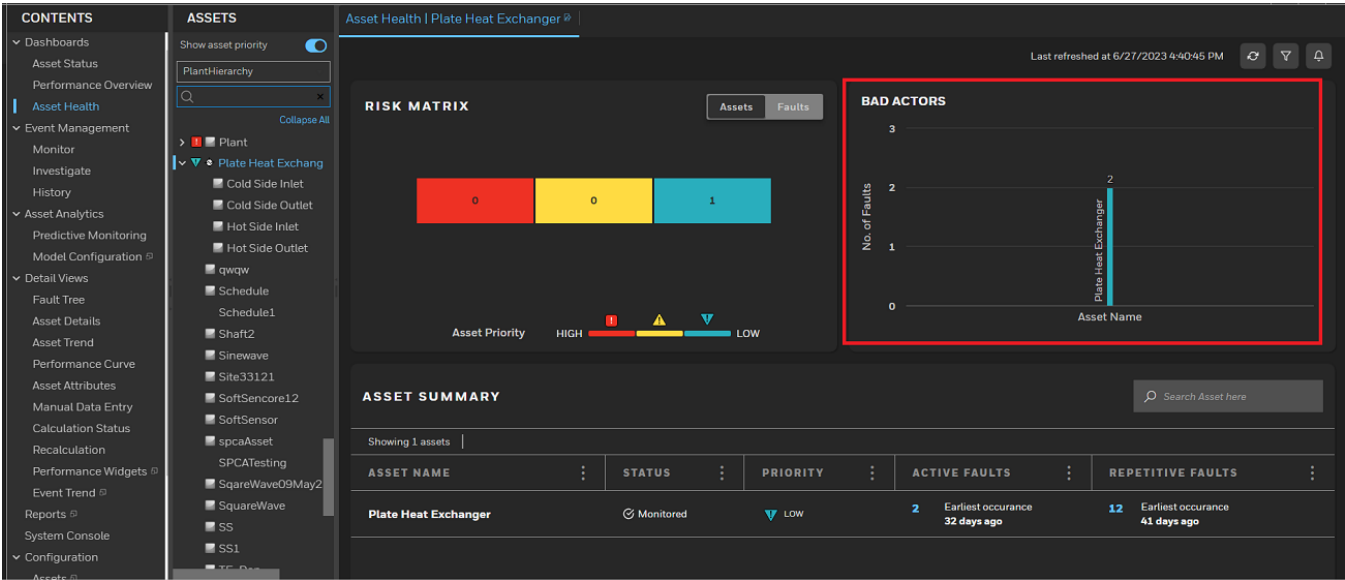
As soon as you open the asset health dashboard, you can see four panes.



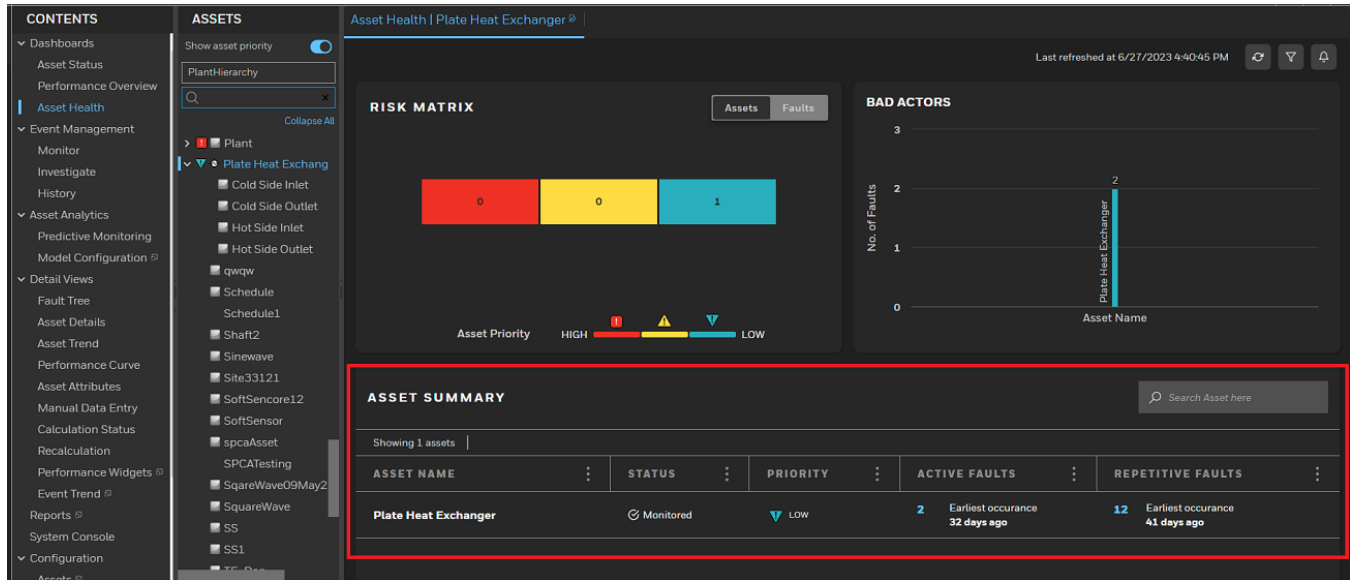
The left pane is used to navigate through the asset hierarchy.



In the right pane, the first block displays the Risk Matrix. You can see the critical assets and critical faults.



Next to the Risk matrix block you can see the Bad Actors section. The bad actors section displays a graphical representation of all the Assets that have the maximum number of faults. The top 10 assets that have the maximum active faults are displayed in this view.



Below the Risk Matrix, you can see the Asset summary section. The Asset summary displays the asset grouped in the Risk Matrix based on the critically of the asset.

5.4.4.1 Global filter

You can narrow down your investigation by using the global filter option.

- 1. Click Filter from the upper-right corner of the application
Filter dialog box appears.
- 2. Type or select Asset class and Asset types from the list.
- 3. Click Apply.
The filter gets applied to the complete asset health dashboard.

For example, to investigate the asset compressor, select compressor from the Asset Class and you can also further narrow down your investigation by selecting asset type in the drop-down list.

5.4.4.2 Refresh

Click Refresh button from the upper-right corner of the page, to view the latest information and this option applies to the complete asset health dashboard

5.4.4.3 Notification

Click Notification from the upper-right corner of the page, the notification pane from the right appears. The notification pane displays all the faults related to the asset that is selected from Risk Matrix, Bad Actors and Asset summary.

NOTE

By default, the Remaining Useful Life value not displayed. To display the value in notification, refer to **Enabling TRL for an Attribute in APM Configuration Guide**.

5.4.5 Identify

5.4.5.1 Risk Matrix

The risk matrix helps to identify assets that have active faults that impacts the health of the asset. At a high level it displays critical assets and critical faults.

The grouping of the assets is based on the priority like high, medium, or low. Each group displays the number of assets that falls in that category. The Fault view shows the total active faults for all the assets. The faults are again categorized as high, medium, and low. It also displays the new and in progress faults.

For example, let us consider that a unit has 10 assets and 5 assets are performing at an optimum level. The other 5 assets have 20 active faults. The faulty assets are categorized based on the asset priority as High, Medium, and Low. For example, 2 assets might be categorized as high, 1 as medium, and 2 as low.

[Switch to Fault View] The fault view displays all the active 20 faults. Let's say, out of the 20 faults 12 are high, 4 are medium, and 4 are low.

Asset Priority

The color represents the criticality levels of the assets, with red for high priority, yellow for medium priority, and blue for low priority. Each priority displays the number of assets that falls in that category.

Fault Priority

Fault Priority: The color represents the criticality levels of the faults, with red for high priority, yellow for medium priority, and blue for low priority.

Fault Status: The number of most recent Active faults with high, medium and low priority are displayed in New. In Progress displays the number of faults that are currently worked on.

Viewing the details for each category

Click the category to view the details of all assets in the Asset Summary section. For example, if you click the High priority block, you get to see all the assets that are categorized as high in the Asset Summary section.

5.4.5.2 Bad Actors

Bad actors are a graphical representation of asset health. The graph indicates the asset name displayed on the x-axis and the active fault count displayed on the y-axis. The fault count considers only the unique faults, and not the potential faults or symptoms. The color indicates the priority parameters of the asset. Click the asset name from the graph to view the details of assets and faults, in the asset summary. Click the asset from the asset summary, to view all the faults and fault details in the notification pane.

5.4.5.3 Asset Summary View

The assets displayed in the asset summary are the assets grouped in the Risk Matrix based on the priority of the asset. From the Risk Matrix block, click the category for which you want to view the details of the assets. The Asset Summary view displays the information of all the assets in the category. You can also search the required asset name in the search box

The Asset Summary pane includes the following details:

- **Asset Name** - Displays the name of the asset.
- **Status** - Displays the status of the asset if it is monitored or unmonitored.
- **Criticality** - Displays the configured criticality of the assets that have active faults, which impacts the health of the plant.
- **Priority** - Helps to prioritize the Asset based on the criticality like high, medium, or low.

- **Active faults** - The recent active faults are displayed in this column. Shows the number of increase or decrease of active faults in the asset. Blue color represents the decrease in faults and red color represents the increase in faults.

ASSET SUMMARY					
Showing 68 assets Search Asset here					
ASSET NAME ↑	STATUS	PRIORITY	ACTIVE FAULTS	REPETITIVE FAULTS	
04-K-101B Propane Refrig Comp	Monitored	LOW	1 Earliest occurrence 118 days ago	0	Earliest occurrence N/A
0721-C-201 Compressor Unit	Monitored	MEDIUM	1 Earliest occurrence 56 days ago	0	Earliest occurrence N/A
0721-C-201 Vapour Recovery Comp	Monitored	MEDIUM	1 Earliest occurrence 56 days ago	29	Earliest occurrence 56 days ago

You can identify the first occurrence of the faults. For example, if the earliest active fault occurred 67 days ago, the active faults display the earliest occurrence as 67 days ago

ASSET SUMMARY					
Showing 25 assets Search Asset here					
ASSET NAME ↓	STATUS	PRIORITY	ACTIVE FAULTS	REPETITIVE FAULTS	
607-PU-1101 Injection Gas Comp	Monitored	MEDIUM	3 Earliest occurrence 67 days ago	77	Earliest occurrence 63 days ago
607-CM-1103 Shaft Seal System(Casing2)	Monitored	MEDIUM	1 Earliest occurrence 89 days ago	10	Earliest occurrence 86 days ago
607-CM-1103 Compressor Unit(Casing2)	Monitored	MEDIUM	1 Earliest occurrence 67 days ago	25	Earliest occurrence 67 days ago
541-ST-101 and 541-C-101	Monitored	MEDIUM	1 Earliest occurrence 89 days ago	10	Earliest occurrence 56 days ago

- **Repetitive faults** - Displays the number of faults that frequently occurred and the fault's first occurrence. For example, you can see that there are 77 repetitive faults and the earliest occurrence of the repetitive fault has occurred 63 days ago.

ASSET SUMMARY					
Showing 25 assets Search Asset here					
ASSET NAME ↓	STATUS	PRIORITY	ACTIVE FAULTS	REPETITIVE FAULTS	
607-PU-1101 Injection Gas Comp	Monitored	MEDIUM	3 Earliest occurrence 67 days ago	77	Earliest occurrence 63 days ago
607-CM-1103 Shaft Seal System(Casing2)	Monitored	MEDIUM	1 Earliest occurrence 89 days ago	10	Earliest occurrence 86 days ago
607-CM-1103 Compressor Unit(Casing2)	Monitored	MEDIUM	1 Earliest occurrence 67 days ago	25	Earliest occurrence 67 days ago
541-ST-101 and 541-C-101	Monitored	MEDIUM	1 Earliest occurrence 89 days ago	10	Earliest occurrence 56 days ago

Click  next to the column Name, to sort the asset from ascending or descending order. You can also click the columns button and select from the list the required columns to display in the asset summary.

Notifications

Click the asset to view more details and the notification pane appears on the right. The notification pane displays all the faults related to that asset. The faults are further classified as New and In Progress, where New displays all the new faults and In Progress displays the faults that are currently handled. You can also view the criticality status, the fault occurrence time and the asset name from which the failure occurs in each fault.

Filters

You can also narrow down your investigation by prioritizing the priority of the faults. Select the faults that you want to investigate first based on their priority, from the filter option. Click Filter to select an option from High, Medium, and Low or select all.

Sort

You can click the sort option to view the faults from Newest to Oldest or Oldest to Newest. By default, the notification pane shows the newest faults at the top of the list.

To View more Information about the Fault

After you sort and filter the faults that you need to investigate, click the selected fault to get more information about fault failure.

Event information

This pane gives you more information about the faults like

- **Cause**- This section explains you about the cause for the failure to occur.
- **Recommendation**- Helps you with few suggestions to rectify the failure.
- **Consequences**- Describes the consequences to be aware of from the failures.

Comments

Displays the comments of prior investigation that are added to the fault.

5.4.6 Investigate

Related Information

Related information section will help you to further investigate the failure that will navigate to the following pages:

- **Event Details** - Navigates you to **Event Monitor** page, the page displays the information about the fault that was selected in the navigation pane.
- **Event Trend** - Navigates you to the **Event Trend** page, the page provides more information on the events, event type, condition, and status of the events that helps to further investigate on the failure.
- **Anomaly Index** - Navigates you to **Anomaly Index** page, the page displays the information of the physical health of an equipment models.
- **Attribute Overview** - Navigates you to **Attribute Overview** page, the page provides a list of the asset attributes and their details, including the current value. You can use this list to review and update asset attribute information.
- **Fault Tree** - Navigates you to **Fault Tree** page, the page displays the graphical representation of the asset and assets and their associated faults and symptoms with performance indicators, to locate the source of a performance issue.

5.4.7 Act

After navigating to required pages, you can take corrective actions to bring back the asset's health to optimum level. You can further view the specific details of an event and determine how the event is to be resolved. Add comments to the event history to record details and information discovered during the investigation, close-out the event once the investigation and any corrective actions are complete. If it is a valid event, and you are the user responsible for the investigation, accept the event and if it is not a valid event, reject the event.

5.4.8 Configuration

You need to mark and identify the main assets in the plant/site to display the information at the main asset level on the asset health dashboard.

“IsMainAsset” attribute need to be marked to 1 against the asset to indicate the main asset.

The attribute gets imported when you import the asset library, and then you need to mark the attribute to 1 for the main asset using excel add-in. The same procedure applies for the migrated system to R532.

5.5 Performance Widget

5.5.1 Introduction

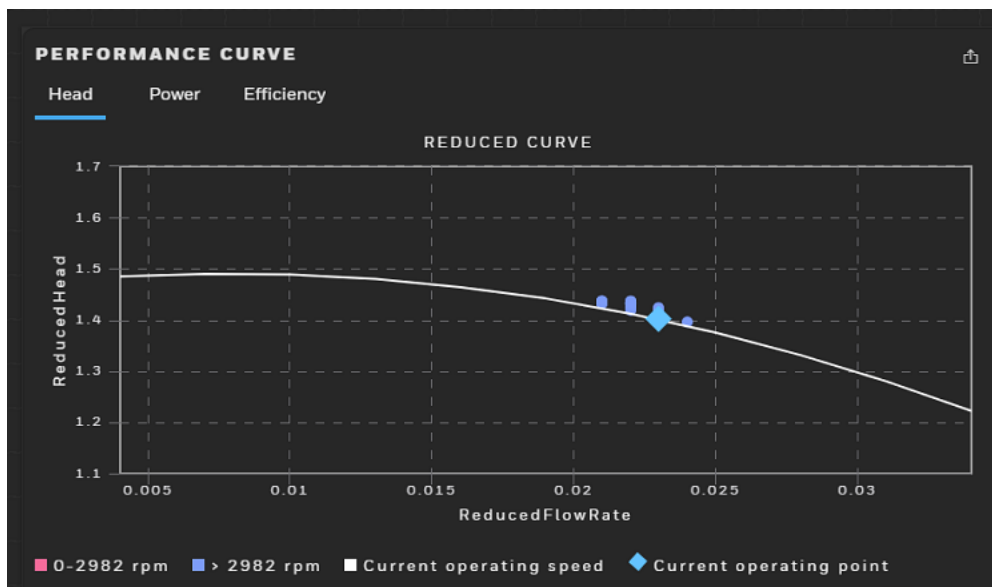
The performance widget displays a list of configurable widgets that aid in analyzing real-time and historical asset performance. To access asset performance data, drag and drop suitable widgets from a predefined list of widgets and configure them using asset-type level or instance level option.

The list of available widgets includes:

- [Performance Curve](#)
- [Monitoring Table](#)
- [Trends](#)

Performance Curve

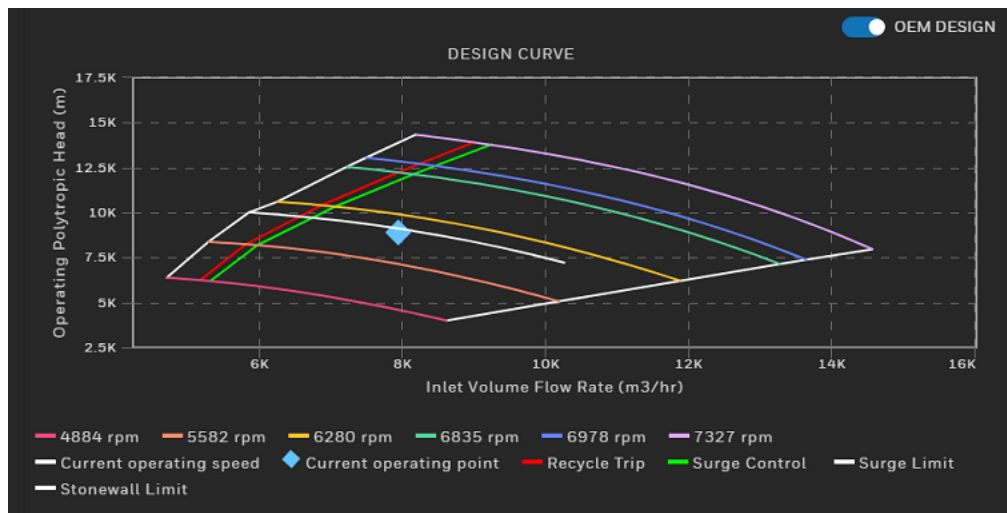
Performance Widget display the expected (Manufacturer) Performance Curve graphs corrected for actual operating conditions.



The legends used in the **DESIGN CURVE** are:

1. **Surge Limit:** The white line on the extreme left in the design curve represents surge. This line represents the locus of all the points on a compressor map at which compressor will surge.
2. **Stonewall Limit:** The white line on the extreme right in the design curve represents stall. This line represents the locus of all the points on a compressor map at which compressor will stall.
3. **Surge Control Line:** The green line represents the surge control. Once the operating point touches this line, anti-
4. **Surge Limit Line or Recycle Trip Line:** The red line represents surge limit or recycle trip. Compressor logic will

trigger emergency shutdown once operating point reaches surge limit or recycle trip.



NOTE
By default, the **OEM DESIGN** toggle button is disabled and the corrected design curve is displayed in the dashboard. Enabling the toggle button, widget will report design performance curve provided by OEM.

NOTE
Refer [Calculation Flow Diagram](#) to estimate **Surge Control Line** and **Surge Limit Line / Recycle Trip Line**.

Monitoring Table

Monitoring Table is the configurable tabular representation of attribute values and properties.

OPERATING PARAMETERS				
Measured Inputs Calculated Outputs Manual Inputs				
ATTRIBUTE	TAG NAME	UOM	ACTUAL	EXPECTED
Suction Pressure	1401P_SUC_PRESS	barg	6.516	
Discharge Pressure	1401P_DISCH_PRESS	barg	34.453	
Flowing Temperature	1401P1001_Temp	C	63.000	

PERFORMANCE KPI				
Measured Inputs Calculated Outputs Manual Inputs				
ATTRIBUTE	UOM	ACTUAL	EXPECTED	REDUCED
Operating Efficiency	%	82.692		
Operating Head	m	354.576		
Operating Shaft Power	kW	3870.489		
Speed	rpm	2987.671		
Volume Flow Rate	m3/hr	4165.931		

The legends used in **MONITORING TABLE** are **Measured Inputs**, **Calculated Inputs**, **Manual Inputs**.

To configure the legends, follow the steps below:

1. Open attribute property mapping sheet.
2. Create and associate property named **Native_Tag**.
3. Enter a value for **Native_Tag**.
4. Below are the representation of legends as per the value in **Native_Tag**:
 - a. **ManualInput**: Use **ManualInput** as value in **Native_Tag** property, if the attribute is having static / manual input value and the values are displayed in orange.
 - b. **Measured Inputs**: If **Native_Tag** property contains any text value other than **ManualInputs**, system will take it as measured inputs and the values is displayed in blue.
 - c. **Calculated Inputs**: If **Native_Tab** property does not have any value mapped, system will take calculated values and the value is displayed in white.

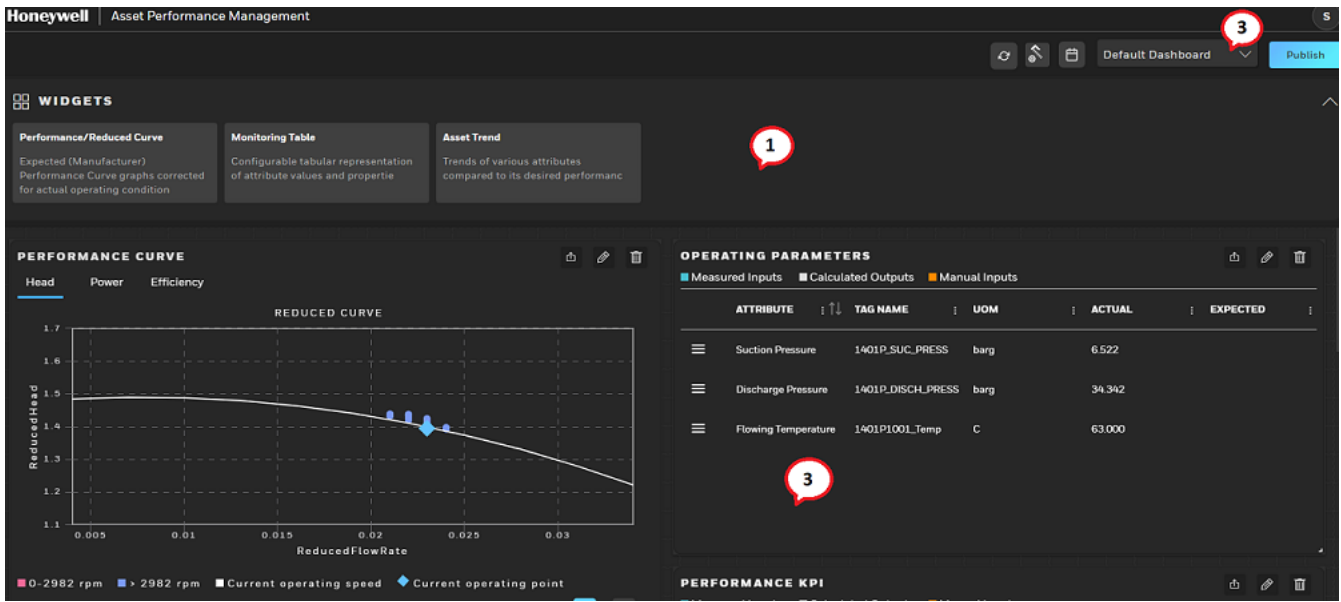
Trends

Trends widget enables user to trend time series data of asset attributes/properties.



5.5.2 Interface

The different sections of performance widget are marked in the below figure.



Button/Number	Name	Description
1	Widgets Pane	The top pane displays the predefined list of widgets to configure them at the asset type or instance level.

Button/Number	Name	Description
2	Center Pane	The center pane displays canvas section where you can drag and drop the widgets to be configured. NOTE You can configure only one widget at a time.
3	Template	List of saved templates for the selected asset. TIP The default template has a star icon on the right side of template name in the drop-down list.
	Time Control	The Time Control displays the historical data of the configured widget. The time control is applicable for only Reduced Curve.
	Asset Type Level UOM	Click the “Asset Type Level UOM” button to display values in type level . By default attribute/property values are reported in instance level . NOTE The Type Level values are displayed for reference and refreshing the page will revert the unit values back to Instance Level.
	Refresh Widget	Refreshes the content displayed in the grid.

5.5.3 Default View

A **Default view** is the standard view of **Performance Dashboard** configured for each asset type. This view is displayed during asset contextualized navigation to configuration section of Performance Dashboard. You can configure a default dashboard at asset type level / asset instance level from default view.

The Trends displayed in default view will have all the attributes associated with the monitoring table.

The default view is configured for the below three asset types:

- Centrifugal compressor with variable speed
- Centrifugal Compressor - IGV
- Reciprocating Compressor
- Centrifugal compressor with constant speed
- Centrifugal pump
- Gas Turbine

NOTE

A gas turbine will have only Current Operating Speed and Current Operating Point plotted in the performance curve.

To display the **Default View** configured for an asset type, follow the steps below:

1. Select any of the predefined asset.
2. The **CONTENT** pane displays a list of different screen options.
3. From the configuration drop-down menu, click **Performance Widgets**.

The **Performance Widgets** is displayed in the new tab with default view of selected asset.

4. To set **Default view** as **Default Dashboard**, click **Publish** and select **Set as default dashboard** checkbox.

NOTE

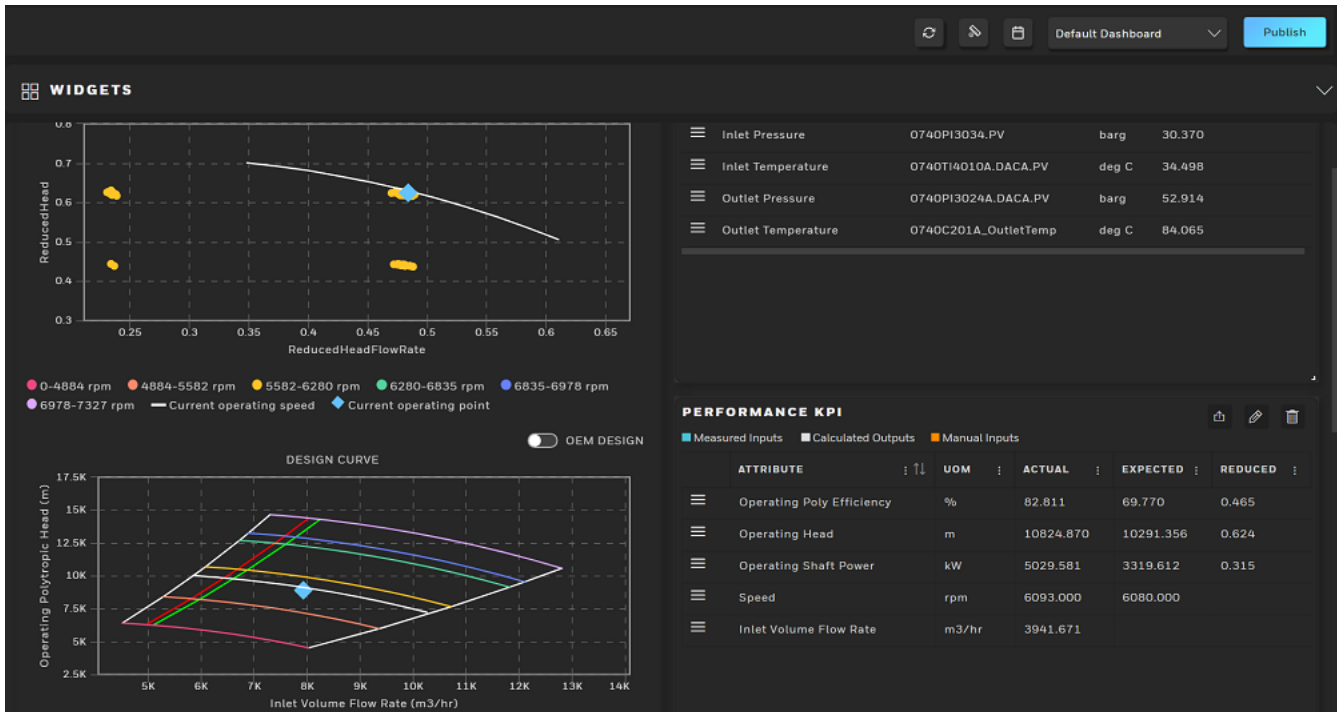
You will not be able to make any changes to the **Default View** and the dashboard saved without selecting the check box is saved as a custom dashboard.

You need to historize the attribute value and carry out a tag mapping of historized tag in attribute property mapping to view the attribute property values besides attribute value .

NOTE

Selecting an asset without a default view takes you to a clean canvas where you can create a performance dashboard for the selected asset.

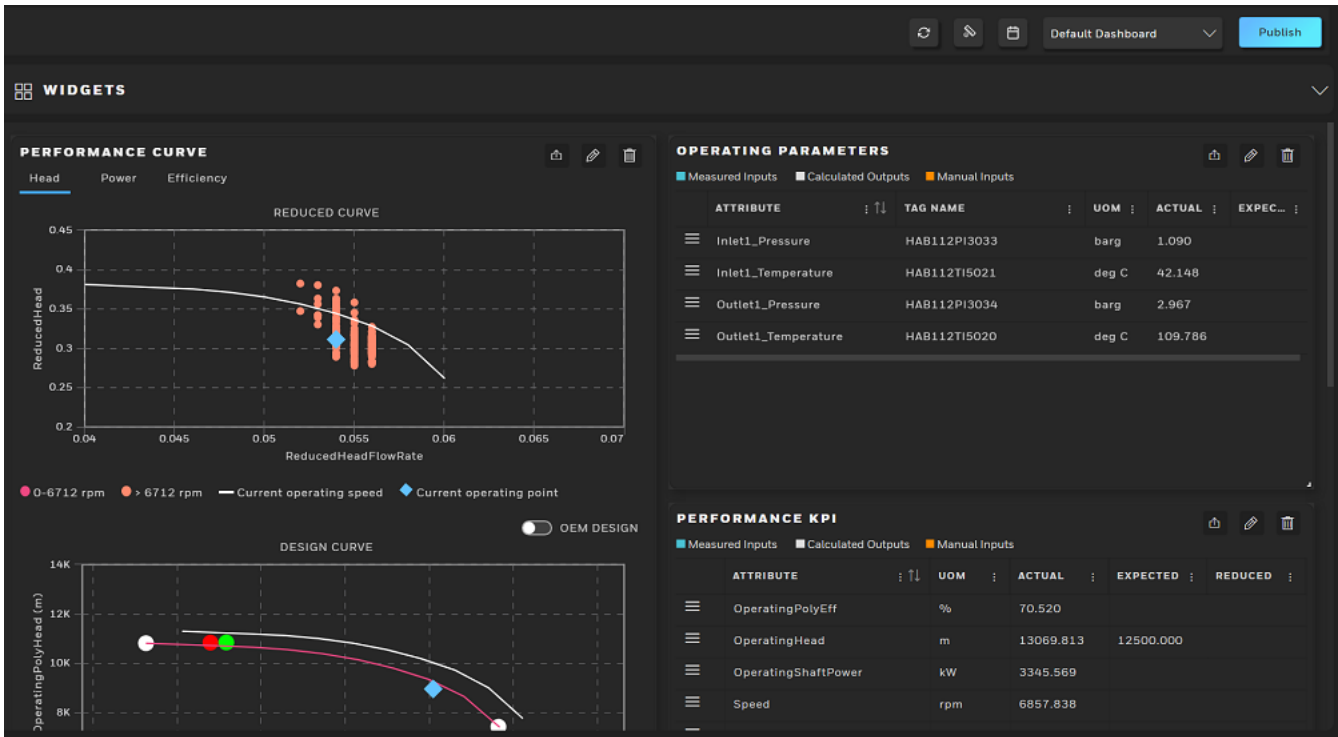
Default View for Compressor Variable Speed



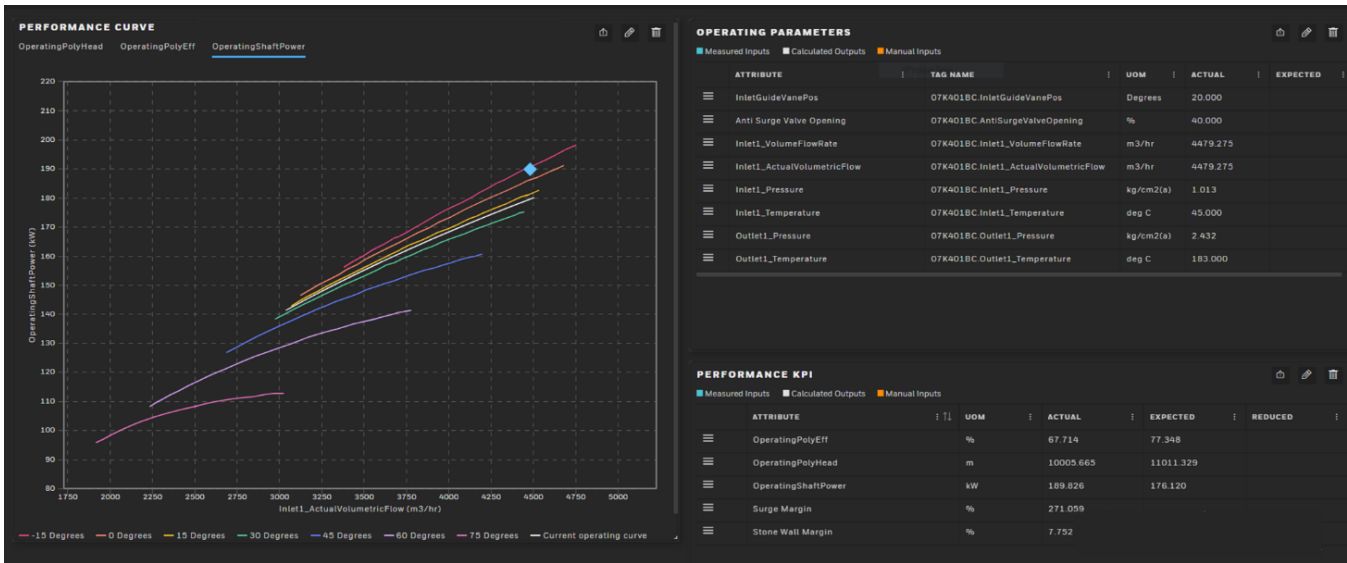
NOTE

Ensure that volume flow rate attribute name (Inlet1_ActualVolumetricFlow or Inlet1_VolumeFlowRate) mentioned in tabular array (poly performance data) is matching with historised actual volume flow rate for that asset

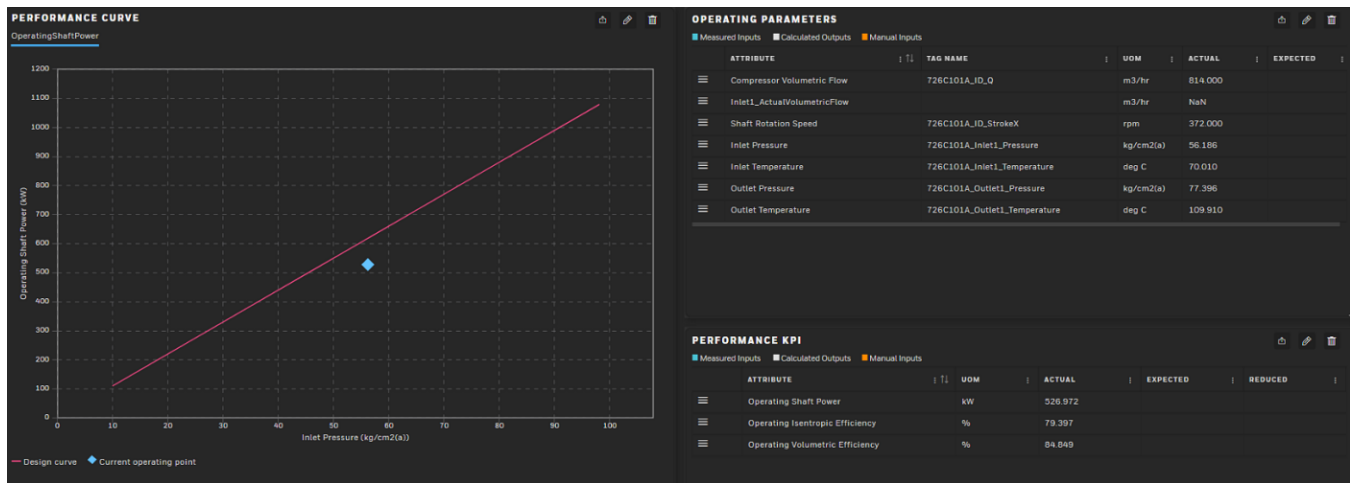
Default View for Compressor Constant Speed



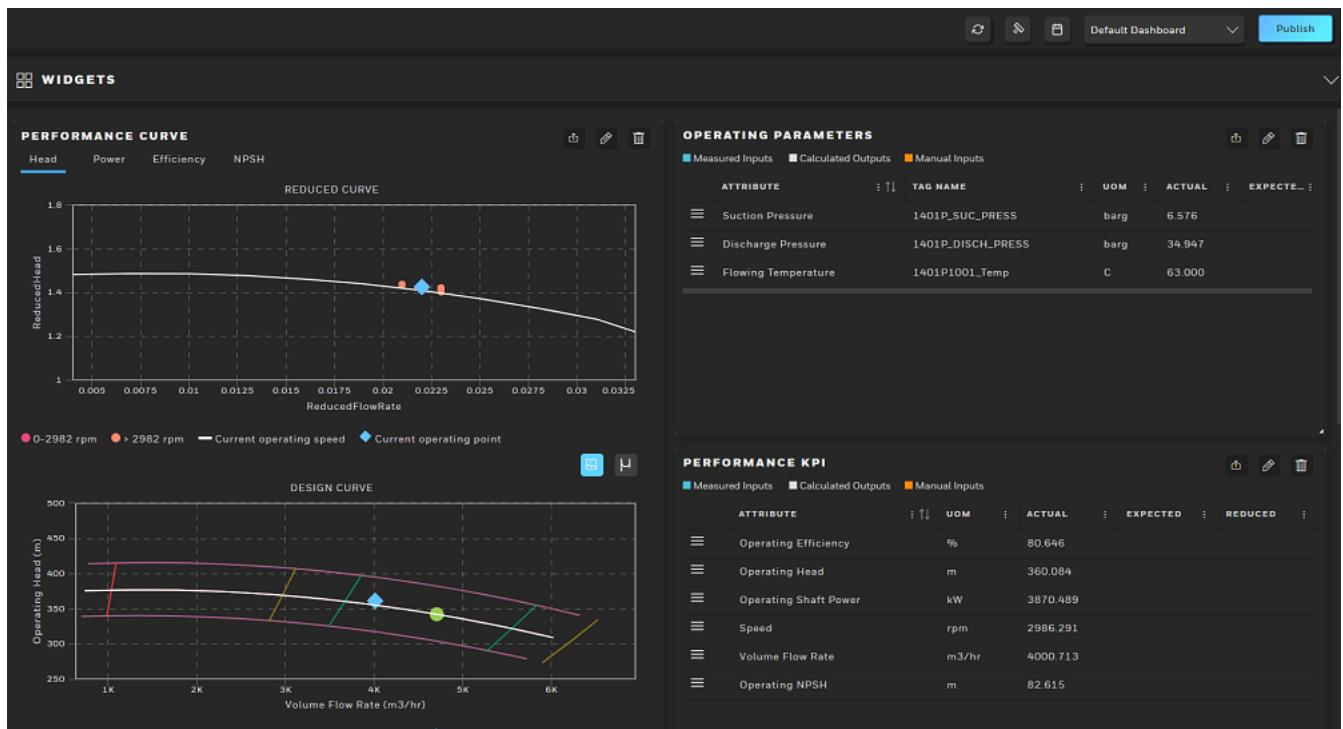
Centrifugal Compressor - IGV



Reciprocating Compressor



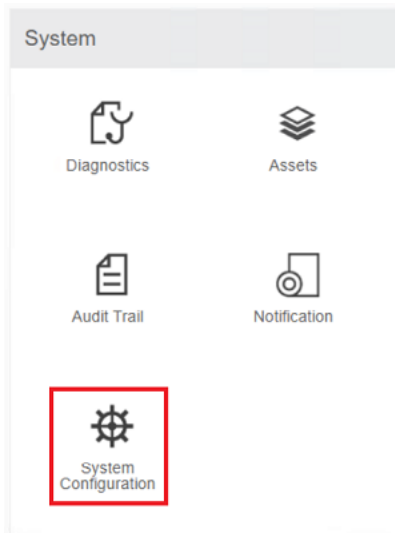
Default View for Pump



5.5.4 Performance Widget Configuration

To configure the performance widget, follow the steps below:

1. Open any web browser.
Type `https://<machine name>:<port>/MES/Default.aspx#/UAS/UASLandingPage.aspx` in the address bar, where the **<machine name>** is fully qualified domain name of the Asset Performance Management Server and **<port>** is the configured SSL port.
The **Honeywell Application** landing page appears.
2. In the **System** area, click **System Configuration**.



- 3. Locate the following setting:
Honeywell.UAS.PerformanceMonWidgetList
- 4. Double-click the Value tab and then type new value.

Widget	Value
Performance Curve	{PerformanceCurve:PERFORMANCE/REDUCED CURVE:Expected (Manufacturer) Performance Curve graphs corrected for actual operating conditions}
Monitoring Table	{MonitoringTable:MONITORING TABLE:Configurable tabular representation of attribute values and properties}
Performance Curve and Monitoring Table	{PerformanceCurve:PERFORMANCE/REDUCED CURVE:Expected (Manufacturer) Performance Curve graphs corrected for actual operating conditions}, {MonitoringTable:MONITORING TABLE:Configurable tabular representation of attribute values and properties}

NOTE
More widgets can be added by separating each widget command with a ",".

- 5. Click **Save**.
- 6. Under the Contents tree, select the **System Console**.
- 7. In the Publish Saved Changes section, click **Publish**.

NOTE
By default both Performance Curve and Monitoring Table widgets will be available.
You can follow the above steps to configure more widgets in the dashboard.

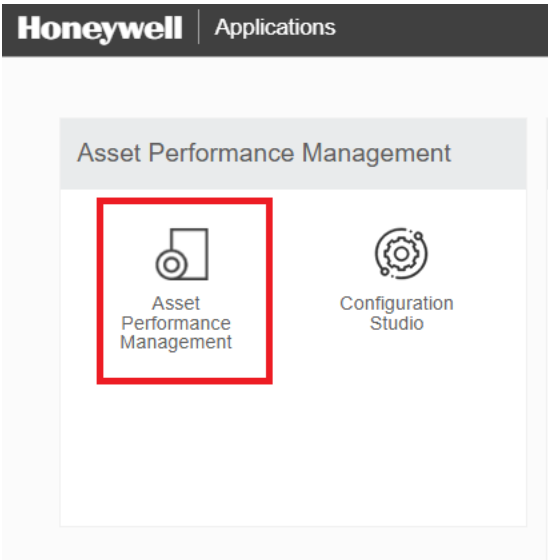
5.5.5 Opening Performance Dashboard

Navigate to the performance widget through the paths mentioned below.

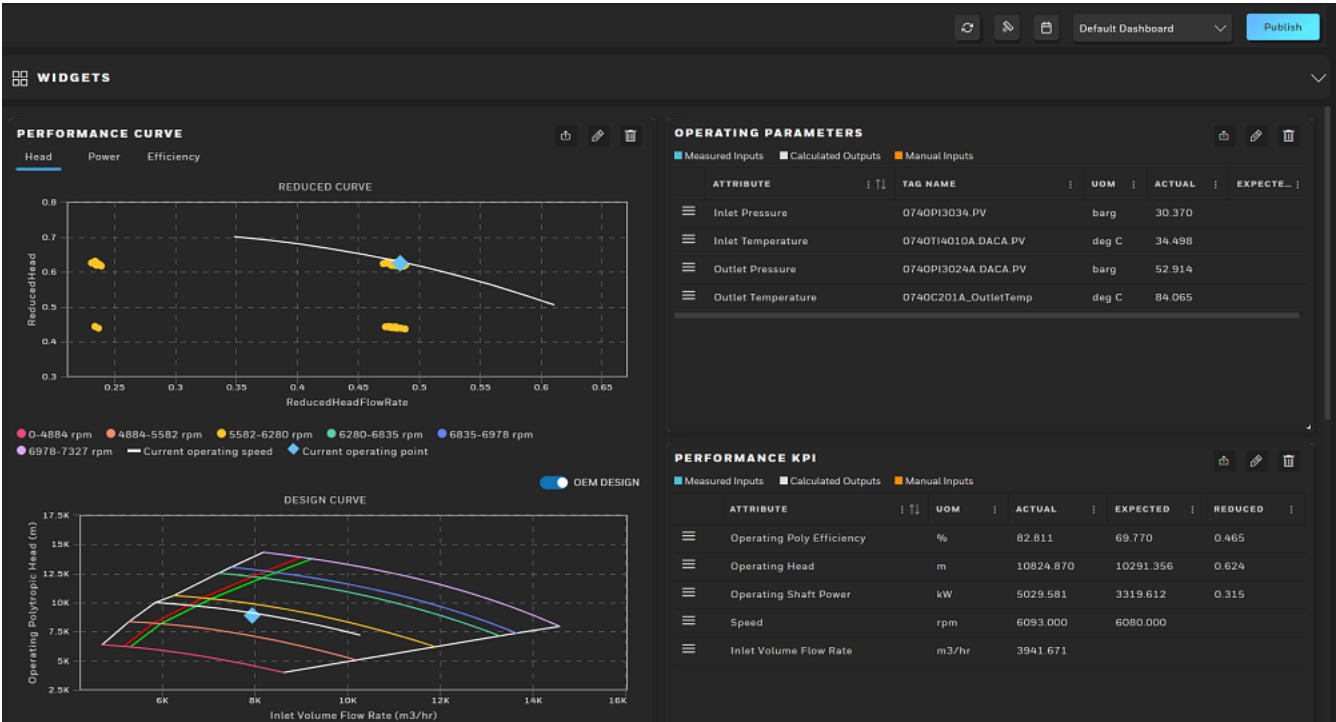
- [Performance Widget](#)
- [Performance Dashboard](#)
- [Hyperlink in Asset Health Dashboard](#)

5.5.5.1 Performance Widget

- 1. Open any web browser.
Type https:// <machine name>:<port>/MES/Default.aspx#/UAS/UASLandingPage.aspx in the address bar, where the <machine name> is fully qualified domain name of the Asset Performance Management server and <port> is the configured SSL port.
The Honeywell Application landing page appears.



- 2. Click **Asset Performance Management**.
- 3. Select an asset for which the performance widget must be displayed.
- 4. The **CONTENTS** pane displays a list of different screen options.
From the **Configuration** drop-down menu, click **Performance Widgets**.
- 5. Default view configured for the selected asset type is displayed in new tab.



NOTE

Selecting an asset without a default view takes you to a clean canvas where you can create a performance dashboard for the selected asset.

6. Drag and drop new widget to the canvas to configure new default / custom dashboard.
7. The created default / custom dashboard can be deleted by selecting the dashboard and clicking delete icon.

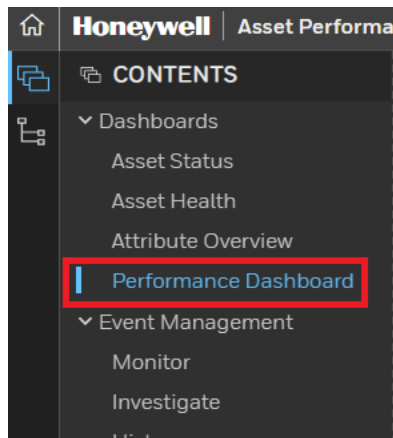
NOTE

Only users with administrative privilege have access to the widget.

5.5.5.2 Performance Dashboard

You can use this method to perform asset contextualized navigation to Performance dashboard. You will have read only access to Performance dashboard.

1. Follow steps 1 to 3 of [Opening Performance Dashboard](#).
2. The **CONTENT** pane displays a list of different screen options.



3. From the **Dashboards** drop-down menu, click **Performance Dashboard**.
4. Default dashboard or latest custom dashboard is displayed in the right pane.



TIP

The default view for the selected asset is marked with a blue star in the saved template dropdown.

NOTE

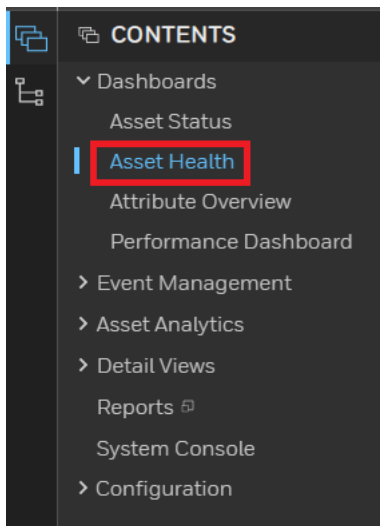
You can use the saved template dropdown to view and export the data available in default and custom dashboard. All other functionality except time control is disabled.

Default view is displayed if both default / custom dashboard is not available.

5.5.5.3 Hyperlink in Asset Health Dashboard

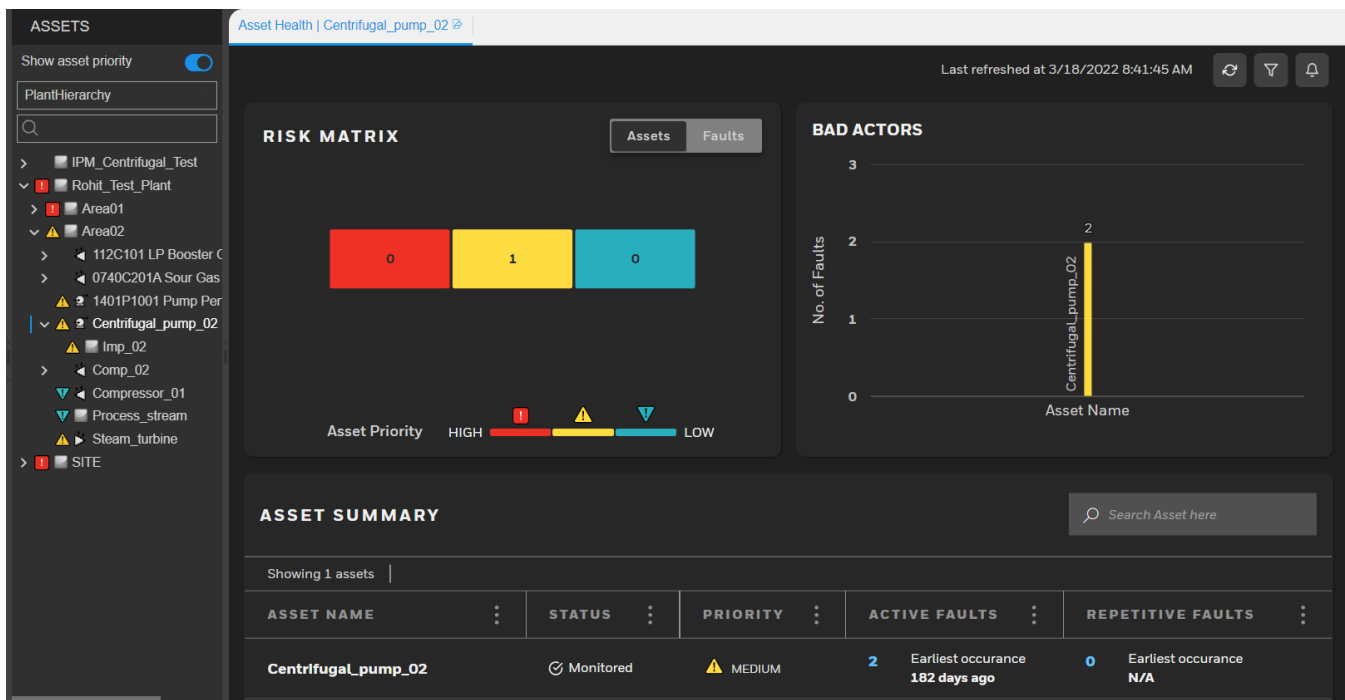
You can view the performance dashboard using the hyperlink in notification panel of Asset Health Dashboard.

1. Follow steps 1 to 2 of [Opening Performance Dashboard](#).
2. The **CONTENT** pane displays a list of different screen options.



3. From the **Dashboards** drop-down menu, click **Asset Health**.

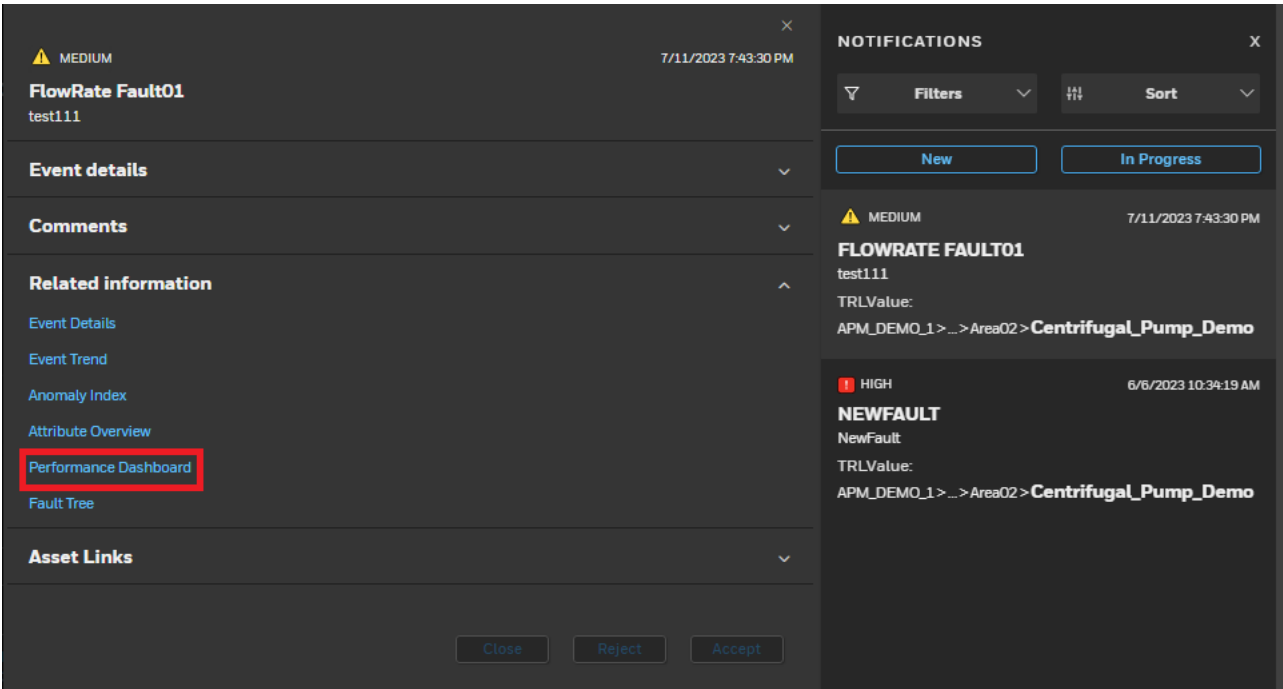
The **Asset Health** page appears.



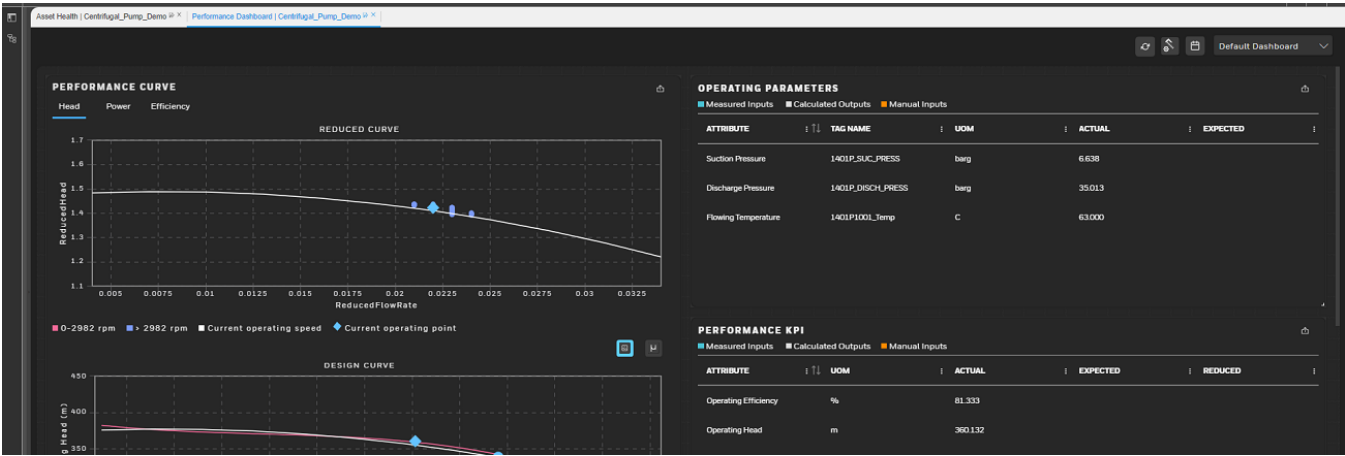
- 4. Select the asset from the asset summary.
- 5. On the top-right corner of the page, click Notification icon .

On the right pane, the notification box appears that displays all the faults related to corresponding asset.

- 6. From the **Notification** panel, select the defect.
- 7. Select **Performance Dashboard** from Related information drop-down.



- 8. The **Performance Dashboard** appears.



NOTE
You can use the saved template dropdown to view and export the data available in default and custom dashboard. All other functionality except time control is disabled.

5.5.6 Using Performance Dashboard

Steps to configure, edit and delete widgets in the **Performance Dashboard**:

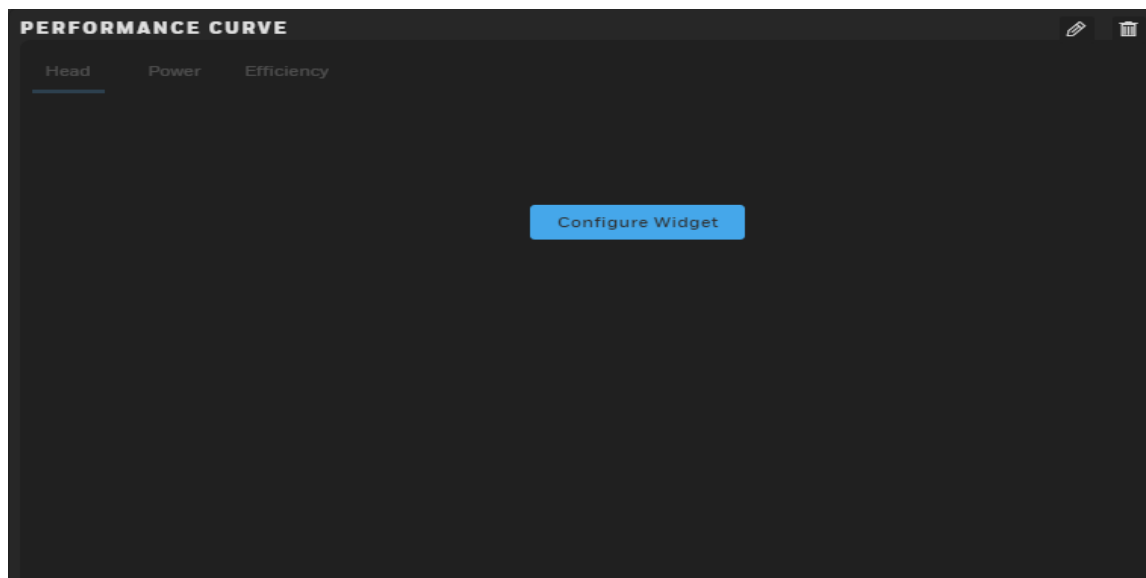
- [Performance / Reduced Curve](#)
- [Custom Curve](#)
- [Overlay Operating Curve and Limit - Centrifugal Pump](#)
- [Monitoring Table](#)
- [Trends](#)
- [Having multiple widgets](#)
- [Edit / Delete a Configured Dashboard](#)

5.5.6.1 Performance / Reduced Curve

To configure a Performance / Reduced Curve, follow the steps below:

1. From the top pane, drag and drop the Performance / Reduced Curve to a blank canvas.
2. Click **Configure Widget**.

The **PERFORMANCE CURVE** window appears.



3. To select filter options, perform the following steps:
 - a. Select the **Asset Type** drop-down to view the available asset types in the selected asset class.
 - b. Select the **Asset Select** drop-down to view the assets available in selected asset type.

PERFORMANCE CURVE - ASSET/ASSET TYPE SELECTION

This page is used to select the asset type / asset for performance curve.

Step 1 of 2

Select Asset Type :-
Centrifugal Pump

Select Asset (Optional) :-
1401P1001 Pump Perf Model

NOTE

The **Asset Type** selection is mandatory.

The **Select Asset** should be changed if the Performance Curve to be applied to a specific asset. If it is not selected, all the assets coming under the Asset type will have the performance curve applied to it.

4. Click **Next**.
5. Select the curves to be added.
 - a. **Reduced Curve:** It provides an overview of the asset performance with the help of trending capabilities of asset attributes with actual and reduced coordinates.
 - b. **Design Curve:** This curve is dynamically plotted based on the current operating conditions.

PERFORMANCE CURVE - CURVE SELECTION

This page is used to select KPI and their respective views for performance curve.

Step 2 of 2

Select - Performance Curve

☒

 Design Curve

☒

 Reduced Curve

☐

 Custom Curve

KPI Selection

☒

 Head

☒

 Power

☒

 Efficiency

Select view option

☒

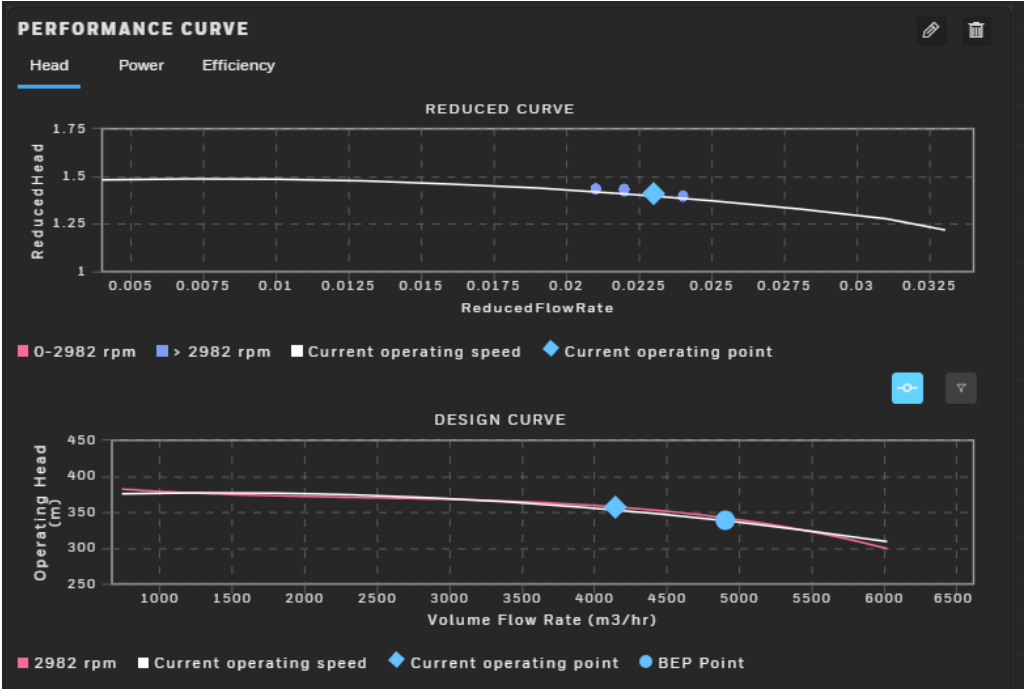
 Tab view

☐

 Non tab view

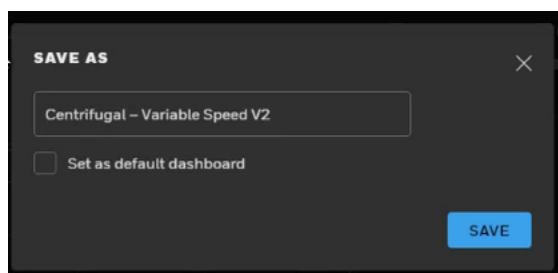
NOTE
Select **Tab view** to see all KPI plotted in three different tabs.
Select **Non tab view** to see all KPI plotted in a single screen.

6. Click **PREVIEW CHANGES**.
7. The **Performance Curve** widget appears.



8. On top-right corner of the screen, click **Publish**.

The **SAVE AS** dialog box appears.



9. Enter a name for the configured dashboard.
10. Select the check box to save the configured widget as **DEFAULT DASHBOARD**. This will overwrite the existing default configured widget and move the existing configured widget to **CUSTOM DASHBOARDS**.
11. Click **SAVE**.

NOTE

Clicking **SAVE** without selecting check box to save the widget under **CUSTOM DASHBOARDS**.

12. The left pane displays the **DEFAULT** and **CUSTOM DASHBOARDS**.
13. The time control is used to view the historical data.

NOTE

The time control is applicable for both **Reduced** and **Design Curve**. For reduced curve the data of selected time range is displayed and for design curve the data for the end time stamp is displayed.

By default the data displayed is for last 3 hours.

NOTE

You can configure Performance Curve for any asset type even if there is no asset configured for that particular asset type.

In this case performance curve will be created with a message "Performance Curve not found".

Configuring Performance Curve for Gas Turbine

1. Follow steps 1 to 4 of configuring [Performance / Reduced Curve](#).
2. Select the **Performance Curve** (key performance indicators) and the **Ambient Temperature** for which the curve should be plotted.

PERFORMANCE CURVE - CURVE SELECTION

This page is used to select the curve for performance curve.

Step 2 of 2

Select - Performance Curve

- ☒ HeatRate
- ☒ Power
- ☐ Efficiency
- ☐ Temperature

Select view option

☒ Tab view ☐ Non tab view

Ambient Temperature

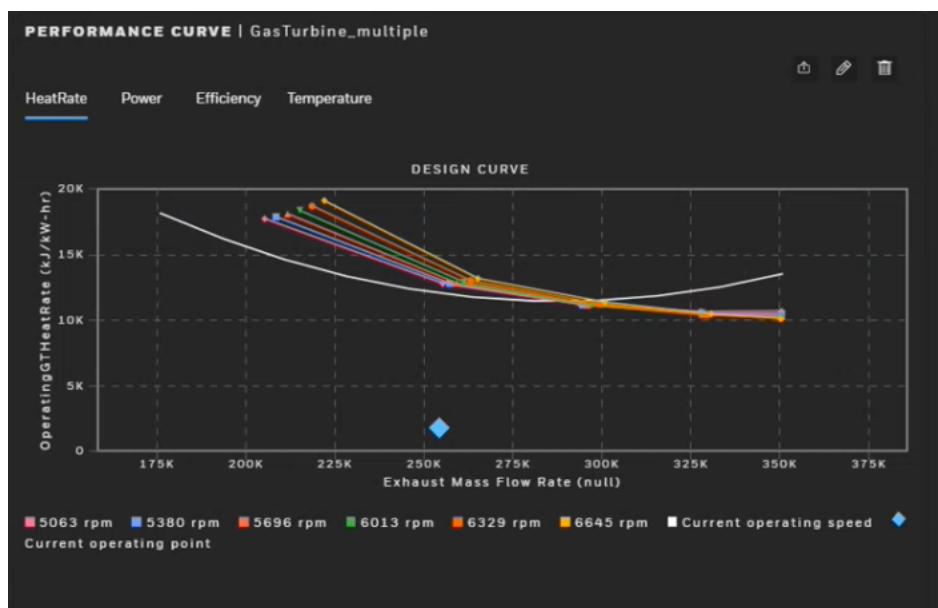
- ☐ 5
- ☐ 42
- ☐ 50

NOTE

Select **Tab view** to see all KPI plotted in three different tabs.

Select **Non tab view** to see all KPI plotted in a single screen.

3. Click **PREVIEW CHANGES**.
4. The **Performance Curve** for gas turbine appears.



5. Follow steps 8 to 12 of configuring [Performance / Reduced Curve](#).

Performance Curve Axis Scalability

The x and y axis scale values of performance curve can be edited as per the requirement. Follow the below steps to edit the scale.

- 1. Open the performance curve for which the axis values need to be updated.
- 2. Double click on the curve.
- 3. **Format Axis** dialog box appears.

Format Axis

Bounds X-Axis

Minimum0.4170000000000004Auto

Maximum0.517Auto

Bounds Y-Axis

Minimum0.522Auto

Maximum0.6419999999999999Auto

Units X-Axis

MajorAuto

Units Y-Axis

MajorAuto

CANCEL

SAVE

- 4. Enter the minimum and maximum values.

NOTE

Click **Auto** to revert it back to default value.

- 5. Click **Save** to update the scale values.
- 6. Click **Publish** to save the curve with new x and y axis scale values.

NOTE

The new scale values can be published only at Asset Instance Level.

5.5.6.2 Custom Curve

To configure a custom curve for an asset, follow the below steps:

- 1. From the APM landing page, select the asset you want to create a custom curve and then click **Attribute Overview**.
- 2. In the attribute overview search bar, search for **Data**.
- 3. Double click on the **Custom Performance Data Input** value field, enter the custom values and then click **Save**.

TIP

Click  icon to add more rows to the table.

4. In the attribute overview search bar, search for **Interpolation**.
5. Double click on the **Interpolation Type** value field, enter the interpolation type value and then click **Save**.

NOTE

For more information on the type of interpolation each value will obtain, refer to [Generic Interpolation](#) in Model Library Guide.

6. Based on the Custom Performance Data Input and Interpolation Type value, the **Custom Performance Data Out** values are generated.
7. From the **CONTENTS** pane, click **Performance Widget**.
8. If the selected asset have default view configured, click the edit  icon of **Performance Curve**.
In case if selected asset do not have a default view configured, drag and drop the **Performance/Reduced Curve** from widget pane to the empty canvas and then click **Configure Widget**.
9. From the ASSET/ASSET TYPE SELECTION pane, select the specific asset for which you want to configure the custom curve and click **Next**.
10. From the CURVE SELECTION pane, select the **Custom Curve** checkbox.
Based on the data available in Attribute Overview page, list of KPI's available is displayed.
11. Select the KPI's to be displayed in the custom curve and click **PREVIEW CHANGES**.
The custom curve is displayed.
12. The current operating curve is generated based on the interpolation type selected and by default the current operating point for the current time stamp is displayed.

NOTE

Use time control to plot the operating point for historical data.

5.5.6.3 Overlay Operating Curve and Limit - Centrifugal Pump

Follow the steps below to configure user defined operating line and operating curves in the default view of a pump:

1. From the APM landing page, select pump from **ASSETS** pane and click **Performance Widgets**.
2. Click the edit  icon of **Performance Curve**.
3. Click **Next** in the ASSET/ASSET TYPE SELECTION pane.
4. Enable **Select - Operating Line** toggle button and select **User Defined** checkbox.
5. Update the following fields:
 - a. Name
 - b. Estimation Type
 - c. Min Value (%)
 - d. Max Value (%)
 - e. Head, Power and Efficiency checkbox
 - f. Color

TIP

Click Add button to add more lines in the design curve and click delete  icon to remove a line.

6. New default view for the pump is displayed as per the values updated in operating line estimation columns. The default values in estimation column is shown below and these values can be updated as required.

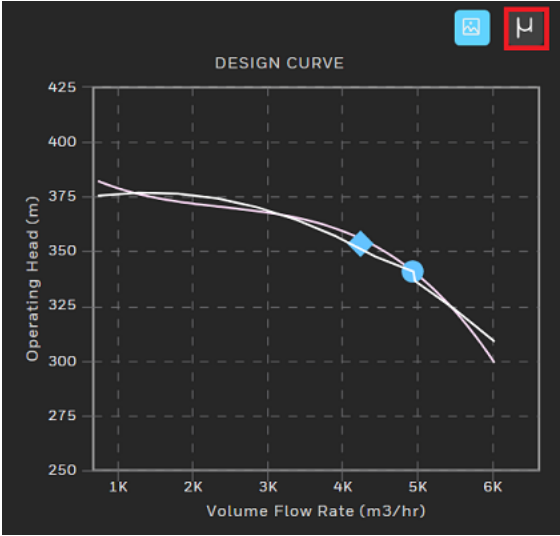
Select - Operating line estimation approach

☐ Default Estimation ☒ User Defined

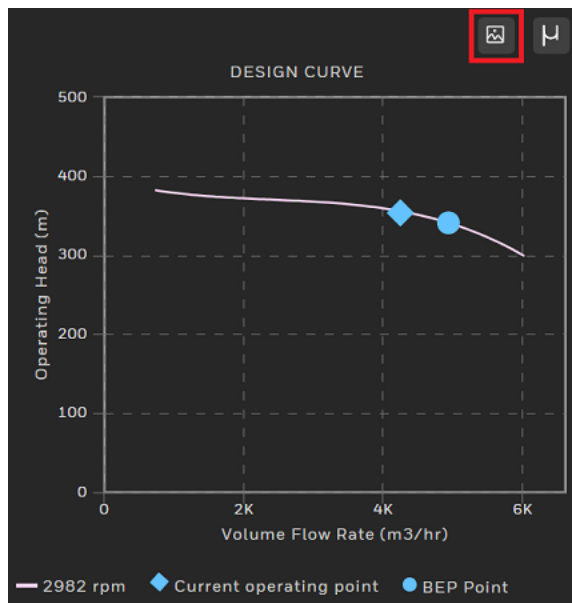
NAME	ESTIMATION TYPE	MIN VALUE(%)	MAX VALUE(%)	HEAD	POWER	EFFICIENCY	COLOR
Preferred line	Op Line	22	18	☒	☐	☐	
Allowable line	Op Line	37	32	☒	☐	☐	
Min flow	Op Line	78	0	☒	☐	☐	
Speed range	Op Curve	5	5	☒	☐	☐	



7. Clicking the Viscosity Correction button will enable the performance model to collect the viscosity corrected design data and operating curve. The button is disabled by default.



8. Disable the Operating Curve button to hide the operating curve and display the historical operating points.

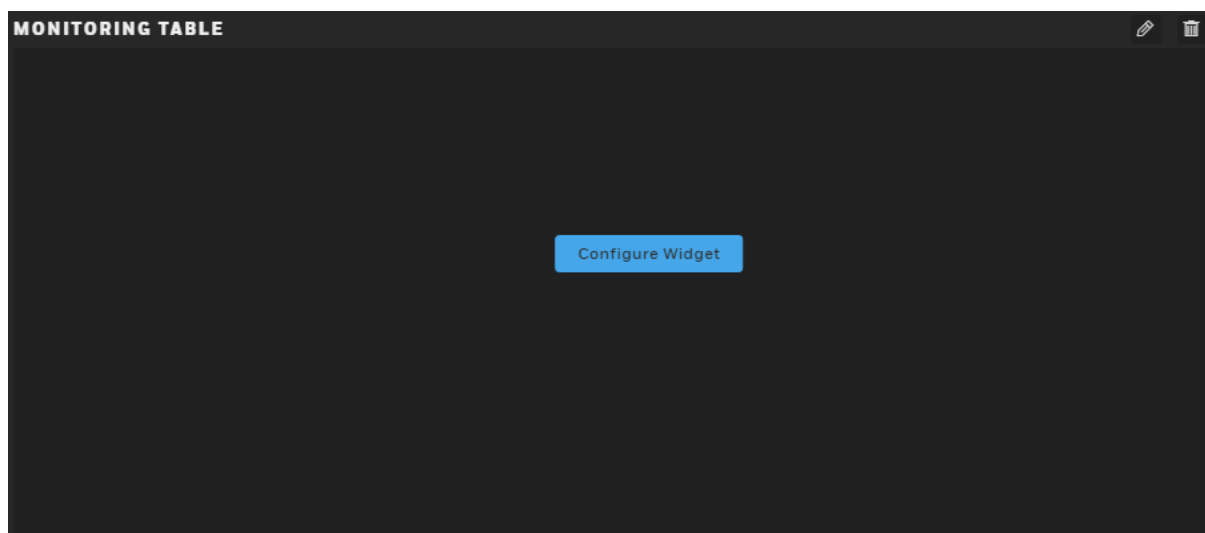


5.5.6.4 Monitoring Table

To configure a Monitoring Table, follow the steps below:

1. From the top pane, drag and drop the Monitoring table to blank canvas.
2. Click **Configure widget**.

The **MONITORING TABLE** window appears.



3. To select filter options, perform the following steps:
 - a. Select the **Asset Type** drop-down to view the available asset types in the selected asset class.
 - b. Select the **Asset Select** drop-to view the assets available in selected asset type.

MONITORING TABLE - ASSET/ASSET TYPE SELECTION

This page is used to select the asset type / asset for monitoring table.

Step 1 of 3

Select Asset Type :-

Centrifugal Pump

Select Asset (Optional) :-

Select an option

NOTE

The **Asset Type** selection is mandatory.

The **Select Asset** should be changed if the Monitoring Table to be applied to a specific asset. If it is not selected, all the assets coming under the Asset type will have the Monitoring Table applied to it.

4. Click **Next**.
5. Select the attribute properties from the given list.

MONITORING TABLE - ATTRIBUTE PROPERTY SELECTION

This page is used to select the columns for monitoring table.

Step 2 of 3

Select attribute properties to be added as columns :-
(Max 5 columns allowed)

☒ Attribute

☒ Tag Name

☒ UOM

☐ Actual

☐ High Limit

☐ Low Limit

☐ Expected

☐ TimeStamp

☐ Quality

☐ Percent Deviation Limit

☐ Deviation LowLimit

☐ Deviation HighLimit

☐ Cleansed

☐ Percent HighLimit

☐ Percent LowLimit

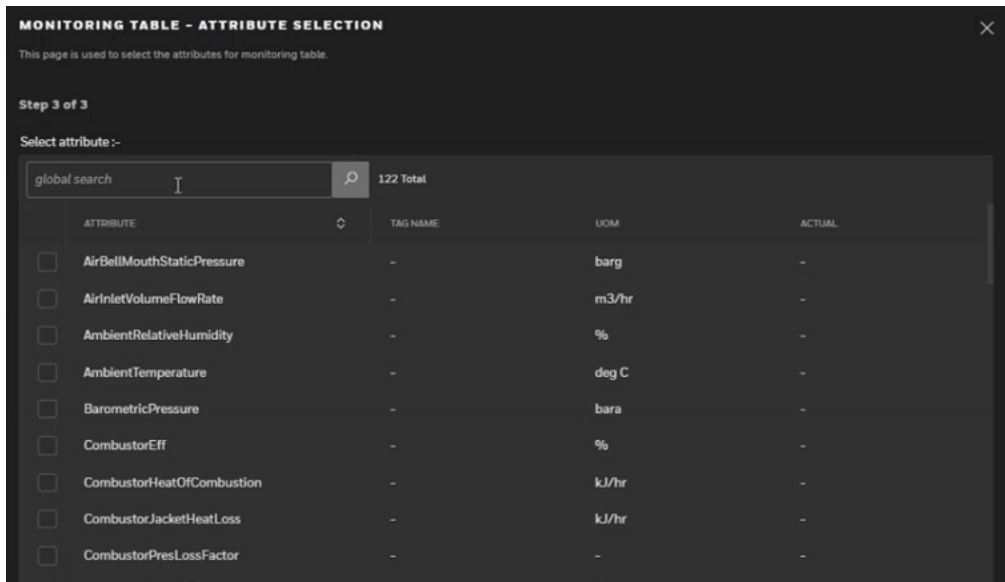
☐ Reduced

NOTE

Only a maximum of 5 attribute properties can be selected at a time and **Attribute** is selected by default.

- 6. Click **SELECT ATTRIBUTE**.
- 7. Tabular report consisting of all attributes and selected attribute properties will be created along with a check box to select / remove attribute before publishing.

TIP
Use the search bar to find and add a particular attribute from the displayed list.



- 8. Based on sequence of attribute selection, order of attribute (top to bottom) will be listed in the table. Attribute selected first will be the topmost entry in the table.

NOTE
The properties of selected attribute will be displayed by default.

- 9. Click **PREVIEW CHANGES**.
- 10. The **Monitoring Table** widget appears.
- 11. Click  icon to change the sequence of selected attribute and the sequence of attribute properties can be updated by dragging the attribute property names left/right.

NOTE
Sequence of all attribute properties except **ATTRIBUTE** can be updated. The position of **ATTRIBUTE** is fixed to display first in the monitoring table.

- 12. Click on the three dots to add more attribute properties into the monitoring table.

	ATTRIBUTE	TAG NAME	UOM	ACTUAL
	GTOutletPressure			0.014
	CompressorInletTemperature			38.904
	CompressorInletPressure			-0.039
	FuelInletTemperature			33.232
	FuelInletMassFlowRate			5459.000
	FuelInletPressure			24.068
	GTOutletTemperature			591.910
	GTMeasuredShaftPower		kW	16427.513

NOTE
Only a maximum of 5 attribute properties can be selected. Deselect a selected attribute property to add another from the list.

13. On top-right corner of the screen, click **Publish**.
SAVE AS dialog box appears.

SAVE AS

Centrifugal - Variable Speed V2

☐ Set as default dashboard

SAVE

14. Enter a name for the configured dashboard.
15. Select the check box to save the configured widget as **DEFAULT DASHBOARD**. This will overwrite the existing default configured widget and move the existing configured widget to **CUSTOM DASHBOARDS**.
16. Click **SAVE**.

NOTE
Clicking **SAVE** without selecting check box to save the widget under **CUSTOM DASHBOARDS**.

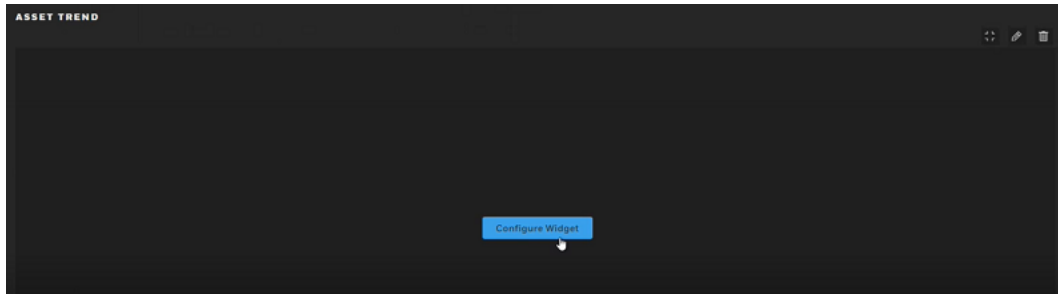
17. The left pane displays the **DEFAULT** and **CUSTOM DASHBOARDS**.

5.5.6.5 Trends

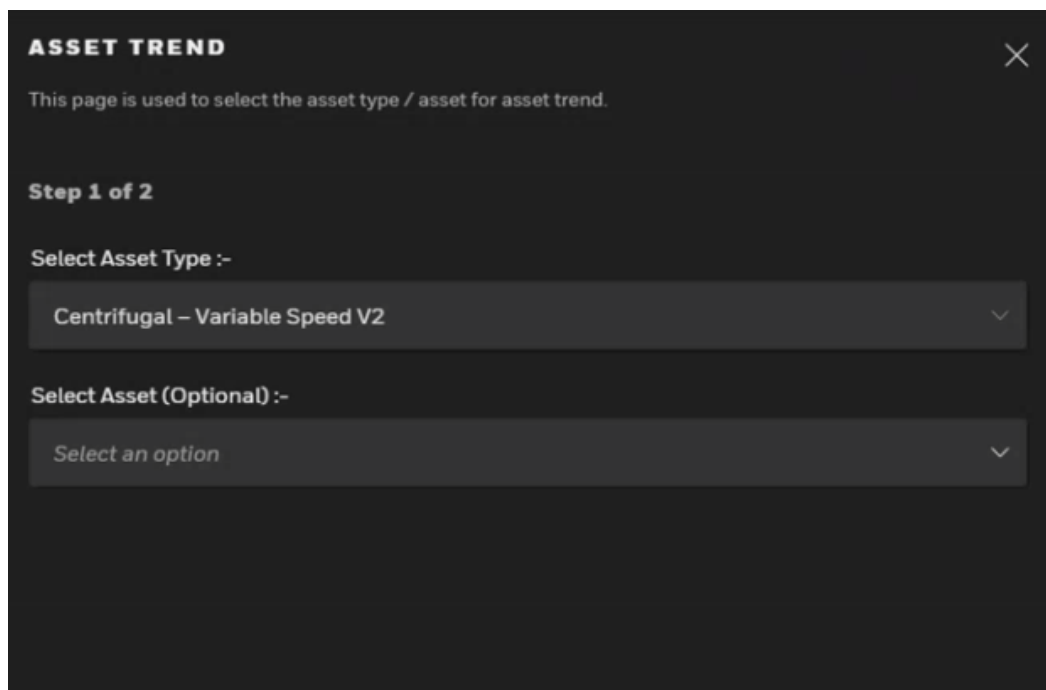
To configure a Trend, follow the steps below:

1. From the top pane, drag and drop the trend to blank canvas.
2. Click **Configure widget**.

The **TRENDS** window appears.



3. To select filter options, perform the following steps:
 - a. Select the **Asset Type** drop-down to view the available asset types in the selected asset class.
 - b. Select the **Asset Select** drop-down to view the assets available in selected asset type.

A screenshot of a dark-themed configuration window titled 'ASSET TREND' with a close button (X) in the top right corner. The text inside says 'This page is used to select the asset type / asset for asset trend.' Below this, it says 'Step 1 of 2'. There are two sections: 'Select Asset Type :-' with a dropdown menu showing 'Centrifugal – Variable Speed V2', and 'Select Asset (Optional) :-' with a dropdown menu showing 'Select an option'.

NOTE

The **Asset Type** selection is mandatory.

The **Select Asset** should be changed if the Trend to be applied to a specific asset. If it is not selected, all the assets coming under the Asset type will have the Trend applied to it.

4. Click **Next**.
5. Select the attribute properties from the given list.

ASSET TREND

×

This page is used to select the properties for asset trend.

Step 2 of 2

Select attribute properties to be trend :-

☒ Actual

☐ Expected

☐ High Limit

☐ Low Limit

☐ Threshold HighLimit

☐ Threshold LowLimit

NOTE

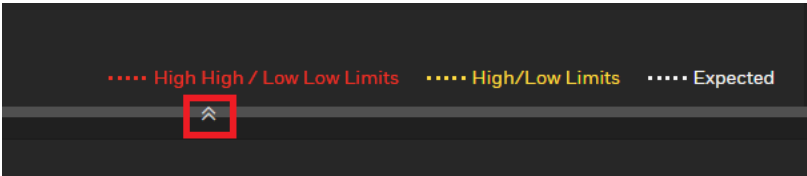
Only a maximum of 5 attribute properties can be selected at a time.

6. Click **PREVIEW AND SELECT ATTRIBUTE**.
7. The configured Trend widget appears giving you the ability to plot the asset attribute values on an x-y trend or a performance curve.

For more information on how to use trend, refer to [Using the Trends](#).



8. Click the arrow highlighted in the below image and select the attributes to be displayed on the trend.



TIP

Use the cursor on the right side of stack plot to scroll between different stack plots.

NOTE

The maximum stack plot that can be configured is 15 and the maximum number of attributes that can be contained in a stack plot is 5.

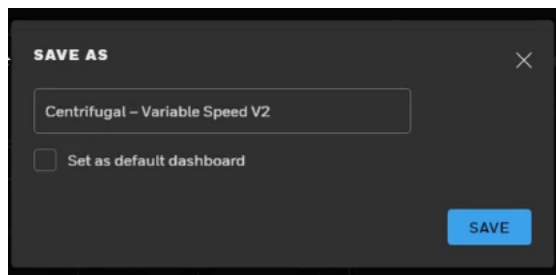
9. Click the maximize and minimize icon  to change the trend display area.

NOTE

For more information on different icons available in the trend , refer to [Using Trend](#).

10. On top-right corner of the screen, click **Publish**.

SAVE AS dialog box appears.



11. Enter a name for the configured dashboard.
12. Select the check box to save the configured widget as **DEFAULT DASHBOARD**. This will overwrite the existing default configured widget and move the existing configured widget to **CUSTOM DASHBOARDS**.
13. Click **SAVE**.

NOTE

Clicking **SAVE** without selecting check box to save the widget under **CUSTOM DASHBOARDS**.

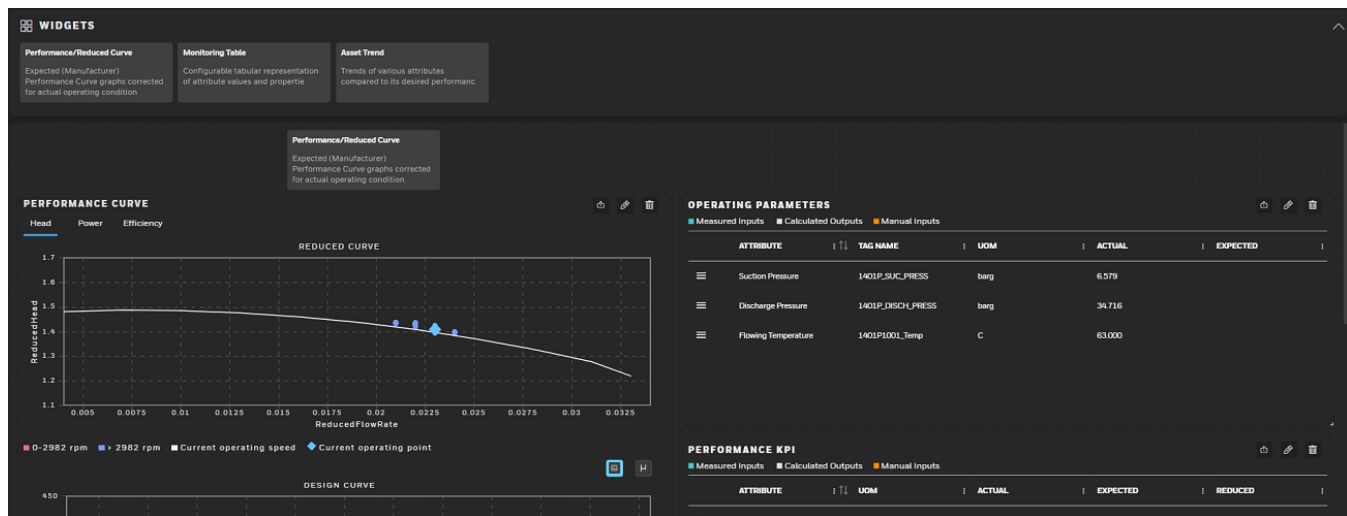
14. The left pane displays the **DEFAULT** and **CUSTOM DASHBOARDS**.
15. You can Delete / Edit the configured trend.

5.5.6.6 Having multiple widgets

You can have multiple widgets in a single canvas by configuring one after another. Below are the steps to configure multiple widgets in a single canvas.

To configure multiple widgets in the same dashboard, follow the steps below:

1. From the top pane, drag and drop a widget into the blank canvas.
2. Click **Configure Widget**.
3. Once the particular widget is configured, click the drop-down on the right corner of widget tab.
It displays a list of widget that you can configure into the same canvas.
4. Drag and drop the available widget outside the existing widget.

**NOTE**

If Performance Curve was configured first, then Monitoring table cannot be dragged and dropped on top of performance curve. It should be dropped outside the performance curve space.

- Click **Configure Widget** on the second added widget.

NOTE

The asset level selection cannot be modified for the second widget. This will be as per the selection in the first widget.

NOTE

Refer Performance/Reduced Curve and Monitoring Table for steps on configuring this widgets.

- Continue following the same steps to add more widgets into the canvas.

5.5.6.7 Edit / Delete a Configured Dashboard

To edit / delete a configured widget, follow the steps below:

- Open the left pane to view the configured default / custom dashboard.
- Select a widget from default / custom dashboard.
The selected widget is displayed on the right side.
- Click the edit  icon to update the configured dashboard.

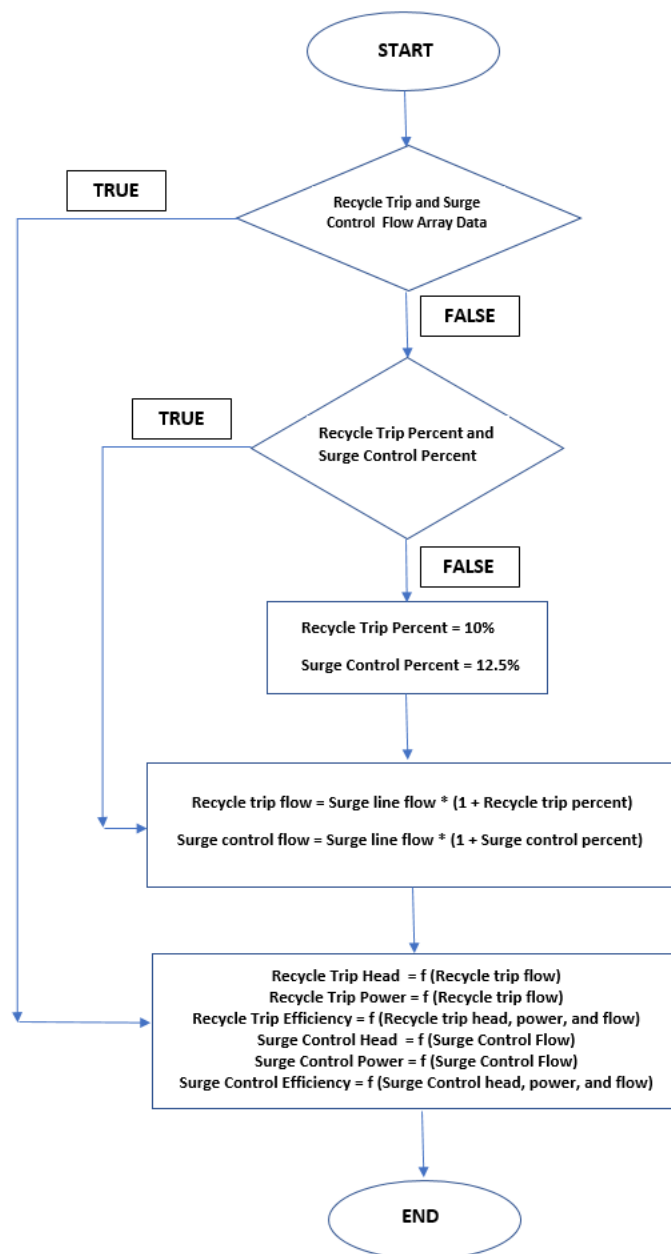
TIP

Initially you can configure at asset instance level and can use the edit option to change it to asset type level, so the widget is applied to every asset under the selected asset type.

- Click the delete  icon to remove the configured dashboard.

5.5.7 Calculation Flow Diagram

The Calculation Flow Diagram to estimate Surge Control Line and Surge Limit Line / Recycle Trip Line is shown below:



EVENT MANAGEMENT

You can view new events (including faults and symptoms) in the following views:

- Event Monitor
- Event Investigation
- Event History
- Fault Tree

The screenshot displays the 'Event Monitor' interface for the asset 'Plate and Frame HeX'. The left sidebar shows the 'CONTENTS' menu with 'Event Management' expanded, highlighting 'Monitor'. The 'ASSETS' panel shows a hierarchy with 'APM_DEMO' expanded, listing 'Pump01' and 'Imp01'. The main panel shows a table of events:

TYPE	PRIORITY	START TIME	NAME	DESCRIPTION
Symptom	High	8/10/2023 11:42:...	ColdSidePress...	Cold Side Pressure Difference RH (H...
Fault	High	8/10/2023 11:42:...	PHE HOT COL...	Plate and Frame Heat Exchanger Ho...
Symptom	High	8/10/2023 11:33:...	ColdSidePress...	Cold Side Pressure Difference RH (H...
Symptom	High	8/10/2023 3:19:3...	ColdSidePress...	Cold Side Pressure Difference RH (H...
Symptom	High	8/10/2023 12:05:...	ColdSidePress...	Cold Side Pressure Difference RH (H...
Symptom	High	8/9/2023 4:19:23...	ColdSidePress...	Cold Side Pressure Difference RH (H...
Symptom	High	8/9/2023 4:04:23...	ColdSidePress...	Cold Side Pressure Difference RH (H...
Symptom	High	8/9/2023 4:02:23...	ColdSidePress...	Cold Side Pressure Difference RH (H...
Symptom	High	8/9/2023 4:00:23...	ColdSidePress...	Cold Side Pressure Difference RH (H...
Symptom	High	8/9/2023 3:58:23...	ColdSidePress...	Cold Side Pressure Difference RH (H...
Symptom	High	8/9/2023 3:50:23...	ColdSidePress...	Cold Side Pressure Difference RH (H...

The bottom of the interface shows pagination: Page 1 of 5.

6.1 Event Management Views

6.1.1 Event Monitor

The monitor and review all New events. From this display you can access the detailed event view for more information, launch event trends, or choose what to do with the event (accept, reject, or close).

By default a maximum of 1000 events are displayed in event monitor page for each asset and active events are prioritized. This limit can be changed from the configuration settings as per the requirement.

Example:

If there are 2000 events (500 active and 1500 inactive), the 500 active events are displayed first even if they are of

older time stamp and the remaining 500 inactive events are displayed as per new to old time stamp order.

NOTE

It is recommended to view event monitoring at Asset Level for drilled down analysis instead of viewing at Site Level. Site level selection may not display older events in the list of 1000 events displayed in event monitoring page.

6.1.2 Event Investigation

Investigate, review, and update all the events you have accepted. From this display you can access the detailed event view for more information, launch event trends, or choose what to do with the event (unaccept, reject, or close). You can also lock the events or add comments as you investigate.

6.1.3 Event History

Review all the events for the selected time period, including those currently accepted or closed. From this display you can access the detailed event view for more information, launch event trends, or choose what to do with the event (unaccept, reject, or close).

A maximum of 1000 events are displayed in event history page for each asset and active events are prioritized.

Example:

If there are 2000 events (500 active and 1500 inactive), the 500 active events are displayed first even if they are of older time stamp and the remaining 500 inactive events are displayed as per new to old time stamp order.

NOTE

It is recommended to view event history at Asset Level for drilled down analysis instead of viewing at Site Level. Site level selection may not display older events in the list of 1000 events displayed in event history page.

This view is useful when analyzing an event for past occurrences or reviewing all events in a historical time frame.

TYPE	PRIORIT	START TIME	NAME	DESCRIPTION	ASSET NAME
Symptom	▼	8/10/2023 11:42:...	ColdSidePress...	Side Pressure Difference RH (H...	Plate and Frame ...
Fault	▼	8/10/2023 11:42:...	PHE HOT COL...	Plate and Frame Heat Exchanger Ho...	Plate and Frame ...
Symptom	▼	8/10/2023 11:33:...	ColdSidePress...	Side Pressure Difference RH (H...	Plate and Frame ...
Fault	▼	8/10/2023 11:33:...	PHE HOT COL...	Plate and Frame Heat Exchanger Ho...	Plate and Frame ...
Symptom	▼	8/10/2023 3:19:3...	ColdSidePress...	Side Pressure Difference RH (H...	Plate and Frame ...
Fault	▼	8/10/2023 3:19:3...	PHE HOT COL...	Plate and Frame Heat Exchanger Ho...	Plate and Frame ...

Button	Name	Description
	Accept	Accepts the event, assigning it to your user account.
	Unaccept	Unaccepts the event, assigning it to your user account.

Button	Name	Description
	Reject	Rejects the event, closing it without action. Use reject in the case of a false positive.
	Close	Closes out the event once actions have been taken to correct the reported issue.
	Add Comment	Adds a comment to the event history of an event still under investigation.
	Lock	Prevents an active event from a status change.
	Export	Exports the currently displayed event list (filtered items) to Excel.
	Show Event Details	Opens the Detailed View pane, showing the details of the currently selected event.
	Insert Event	Opens the Insert Event dialog box and allows you to manually enter an event into the system.
	Edit Event Details	To edit an existing event.
	Filter to Time Range	Applies the filtered time range once a new start and end time have been selected.
	Event Filter	Applies the filter to select a specific fault or event.

6.1.3.1 To change the time range in the event history view

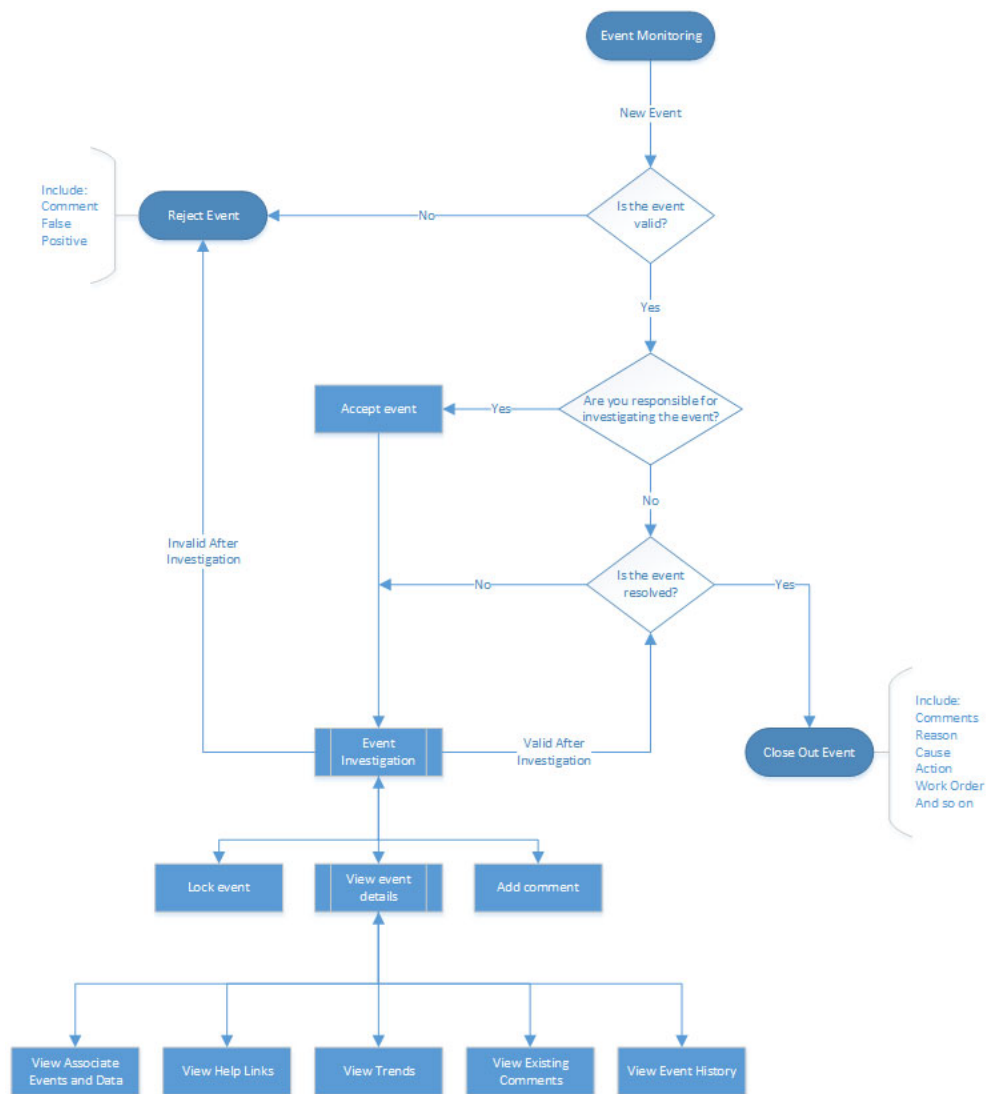
- In the start time box, do any of the following:
 - Type the start time in the format MM/DD/YYYY hh:mm AM/PM.
For example, to enter December thirteenth at midnight, type 12/13/2016 12:00 AM.
 - Click the calendar icon to open the date picker, and then select the date from the calendar.
 - Click the time icon to open the time picker, and then select the time.
- In the end time box, do any of the following:
 - Type the end time in the format MM/DD/YYYY hh:mm AM/PM.
For example, to enter December thirteenth at midnight, type 12/13/2016 12:00 AM.
 - Click the calendar icon to open the date picker, and then select the date from the calendar.
 - Review the Associated Data (for example, attributes, tagname and properties related to the event) listed in the Associated Data table.
- Click the Filter to Time Range  button to apply the time range.

6.1.4 Filters in Event Management

Each view of the event viewer is filtered for a specific view of the events:

- Monitor Faults: New events
 - Investigate Faults: Accepted events (by current user)
 - Event History: Specific time range
- In addition to these events, you can add custom filters.

6.2 Event Management Work-flow



The basic event management work-flow has two primary tasks:

- Event Monitoring: Where New events appear and initially processed.
- Event Investigation: Where events that require an in-depth look are investigated and resolved.

6.2.1 Event Monitoring

As faults, symptoms, or other events occur, they appear on the event grid. From the event monitor, you can view the details of the event, determine if it is a valid event (instead of a false positive, for example), and accept them for further investigation.

1. Open the event view to Monitor Faults, as described in [To open the event viewer](#).
2. For each event, do one of the following:
 - a. If it is not a valid event, reject the event, as described in [To reject an event](#). You can view the event details in the event monitoring view to assist in making this determination. (See [To view the event details](#).)
For example, if the Interval setting in a rule configuration is too low, events can appear too frequently for the related fault. These events can be rejected as a false positives until the configuration error is corrected.
 - b. If it is a valid event, and you are the user responsible for the investigation, accept the event, as described in [To accept or unaccept an event](#).
 - c. If it is a valid event, but you are not the user responsible for the investigation, take no action unless the issue is already resolved. If the issue is resolved, see [To close events](#).

To display both variables used in the expression and associated data in the event monitor, follow the below steps:

1. On the home page of Intuition, choose **Configuration > Application**.
2. Locate the following setting:

Key	Description
Honeywell.UAS.Events.IncludeVariablesToAssociatedData	If the is set as true the selected associated data and variables used in the expression are displayed in the associate data column of the event monitor. if the value is set as false either the expression or associated data is displayed in the associate data column of the event monitor based on the selection.

3. Click **Save**.

6.2.2 Event Investigation

During the event investigation, you can view the specific details of the event, and determine how the event is to be resolved. Add comments to the event history to record details and information discovered during the investigation, and close-out the event once the investigation and any corrective actions are complete.

1. If you have accepted the event, open event view to Investigate Faults, as described in [To open the event viewer](#).
2. To prevent the system from changing the status of an active event (for example, from active to inactive) or to prevent the same event from coming in again, lock the event, as described in [To lock an event](#).
3. To record information as you investigate the event, add a comment, as described in [To add a comment](#).
4. To view more information about the event, view the event details, as described in [To view the event details](#).
5. If after investigation the event is determined to be invalid, reject the event, as described in [To reject an event](#).
6. Once the event is resolved, close the event, as described in [To close events](#).

6.3 Event Management Example

6.3.1 Compressor Vibration Increased

Event Monitoring

The overall vibration for a compressor is being monitored in Asset Performance Management. An increase in the compressor's vibrations from 21 microns peak-to-peak to 26 triggers an event and notifies the maintenance engineer, who accepts the event for investigation.

Event Investigation

The maintenance engineer reviews the event data and requests a vibration analyst to do further analysis. He also uses the event details to launch the asset event tree and performance trend to see the impact of the problem.

The analysis by the vibration analyst concludes that the frequency changes is due to potential unbalance/mechanical looseness in the rotor (dominant peaks occurring at 1x and 2x, together with a cluster of sub-synchronous frequencies of 0.3x, 0.5x, 0.8x, 1.5x and 2.5x at the discharge end of the radial bearing).

The results of the analysis are added to the event by the maintenance engineer as a comment of the type Note, recording them in the event history. He also adds a comment with the type Action, recommending an inspection of the compressor as soon as possible.

Close-out

- **Reason:** Rotordynamic malfunction.
- **Cause:** Dirt or debris passing through the oil film.
- **Action:** Flushed oil system and replaced duplex oil filters.

6.4 Using the Event Viewer

6.4.1 Opening the Event Viewer

- The event view opens to the selected view.

6.4.2 Interface

Tool-bar

Tool-bar buttons are enabled or disabled based on the actions available for the currently selected event. If multiple events are selected, they are enabled or disabled based on the actions available for the topmost selected event.

Button	Name	Description
	Accept	Accepts the event, assigning it to your user account.
	Reject	Rejects the event, closing it without action. Use reject in the case of a false positive.
	Close	Closes out the event once actions have been taken to correct the reported issue.
	Add Comment	Adds a comment to the event history of an event still under investigation.
	Lock	Prevents an active event from a status change.
	Export	Exports the currently displayed event list (filtered items) to Excel.
	Show Event Details	Opens the Detailed View pane, showing the details of the currently selected event.
	Insert Event	Opens the Insert Event dialog box and allows you to manually enter an event into the system.
	Edit Event View	To edit an existing event.
 	Resume Pause	Starts or pauses the event viewer refresh. If status is Paused, then the Resume button is active. If status is Running, then the Pause button is active. If the connection is interrupted, the status shows as Reconnecting.
	Acknowledge	To move an event to the investigation stage without re-triggering the fault. For information on how to enable the acknowledge option, refer to Event Acknowledge Button .
	CMMS Notification	Manually generate CMMS XML files to create work orders.
	Event Filter	Applies the filter to select a specific fault or event.

Event Grid

The Event Grid shows a list of events based on the current filters (see Filters for details on changing the filter settings). You can show or hide columns as required.

TIP

The selected event is highlighted in blue color.

Column Name	Description	Visible by Default
Type	The type of event. For example, fault or symptom.	Yes
Priority	The priority of the current event, indicated by the Asset Status icon (see Contents Tree). The tooltip for the icon lists the asset status and priority (from 1-30).	Yes
Asset Name	The unique name of the asset connected to the event.	Yes
Start Time	The time the event started. Time shown in the regional server time. NOTE To display the time with milliseconds, refer to Configuring Event Viewer Time to Milliseconds .	Yes
Asset Name	The unique name of the event.	Yes
Description	The description of the event.	Yes
TRL Value	The time to reach the limit.	
Recommendation	The recommendations related to the event.	No
Associated Data	The data associated to the event. NOTE The Associated Data is hidden in the UI,	No
Event Status	The current status of the event: Active or Inactive.	Yes
Condition	The current condition (managed by user) of the event: New, Accepted, Rejected, Locked, or Closed.	Yes
Locked	Locks the event to prevent the creation the same event.	Yes
End Time	The time the event ended/reset, if applicable. Time shown in the regional server time. NOTE To display the time with milliseconds, refer to Configuring Event Viewer Time to Milliseconds .	No
Asset Criticality	The criticality of the asset: Major, Minor, Safety.	No
Path	The Asset Performance Management path for the asset and event.	No
Event Severity	Indicates how severe the problem is, at the asset type level(from 1 to 6).	No
Asset Status	The status of the asset. For example, Monitor, Unmonitored, Out of Service, Unlicensed.	No
Event Category	The asset category.	No

Column Name	Description	Visible by Default
Owner	The event owner.	No

NOTE

By default, the TRL and Remaining Useful Life value columns is disabled. To enable this column, refer to **Enabling TRL for an Attribute in APM Configuration Guide**.

In the Monitor Faults view, the events update in real-time. If you click on a row in the grid, the update pauses automatically for 30 seconds (default).

TIP

The Event Viewer shows a limited number items at a time. To view additional items, use the paging control.

To Page the Event Grid

- 1. To view the next page of items, click the Page Forward  button.
- 2. The view the previous page of items, click the Page Backwards  button.
- 3. To jump to the last page, click the Last Page  button.
- 4. To jump to the first page, click the First Page  button.
- 5. To view a specific page, in the Page box, type the page number, and then press Enter.

6.4.3 Editing Columns in the Event Viewer

Show or Hide the Columns

- 1. On any column in the event viewer, click the down arrow to open the dialog box.
- 2. Point to Columns, and then select or clear the check boxes to show or hide the columns.

Sorting the Columns

- 1. On any column in the event viewer, click the three dots  to open the dialog box.
- 2. Do one of the following:
 - a. Click **Sort Ascending** to sort by ascending values.
 - b. Click **Sort Descending** to sort by descending values.

Resizing the Columns

- 1. Hover your mouse over the column boundary until the resize icon appears.
- 2. Click and drag the column edge to the required size.

Change the Column Order

1. Click any column in the event viewer.
2. Drag the column over the column headers and drop in the new position.

6.4.4 Filters in Event Viewer

Below are the set of filter functionality available in the event viewer.

Filter out a Specific Fault or Event

1. Open **Event Monitor** and click **Event Filter** drop-down.
2. Below are the list of fault and event in the drop-down.
 - a. **All Event**: Selecting this filter enables you to see all events for the default event count configured in the event monitor.
 - b. **All Fault**: Selecting this filter will enable you to see active faults for the default event count configured in event monitor.
 - c. **Active Faults**: Selecting this filter will enable you to see active faults for the default event count configured in event monitor.
 - d. **Active Events**: Selecting this filter will enable you to see active events for the default event count configured in event monitor.
 - e. **Repetitive Fault**: Selecting this filter will enable you to see faults that's repetitive in nature for the default event count configured in event monitor.
3. Select an **Event or Fault** from the list.

Adding a Filter

1. On the column on which you are applying the filter, click the down arrow to open the dialog box.
2. Point to **Filter**.
3. The filter options vary depending on the column. See one of the following for the options, based on the type of column:
 - a. To apply a filter on a value or string column.
 - b. To apply a filter on a multiple selection column.
 - c. To apply a filter on a single selection column.
4. Click **Filter**.

Apply a Filter on a Value or String Column

1. Do any of the following:
 - a. From the drop-down list, select the filter condition:
 - a. Is equal to
 - b. Is not equal to
 - c. Is greater than or equal to
 - d. Is greater than
 - e. Is less than or equal to

- f. Is less than
 - b. In the box, type the value for the filter condition.
2. To add an additional filter, do all of the following:
 - a. select the filter operator:
 - a. **And**: events must match both conditions to be displayed.
 - b. **Or**: events can match either condition to be displayed.
 - b. From the drop-down list, select the filter condition.
 - c. In the box, type the value for the filter condition.

Apply a Filter on Multiple Columns Selected

From the options listed, select all the values to show in the filter.

For example, the options available in the Status columns are Select All, Active, or Inactive.

Applying a Filter on Single Column Selected

From the options listed, select the value to show in the filter.

For example, the options available in the Locked column are Locked or Unlocked.

6.4.5 Configuring Event Viewer Time to Milliseconds

Follow the below steps to display the start and end time with milliseconds for all the events in the Event Management page:

1. Open the Application Configuration page (**Gear icon > System > System Configuration > Application**).
2. Search for the below key and update the key value to **True**:
Honeywell.UAS.Events.IncludeMillisecondsInEvents

NOTE

By default, the value is set to **False**.

3. Click **Save**.
4. Refresh the event management page to view the start and end time with milliseconds for all the events in the page.

6.4.6 Instance Level UoM in Event Viewer

Follow the below steps to display the values and units of an event at Instance Level,:

1. Open any web browser.
 Type `https://<machine name>:<port>/MES/Default.aspx#/UAS/UASLandingPage.aspx` in the address bar, where the **<machine name>** is fully qualified domain name of the Asset Performance Management Server and **<port>** is the configured SSL port.
 The **Honeywell Application** landing page appears.
2. In the **System** area, click **System Configuration**.

3. Locate and update the below key value:

Key	Description
Honeywell.UAS.EventViewer.EnableInstanceLevelUOMForEventMonitor	Update the key value to True to display the event monitor values at instance level Uom. Update the key value to False to display the event monitor values at type level UoM.

NOTE

The associated data column is hidden if instance level UoM for event monitor is enabled. To view the associated data for instance level UoM, export the events to an **Excel**.

6.4.7 Accept or Decline an Event

Follow the below steps to accept or decline an event in an event viewer window:

1. To accept an event, select an unaccepted event in the event in the Grid, and then click the Accept  button.

NOTE

The Accept button cannot be selected if the event has already been accepted.

2. To decline an event, select an event you've accepted in the event in the Grid, and then click the decline  button.

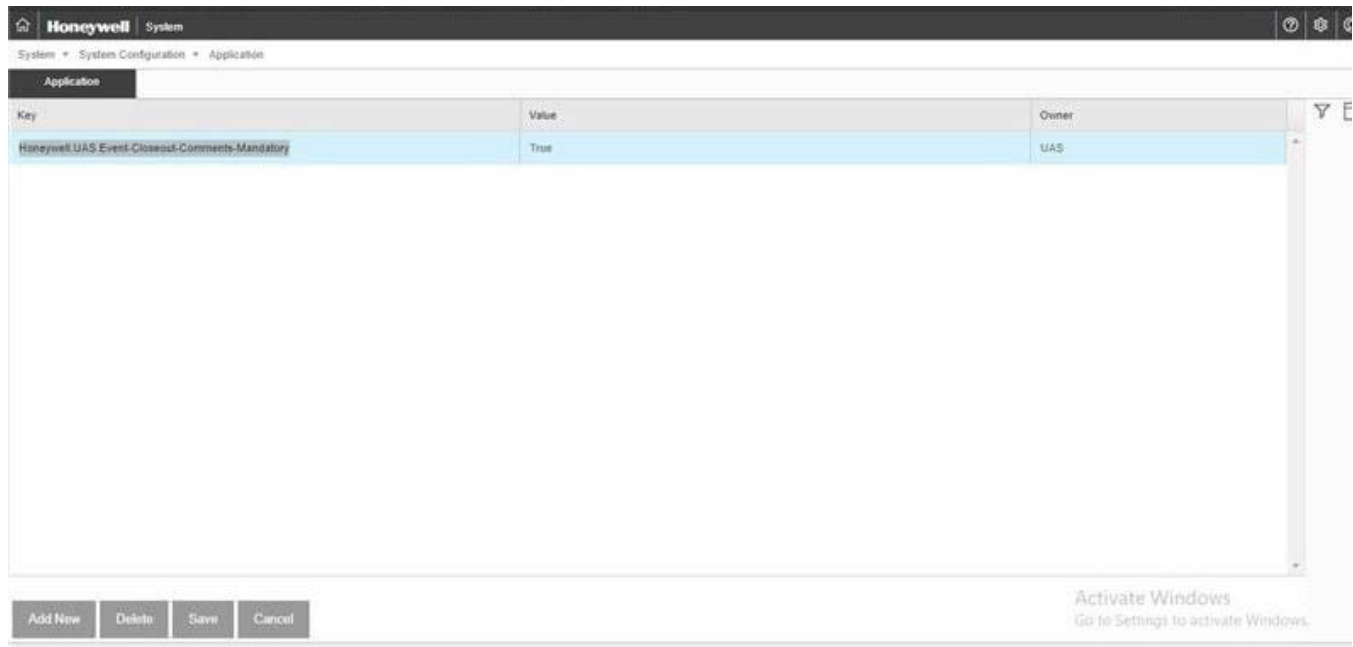
6.4.8 Enabling Event Closeout and Reject Comment

NOTE

The following steps are applicable only for Closing or rejecting the events.

- 1. Open the Application Configuration page (**Gear icon > System > System Configuration> Application**).
- 2. Click **Add New**, in the Lower-right corner of the page.

A dialogue box appears.



- 3. Add the below key and update the value as given below:

Key	Value	Interface
Honeywell.UAS.Event-Comment-isMandatory	True	Close-out comments will be mandatory on Event Pages.
	False	Close-out comments will be optional on Event Pages.

NOTE

When you do not apply any setting in the application configuration page, it is considered as no setting applied, that makes the close-out comments mandatory in Event Page.

- 4. Click **Save**.

6.4.9 Close Out Event

You can close one or more events at a time in an event viewer. If multiple events are selected, the information entered is saved to all the selected events.

1. Select the event(s) in the Grid, and then click the Close  button.
2. In the **Close Out** dialog box, on the **Basic** tab, do all of the following:
 - a. **Category**: Type or select the reason category for the event.
For example, Performance, Safety, or Maintenance.

NOTE

The dialog box will have a predefined list only if the previous selection was predefined. In case a custom information was entered the next dialog box should be entered with custom information.

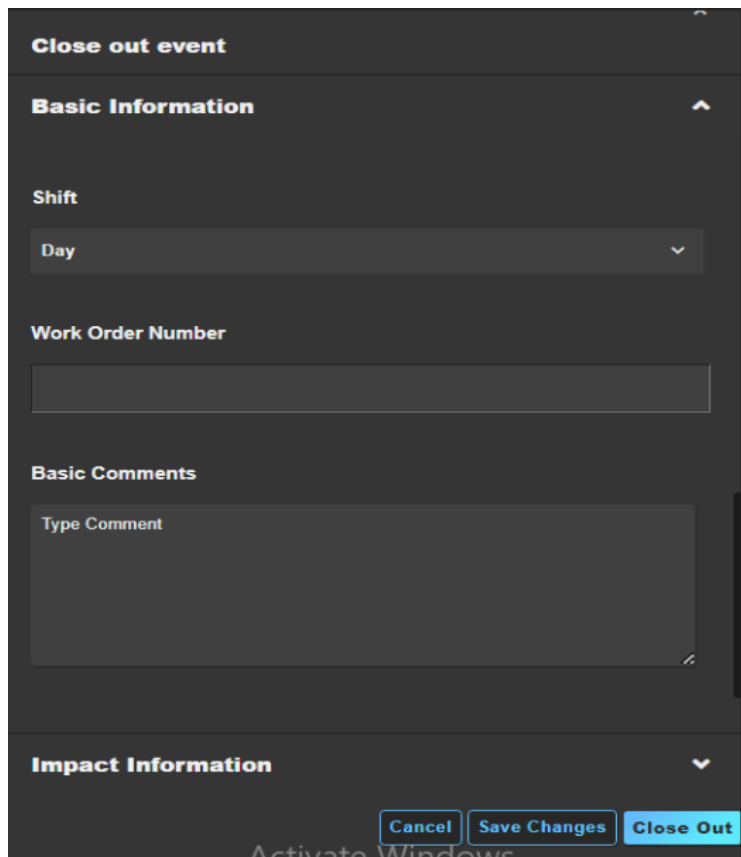
For example, if user enter a custom information in **Category**, then **Reason** and **Cause** wont have a predefined value to select. It should be entered with custom information.

- b. **Reason**: Type or select the reason for the event.
- c. **Cause**: Type or select the cause of the event.
- d. **Corrective Action**: Type or select the action taken to resolve the event
- e. **Recommendation** (Required): Type any additional comments related to the close-out of the event.
- f. **Shift**: Select the name of the shift responsible for closing the event.

NOTE

The shift selection is retained irrespective of the session.

- g. **Work Order Number**: Type the work order number, if any, related to closing the event.
- h. **Basic Comments**: To review previous comments related to the event.



The screenshot shows the 'Close out event' dialog box with the 'Basic Information' tab selected. The dialog has a dark theme. It contains the following fields:

- Shift**: A dropdown menu with 'Day' selected.
- Work Order Number**: A text input field.
- Basic Comments**: A large text area with the placeholder 'Type Comment'.
- Impact Information**: A section header with a dropdown arrow.

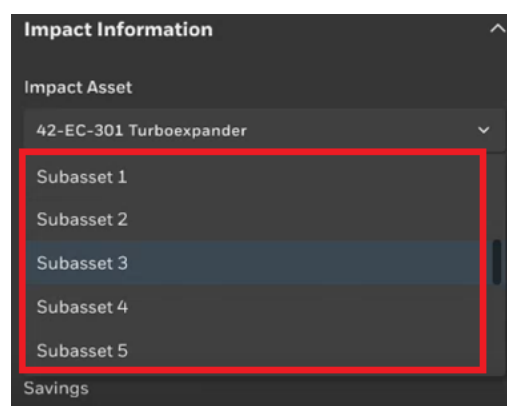
At the bottom of the dialog are three buttons: 'Cancel', 'Save Changes', and 'Close Out'. A watermark 'Activate Windows' is visible at the bottom of the screen.

3. (Optional) If there are details for the monetary impact related to the event, select the Impact tab, and then do all of the following:
- Impact Asset:** Type the name of the asset that was the root cause of the event.

NOTE

By default the asset to which the close out have been triggered will be displayed here.

User can have a predefined list of components added to **Impact Asset** by updating the value tab of **Honeywell.UAS.EventViewer.CloseOut.ImpactComponentsList** from system configuration settings.



- Impact Type:** Type or select an impact type from the list. By default, the following impact types are available:
 - Low Quality
 - Maintenance
 - Over Production
 - Shutdown
 - Slowdown

NOTE

The status Not Applicable is used when there is no associated impact. When Not Applicable is selected an impact comment is not required and the cost fields are disabled. To enter an associated cost, select one of the other Impact Types.

- Dollar Saved (USD):** Savings achieved by the early triggering of an event avoiding downtime.
- Downtime Avoided (Hours):** Type the potential downtime that was avoided because of early detection of a fault..
- Duration (Hours):** Type the duration for the event impact.
- Production Cost:** Type the cost for any impact to the production associated with the event.
- Maintenance Cost:** Type the total for any maintenance cost associated with the event.
- Yield Cost:** Type the cost for any impact to the yield associated with the event.

NOTE

If there is a savings associated with the event for any of the cost fields, enter the savings total as a negative value.

- i. (Required) In the **Impact Comment** box, type a description of the impact and related costs or any other details related to the impact.

Close out event

Basic Information

Impact Information

Dollar Saved [USD] Downtime Avoided [Hours]

Cost

Production Cost [USD] Maintenance Cost [USD]

Yield Cost [USD]

Impact Comments

Type Comment

Cancel Save Changes Close Out

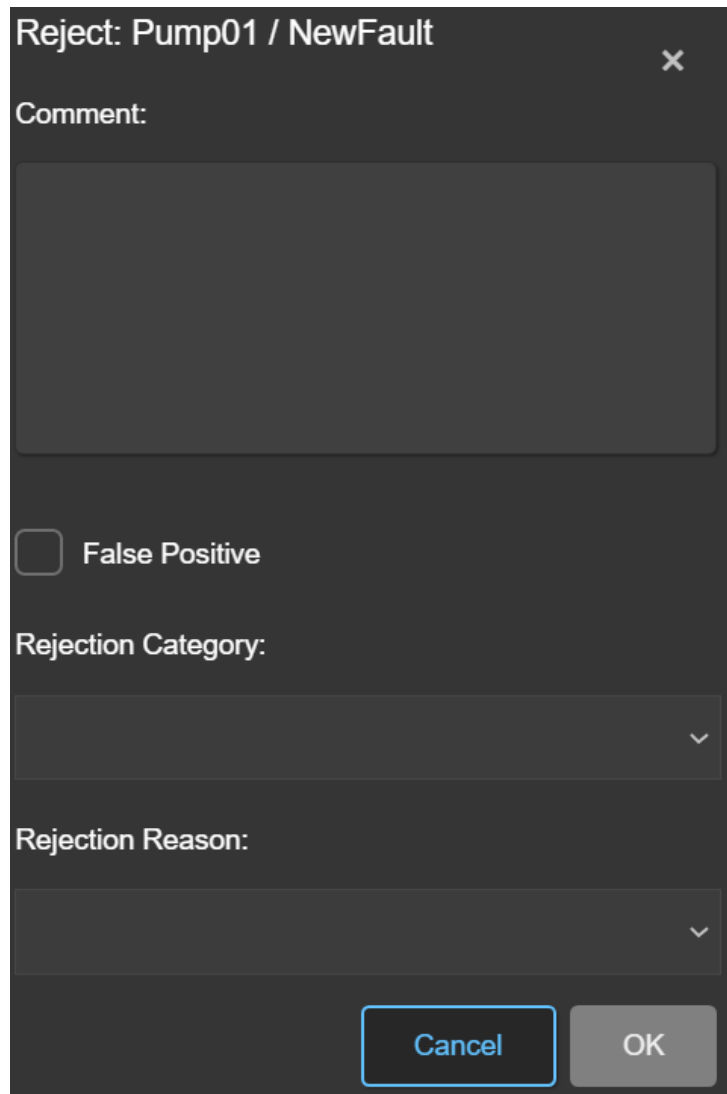
- 4. Do one of the following:
 - a. If the event is resolved, click **Close Out** to close the event.
 - b. To continue making changes, click **Save Changes** to save the current close-out details.

6.4.10 **Reject an Event**

Follow the below steps to reject an event in event viewer window:

1. Select the event in the Grid, and then click the Reject  button.

The **Reject** dialog box appears.



The image shows a 'Reject' dialog box with a dark gray background. At the top, the title bar reads 'Reject: Pump01 / NewFault' with a close button (X) on the right. Below the title bar is a 'Comment:' label followed by a large, empty text area. Underneath the text area is a checkbox labeled 'False Positive'. Below the checkbox is a 'Rejection Category:' label followed by a drop-down menu. Below the drop-down menu is a 'Rejection Reason:' label followed by another drop-down menu. At the bottom of the dialog box are two buttons: 'Cancel' (highlighted with a blue border) and 'OK'.

2. In the **Reject** dialog box, do all of the following:
 - a. In the **Comment** box, provide details for the rejection.
 - b. If the event is a false positive (the condition flagged by the event did not actually occur or has been determined to be invalid by the user), select the **False Positive** box.
 - c. Selecting **False Positive** box, enables **Rejection Category** and **Rejection Reason**.
 - a. From the **Rejection Category** drop-down list, select the required category:
 - a. Configuration Problems
 - b. Data Problems
 - c. Infrastructure Problems
 - d. Model Problems
 - e. Unknown Operational differences
 - f. Other
 - b. From the **Rejection Reason** drop-down list, select the required reason:

- a. Bad Tuning
- b. Default Reason
- c. General Configuration Problems
- d. General Data Problems
- e. General Infrastructure Problems

- d. Click **OK**.

Accepting a Rejected Event

Accepting the rejected event from **Event History** window will bring back the event to **Event Investigate** and **Event Monitor** window with **Event Status** as **Inactive**.

Follow the below steps to enable the functionality to accept a rejected event:

1. Open the Application Configuration page (**Gear icon > System > System Configuration > Application**).
2. Search for the below key and update the key value to **True**:
Honeywell.UAS.EnableRejectEventAccept

NOTE

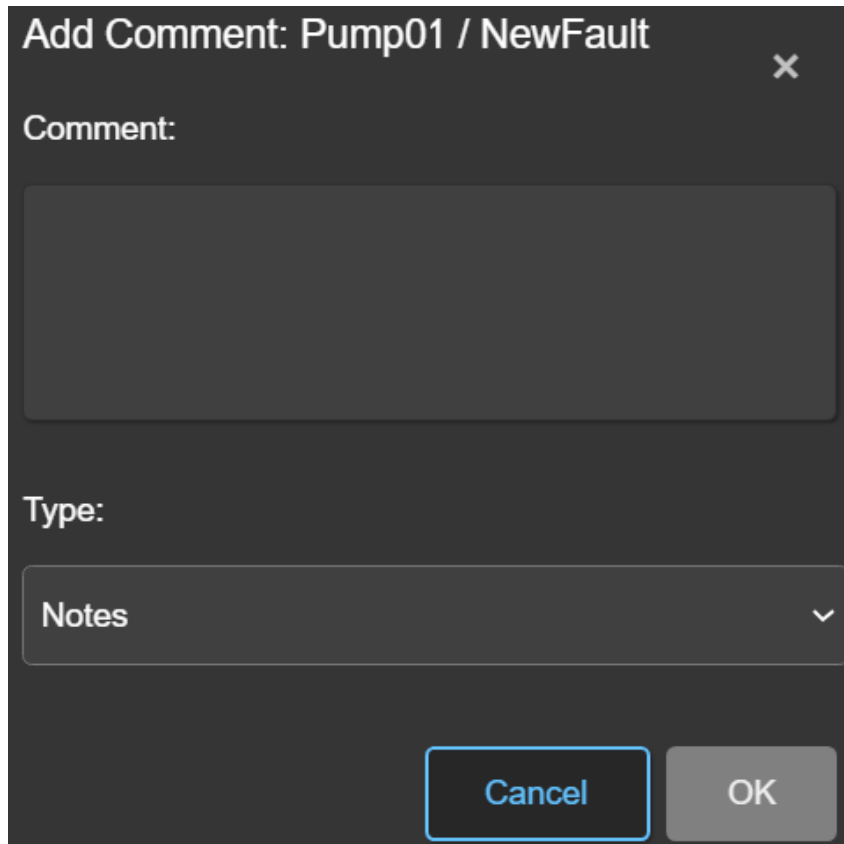
By default, the value is set to **False**.

3. Click **Save**.
4. Open **System Console** under **Default Views** drop-down.
5. Click **Publish**.
6. The reject event accept functionality will be enabled once the changes are publish successfully.

6.4.11 Add a Comment

Follow the below steps to add a comment to an event in the Event Viewer window.

1. Select the event in the Grid, and then click the Add Comment  button.



The image shows a dark-themed dialog box titled "Add Comment: Pump01 / NewFault" with a close button (X) in the top right corner. Inside the dialog, there is a "Comment:" label followed by a large, empty text input field. Below this is a "Type:" label followed by a dropdown menu currently showing "Notes" with a downward arrow. At the bottom of the dialog are two buttons: "Cancel" and "OK".

2. In the **Add Comment** dialog box, in the comment field, enter the information about the event.
3. From the Type list, select the type of comment you are adding:
 - a. Notes
 - b. Inspection
 - c. Observation
 - d. Action
4. Click **OK**.

6.4.12 Lock an Event

Locking an event in event viewer window prevents the same event from reoccurring and also prevents the system from changing the status of an active event. For example, if the event no longer meets the conditions specified, the system can no longer change the status from Active to Inactive.

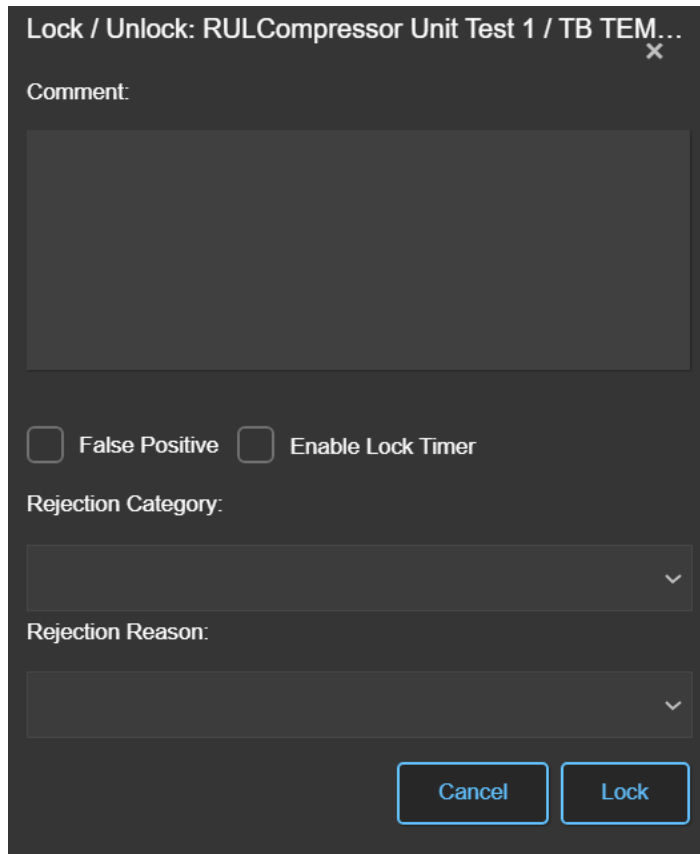
NOTE

Only active events can be locked or unlocked.

Follow the below steps to lock an event:

1. Select the event in the Grid, and then click the Lock  button.

The **Lock/ Unlock** dialog box appears.



The dialog box is titled "Lock / Unlock: RULCompressor Unit Test 1 / TB TEM...". It contains a "Comment:" label above a large text area. Below the text area are two checkboxes: "False Positive" and "Enable Lock Timer". Under these are two drop-down menus labeled "Rejection Category:" and "Rejection Reason:". At the bottom right are "Cancel" and "Lock" buttons.

2. In the Lock/Unlock dialog box, in the comment field, enter the reason for locking the event.
 - a. In the **Comment** box, enter the reason for locking the event.
 - b. If the event is a false positive (the condition flagged by the event did not actually occur or has been determined to be invalid by the user), select the **False Positive** box.
 - c. Selecting **False Positive** box, enables **Rejection Category** and **Rejection Reason**.
 - a. From the **Rejection Category** drop-down list, select the required category:
 - a. Configuration Problems
 - b. Data Problems
 - c. Infrastructure Problems
 - d. Model Problems
 - e. Unknown Operational differences
 - f. Other
 - b. From the **Rejection Reason** drop-down list, select the required reason:
 - a. Bad Tuning
 - b. Default Reason
 - c. General Configuration Problems
 - d. General Data Problems
 - e. General Infrastructure Problems
 - d. Select the **Enable Lock Timer** checkbox to displays the **Expiry Date** column, enabling you to set the lock expiry date.

The event will be unlocked automatically after the selected time interval.

ATTENTION

The minimum lock period must be 5 minutes. For information on configuring lock period, refer to [Configuring Lock Timer](#).

TIP

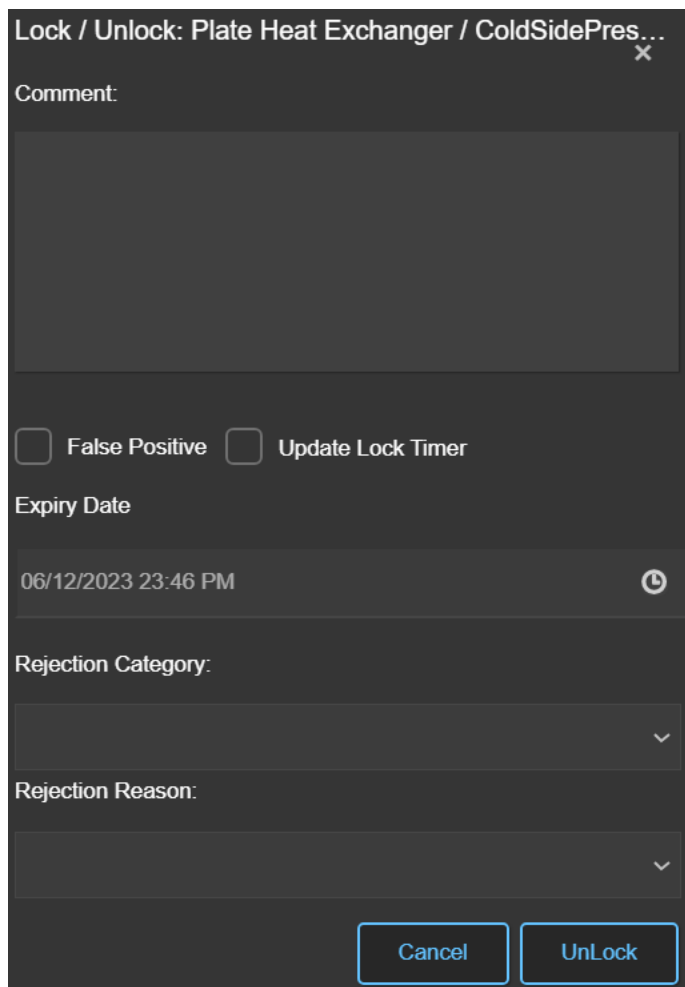
Click **Now** checkbox to select a time period before or after the current time.

- e. Click **Lock**.

Follow the below steps to unlock an event:

1. Select a locked event and click the unlock  button.

The **Lock/ Unlock** dialog box appears.



Lock / Unlock: Plate Heat Exchanger / ColdSidePres...

Comment:

☐ False Positive ☐ Update Lock Timer

Expiry Date

06/12/2023 23:46 PM

Rejection Category:

Rejection Reason:

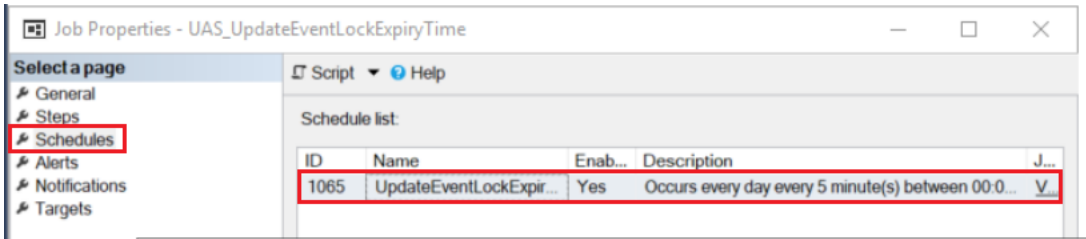
Cancel UnLock

2. In the **Comment** box, enter the reason for unlocking the event.
 3. Select the **Update Lock Timer** checkbox to update the lock time period and then click **Lock** to update the lock time of the event
- Or
- Click **UnLock** to unlock the event before reaching the auto unlock period.

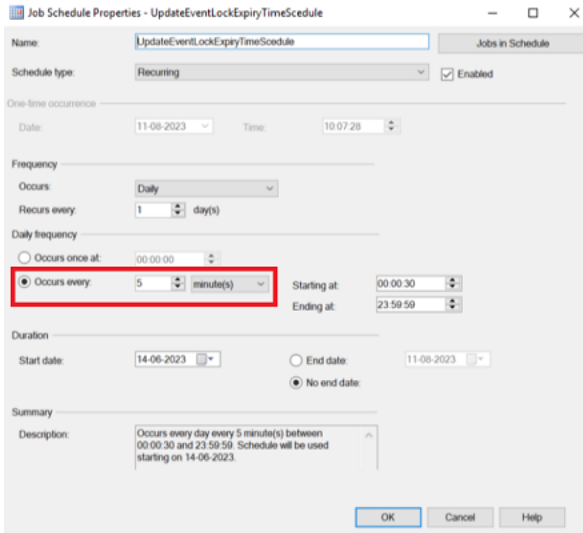
6.4.12.1 Configuring Lock Timer

Perform the below steps in database server to configure minimum lock timer in Event Monitor:

- 1. Open SQL Server Management Studio
- 2. Under Object Explorer pane, right click **UASUpdateEventLockExpiryTime** (SQL Server Agent > Jobs)
- 3. Select **Job Properties**.
Job properties window appears on the right side.
- 4. Click **Schedule** and double click the scheduled job.



- 5. Update the **Occurs every** column value of **Daily frequency** to 5 minutes



- 6. Click **Ok**.

6.4.13 Export Events to Excel

Follow the below steps to export all the event for the selected asset to an Excel:

- 1. Click the Export  button.
- 2. When prompted, save the Excel file.

NOTE

The Excel file contains only the columns currently displayed in the Event Viewer.

6.4.14 Viewing Event Details

As you are investigating the causes for an event, it can be useful to view more information about the event than is shown in the Event Grid. Viewing the event details shows you associated events and data in the system, any help links related to the event, a basic event trend, any existing comments related to the event, and the more recent entries in the event history.

Event history section contains information such as start date, end date, closed date, user name, comments, reason category, reason, cause, and recommendations of historical data.

TIP

Click export  icon to download all historical data for the selected asset into an excel sheet.

NOTE

The event details have quick access to a number of the asset analysis tools, such as the Asset Event Tree and the Trends. For more information on the asset analysis tools, see “Asset Analysis” on page 60.

To View the Event Details

1. Select the event in the Grid, and then click the Show Event Detail  button.
The Event Details pane opens on the bottom of the viewer.
2. On the **Details** tab, do any of the following as required:
 - a. Review any associated events (for example, if this is a fault with associated symptoms) listed in the Associated Events table.
 - b. To view the Asset Event Tree for the selected event, click Show Asset Event Tree. (See [Fault Tree](#) for more information.)
 - c. Review the Associated Data (for example, attributes, tag name and properties related to the event) listed in the Associated Data table.
 - d. To view the trend for the asset for the event, click Show Trends. (See [Using the Trends](#) for more information.)
3. On the **Help** tab, review either of the following:
 - a. **Recommendations**: This field displays any recommended actions (configured on the system) for the event.
 - b. **Associated Links**: This table lists any links associated to the event, such as documentation or procedures.
4. On the **Trends** tab, review the trend.

NOTE

The trend displayed on the event details Trend tab is simplified. To view more detail, use the Performance Trend, as described in Using Performance Trends.

5. On the **Comments** tab, review the comment history for the event.
6. On the **History** tab, review the event history, which includes previous occurrences of the same event.

6.4.15 Insert an Event

If you need to enter an event into the system for a condition not previously configured, or because an event was not recorded when it should have occurred, you can insert an event from the **Event Monitor** or **Event History** views.

NOTE

If you are inserting an event already configured on the system, the inserted event must occur after the most recent recorded occurrence of the event.

1. On the toolbar, click the Insert Event  button.
2. In the **Insert Event** dialog box, from the **Event Type** list, select the type of event you are adding:
 - a. **Fault**: to insert previously configured fault event.
 - b. **Symptom**: to insert a previously configured symptom event.
 - c. **Custom**: to insert a previously configured custom event or to create, configure, and insert a new event.
3. In the Name box, type and/or select the name of the event.

As you type a name, the system will show a list of the previously configured events that match the pattern of the name, filtered to the selected event type.
4. If you are adding an event that has not yet been configured on the system, do all of the following:
 - a. From the **Severity** list, select the severity for the new event.
 - b. In the **Description** box, type the description of the event.
 - c. In the **Recommendation** box, type recommended course of action when the event occurs.

If you have selected a previously configured event, these settings display the previously configured details and cannot be changed.
5. From the **Status** list, select the status of the event you are adding:
 - a. **Active**
 - b. **Inactive**

NOTE

For an inactive event to be accepted by the system, there must be a corresponding active event occurring before the event you are adding. If the event is not active at the time of the new event, the inactive event is not created. If you are entering a record for an event that activated and then inactivated without either being recorded by the system, enter the active event first and then the inactive.

 - c. **Potential** (faults only)
6. For Active or Potential, in the **Start Date** box, enter the time and date at which the event occurred.
7. For Inactive, in the **End Date** box, enter the time and date at which the event occurred.
8. Click **Insert**.

6.4.16 Event Acknowledge Button

Enabling Event Acknowledge Button

Follow the below steps to enable the event acknowledge  button in the event monitor window:

1. Open the Application Configuration page (**Gear icon > System > System Configuration > Application**).
2. Search for the below key and update the key value to **True**:
Honeywell.UAS.enableAcknowledgeOptionInEventViewer

NOTE

By default, the value is set to **False**.

- 3. Click **Save**.
- 4. Refresh the event monitor page to display the **Acknowledge** button.

Event Acknowledge

The acknowledge button will move the event to **Event Investigate** window with the status as **Inactive** and no event **End Time**. If you unacknowledge the event from Investigate window, the event will move back to Event Monitor window as a normal event with **Active Status**.

Follow the below steps to acknowledge an event:

- 1. Open the event monitor page for an asset.
- 2. Select an event and click acknowledge  icon.
- 3. The acknowledged event can be viewed from the event investigate page, event history, and the in-progress section of the asset health dashboard notification panel.

6.4.17 Play or Resume Event Updates in the Event Viewer

- 1. If the updates are currently active, click the Pause  button to pause the updates until you click resume.
- 2. If the updates are currently paused, click the Resume  button to resume real-time updates.

6.4.18 Generate CMMS Notification

Follow the below steps to generate a CMMS notification manually:

- 1. Click **CMMS Notification**  button.
Generate CMMS Notification pane appears right side.
- 2. Select an **Event Type** from the dropdown.

NOTE
By default, **Fault** is selected as the event type.

- 3. Select an **Event Name** from the dropdown.

NOTE

- Only the event names available for the selected asset are displayed in the dropdown.
- A notification can be generated only with the event name, remaining fields are not mandatory.

4. Below fields are automatically updated according to the pre-configured data of the selected event name.
- a. Organization ID
 - b. Enterprise ID
 - c. Site ID
 - d. Location ID
 - e. Meter ID

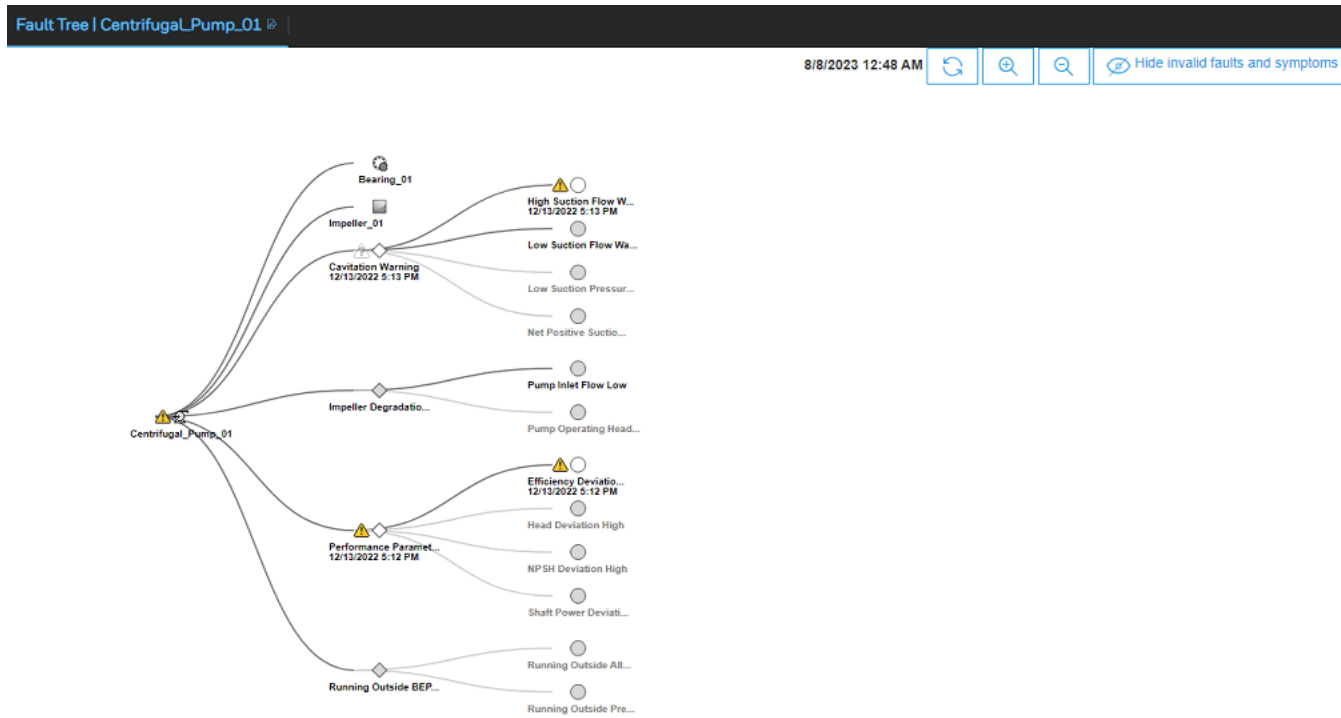
NOTE

You can update the existing data as per the requirement.

5. Select the **Problem Group / Cause Group**, **Problem ID / Cause ID**, and **Notification Type** from the dropdown.
6. Click **Generate Notification**.
7. By default the generated notification will be available in the below location:
<Install Location>:\ProgramData\Honeywell\AssetManager\CMMSXML
8. To update the save location of CMMS Notification files, navigate to the below path in Web Server and update the location in **web.config** file
<Install Location>:\Program Files (x86)\Honeywell\MES\AssetManager\WebServer\UI

6.5 Fault Tree

The Fault Tree is a graphical display of your system assets and associated faults and symptoms with performance indicators. You can use the Fault Tree to locate the source of a performance issue.



The below table describes the Interface of the Fault Tree:

NOTE

Fault or Symptom becomes invalid when it does not have supporting instruments/sensors connected to the asset.

TIP

- By default, you can see all valid and invalid faults and symptoms.
- If you want to hide the invalid faults and symptoms in fault tree page, on the top right corner you have to click **Hide invalid faults & symptoms** button.
- Also you can see the number of Invalid faults and Invalid symptoms.

Tree components	Status	Interface
Faults	Valid	Continuous line in black
Symptoms		
Faults	Invalid	Continuous line in light gray
Symptoms		

To open the fault tree

1. On the asset tree, select an asset to set the context.
2. On the Contents tree, under **Detail Views**, select **Fault Tree**.

TIP

Hover the mouse over the name of an attribute to display its expression in a tooltip.

To open a trend from the fault tree

1. On the Fault Tree, right-click the name of the asset.
 2. Click **Open Trend**.
- The trend for the asset appears in a new tab.

To open an event trend from the fault tree

1. On the Fault Tree, right-click the name of the asset/fault.
 2. Click **Open Event Trend**.
- The trend for the selected asset/fault appears in a new tab.

To show child assets or attributes

On the Fault Tree, click the icon next to the parent asset.

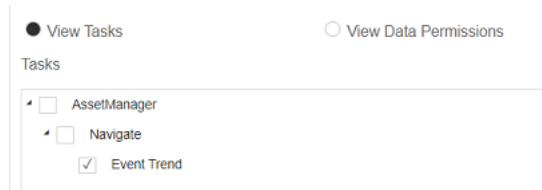
The tree expands to show the child assets or attributes for the selected asset.

6.6 Using the Trend

NOTE
Not all the features are supported on Internet Explorer browser currently. We recommend using Google Chrome for a better experience.

ATTENTION
You will lose the old Asset Trend saved configurations while migrating to new Trends page. Create the same saved configuration in New Trends page and we recommend you use the X-Y plot for tabular arrays to visualize trends in Performance Widget.

ATTENTION
To enable trends feature, open the APM application, open the Security page (**Gear icon > System > System Configuration > Security**) and then enable the Event Trend security role.



To open the Trend

From the **ASSETS** pane, select an asset and click **Trends** under **CONTENTS** pane to display the trend for the selected asset.

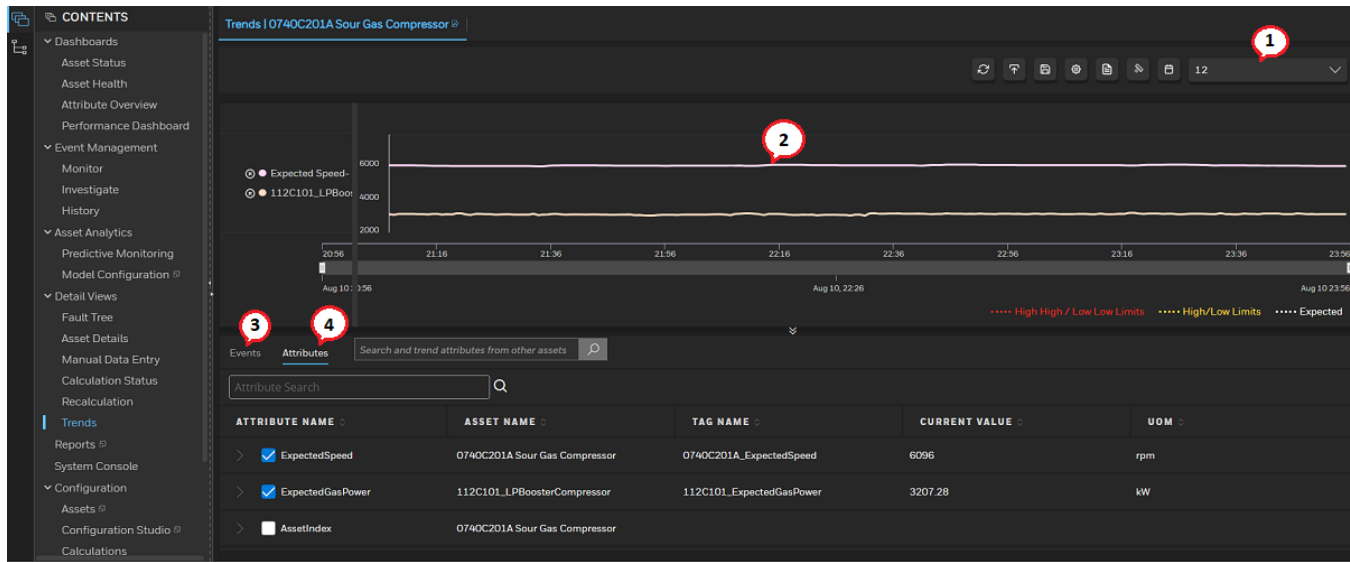
To open the Event Trend

From the **Event Monitor** page, right click any event and select **Event Trend**. The event trend page appears. You can also open the Event Trend from Event Investigation or Event History page by right clicking any event and selecting Event Trend.

6.6.1 Interface

The interface displays the event trend data for the selected event, associated attributes and the selected date range.

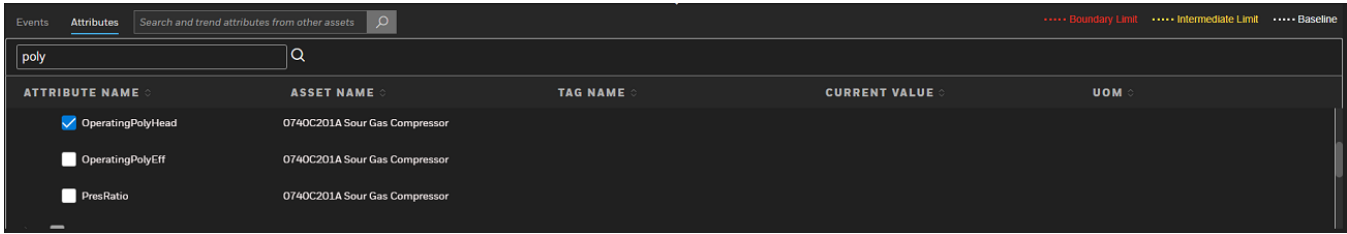
The trend displayed will be empty if there are no trend template configured for the selected asset.



Button	Name	Description
	Refresh	Refreshes the content displayed in the grid and the event trend starts considering the current time
	Export as JPG	To export a JPG image of the displayed trend.
	Save Changes	Click save button to save any changes made to the displayed event trend.
	Settings	Click the trend settings button to update the displayed trend settings.
	Show or Hide Details	Click the button to expand and compress the detail section from the bottom of the page.
	Asset Type Level UOM	Click Asset Type Level UOM button to display the trend at Type Level. By default the units are displayed at Instance Level. NOTE All the unit values available in the selected trend are updated as per the Asset Type Level UOM check box selection. The Type Level values are displayed for reference and refreshing the page will revert the unit values back to Instance Level.
	Time Range	You can select a custom date range from the drop down menu of time picker in the top right corner of the page to view the event trend data for the required time period. NOTE The default time range of 3 hours is displayed in both Chrono view and trend for the selected attributes.
1	Template Selection	Select a configured trend template from the list to display the hairline and values in the grid. Click  icon to remove a configured template. An empty canvas is displayed in case of opening an asset with no configured trend template.
2	Trend Graph	Hover the cursor on the displayed graph to view the data of the preferred time and date. The current value of the selected data point from the high chart is displayed next to the Hairline and also the same data is displayed in the current value column from the attributes grid. The dynamic hairline gray in color and all the fixed hairline will have different colors. Timestamp is visible on both fixed and dynamic hairline. NOTE Right-click on an event from EVENT MONITOR page and select Open Event Trend will display the trend with hairline plotted as per the event start time in event monitor page.
3	Event Information	The event information like, Event type, priority, status, condition, locked status, start time and end time are displayed at the bottom page of the interface.

Button	Name	Description
4	Attributes Information	You can search for the required attributes and events columns data from the global search box in attributes tab.

6.6.2 Attributes



Select the required attributes for the trend from the attributes listed under Attributes tab or drag and drop the attributes to the chart to merge and view the consolidated current value of the attributes.

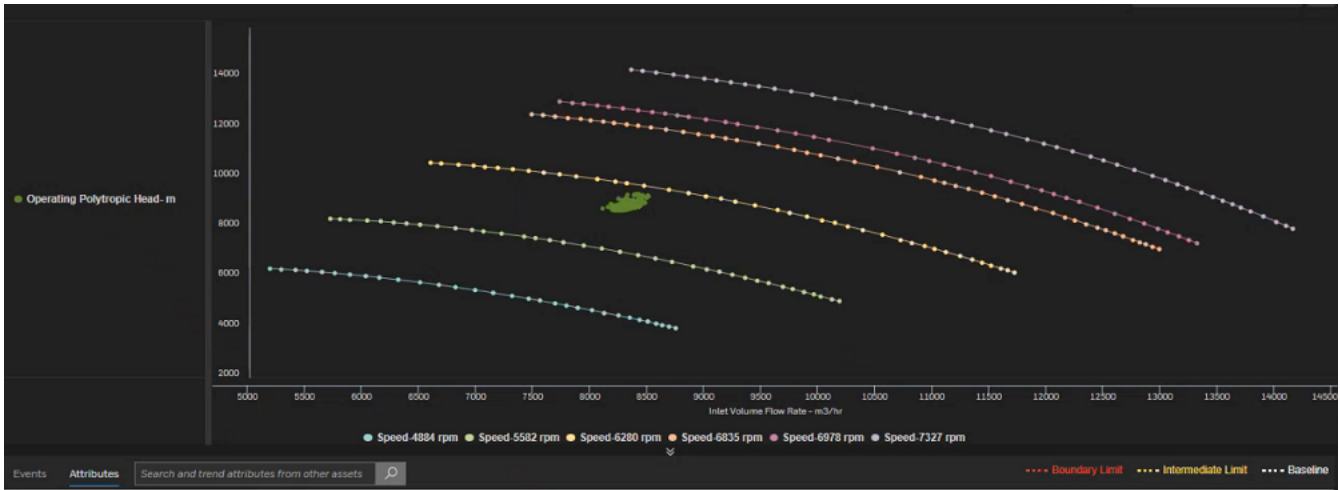
The attributes can be configured from calculation configuration page. This can be done during event Fault or Symptom configuration, excel add-in, or via edit action.

NOTE
By default only the attributes that can be trended is displayed.

Hover the cursor on the displayed graph to view the data of the preferred time and day. If you select data point from the displayed graph the same data will be shown in the current value column for the selected attributes.

TIP
The tags, current value and UoM provides information about the attribute.

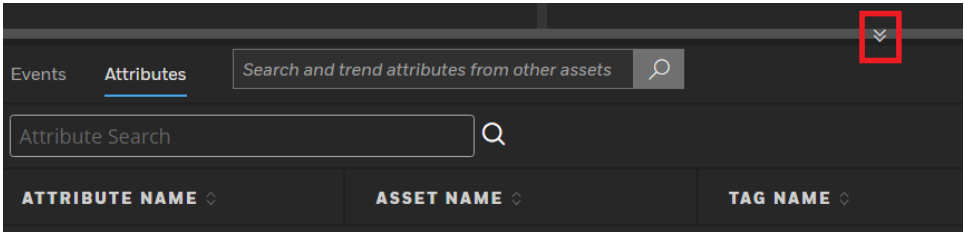
Now you can plot XY and XYZ trends on the Event Trend page and the XY plot is always displayed in datamarkers.



NOTE
You cannot have XY and XYZ plot in same trend. If XYZ trend is plotted, all other stack plots must be of XYZ trend.
You can create a maximum of 15 stack plots in XYZ trend.

TIP

Click the highlighted arrow to increase the chart area.



6.6.3 Events

EVENT NAME ↑↓	START TIME ↑↓	END TIME ↑↓	TYPE ↑↓	PRIORITY ↑↓	STATUS ↑↓	CONDITION ↑↓	LOCKED ↑↓
Compressor_Health_Degradation	06/05/2023 5:08:36 AM	06/06/2023 8:41:53 AM	FAULT	Low	InActive	Closed	false
Flow_Outside_Surge_Line	06/05/2023 4:39:16 AM	06/05/2023 5:08:36 AM	SYMPTOM	Medium	InActive	Closed	false
Surge_Margin_L	06/05/2023 4:39:16 AM	06/05/2023 5:08:36 AM	SYMPTOM	Medium	InActive	Closed	false
Flow_Outside_Stonewall	11/16/2022 10:48:38 PM	11/17/2022 4:19:24 PM	SYMPTOM	Medium	InActive	New	false

The event history for the selected date and time range is displayed in this tab and selecting the event trend is displayed in a new tab.

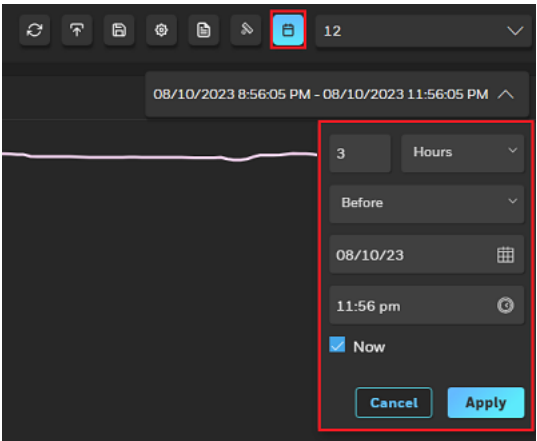
NOTE

The instance of the event that appeared in the past within the selected date and time is displayed.

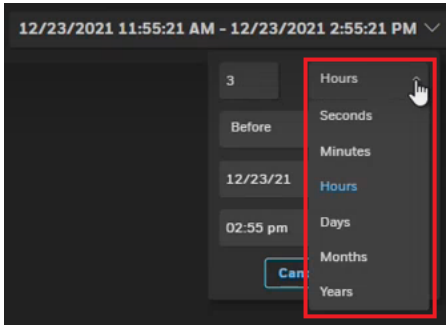
The columns of start time, end time, event type, priority, status, condition, and locked status help to debug the faults or symptom. The legend indicates the Active, Potential and inactive of the fault in the chart.

6.6.4 Date Range Picker

The date range picker is used to select a custom date and time range for the displayed trend. By default time range of 3 hours is displayed in both Chrono view and trend for the selected attributes.



1. Click  button to display the date and time.
2. Select the date and time range in the time picker to display the events.
3. The selected date and time act as reference and you can view data before or after the reference period. You can select the range of data to be displayed as shown in the figure below.



4. Click backward button to view the hours before the time selected and forward button for the time ahead. The default time range of 3 hours displayed in the Date range picker.

Chronological View of Date and Time

Chrono view shows the event history of the selected event & data\time. By default, the chrono view shows complete trend data for the selected data\time range. By moving the chrono view marker, you can Zoom-in & Zoom-out of the trend data for the required date\time range.

Updating Default Trend Settings from System Settings

Follow the below steps to update the default time of event trend:

1. Open the Application Configuration page (**Gear icon > System > System Configuration> Application**).
2. Click **Filter** icon in the top right corner of the page. A dialogue box appears.
3. Select the check box beside **Key**, and select Contains from the drop-down list, enter Value (for example, Trend, etc.) in the box.
4. Select the check box beside Owner, and select Equals from the drop-down list, enter Value (for example UAS etc.) in the box.
5. Click **Ok**.

Update the value of keys in the below table:

Key	Description
Honeywell.UAS.Events.Trends.HistorianDataType	To change the event trend historian data type setting value from Interpolative to Raw . Double-click on the value pane and enter either Interpolative or Raw.
Honeywell.UAS.Events.Trends.HistorianDataType	To change the event trends maxpoints. Double-click on the value pane to enter the number of maxpoints.
Honeywell.UAS.Events.DefaultDateRangeBeforeEventTimeSeconds	The event trend of a time range before the event start time is displayed. By default the trend before 2 hours of event start time is displayed.
Honeywell.UAS.Events.DefaultDateRangeAfterEventTimeSeconds	The event trend of a time range after the event start time is displayed. By default the trend after 3 hours of event start time is displayed.

Key	Description
	NOTE Initially, the trend for time range co nfigured at event level is displayed and if event level is not configured then the trend for time range at pr oject level (Time range configured in the system configuration) is disp layed. In the absence of both event and p roject level, the default time range trend is displayed

NOTE
For information on configuring default time using Bulk Model Configuration, refer [Event Details View](#) of Asset Performance Management Configuration Guide.

Comparison of the Attribute Values Across Different Time Slices

You can compare the attributes values by selecting data points of required time and date from the graph. The selected data points are displayed in the attributes column with different legends. The maximum data points that can be selected from the graph for the required attributes are five.



6.6.5 Trend for the Events

Stack Trend



Stack trend refers to the single graph displayed for each attributes selected. Each attributes that are selected from the attribute grid has an independent graph displayed in the chart.

NOTE
You can have a maximum of ten stack plots and each stack plot can have a maximum of five attributes.

TIP
Use the scroll bar to switch between different stack plots.

Single Trend



Single trend shows different graph in single row for the selected attributes. Drag and drop the attributes to the chart to merge the attributes and view a consolidated graph of the different attributes selected.

The legends that appear on the left corner of the chronological view show the type of event that is triggered.

Trend Settings (Update)

- **Target number of data points:** Configure the number of the data points for the trend display, By default 200 is selected and you can also enter the desired number of data points from the range between 100 to 2000 inclusive.
- **Decimal Precision:** The precision of data points on the trend.
- **Good quality points:** You can configure to show only Good quality points in the trend. By default, it shows all the data points.
- **Aggregation Type:** You can configure to show either Raw data or Interpolate data type for the trend. By default, interpolate data type is selected for the trend.
- **Enable Datamarkers:** To display the data points without connected lines.
- **Digital State Data:** Multi-state digital data enables the plotting of digital state enumeration numbers as stepped interpolation and displays associated text in event trend.

For information on how to configure multi-state digital data, refer to [Multi-state Digital Data](#).

NOTE

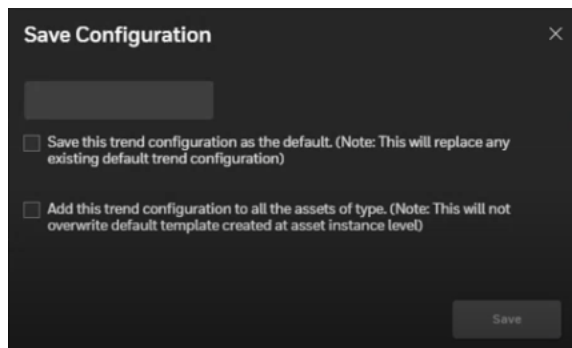
Digital State Data is enabled by default, if an attribute from the configured array data is selected.

6.6.6 Save Trend Configuration and Templatize

You can now save the Event trend configuration and also templatize, the templates are applicable to the same events from the asset belonging to the same asset type. The template saves a lot of time by reusing without configuring the trend setting again. You can also set one of the required configurations as default configuration. The default configuration is displayed every time when you launch the event trend page.

To save the Trend configuration as template:

1. Configure the Trend setting that you want to save.
2. Click **Save** button.
The **Save Configuration** pop-up appears.
3. Enter the configuration name in the **Save Configuration** box.

**NOTE**

Maximum of 50 characters including special characters are allowed in the template name.

4. Click the first checkbox, if you want to set the trend configuration as default.
Click the second checkbox, if you want to create a template for a particular event that will be applicable to all assets belonging to the asset type.
5. Click **Save**.

Below are the trend settings that can be saved and templated

- Date/time configuration.
- Number of data points.
- Historical data type.
- Enriched attribute list (Stack Trend and Single trend).

NOTE

Maximum of 50 characters including special characters are allowed in the template name.

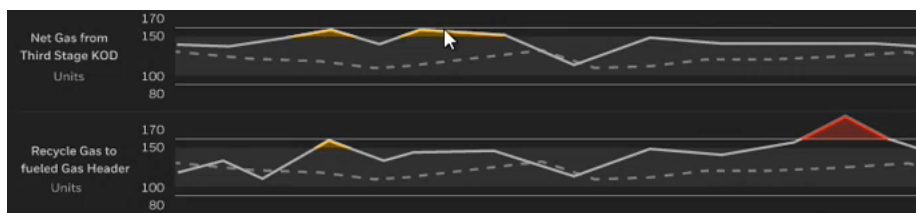
The saved templates enable you to edit the configuration and save the template back.

6.6.7 Threshold Limit View

This display shows you the normal operating range, threshold high-low limit and the high-low limit of an asset.

Below are the list of shaded area that is displayed to you:

1. **Yellow Colour:** This shaded region shows the equipment has crossed the normal operating condition and acts as a warning.
This region is between the normal operating range to the threshold high-low limit and high-low limit.
2. **Red Colour:** This shaded region shows that equipment has been operating outside the limits and needs immediate attention.
3. **Grey Colour:** The grey region is the normal operating limit of equipment.



NOTE

This feature will be disabled if a single plot is having multiple attributes. To enable this view you need to select any one attribute.

The normal, threshold and critical value should be configured for each attribute to have this feature displayed.

6.6.8 Plot Fault Overlay

Plot fault overlay highlights the time interval when a fault or associated fault symptoms is triggered for an equipment.

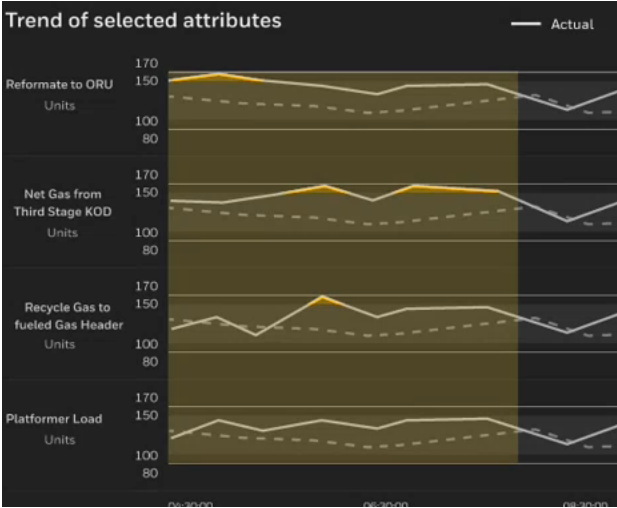
To view the **Plot Fault Overlay** follow the steps given below:

- 1. Open **Event Monitor**.
- 2. Right click on an event and select **Event Trend**.
- 3. Choose a time range on top right corner.
- 4. Click on the event or any of the associated symptoms from **Event Trends**.
- 5. Enable event marker toggle button “Show event markers”



Events Trends		Associated Attributes	
EVENT NAME	EVENT START TIME	EVENT END TIME	TYPE
Feed pump efficiency loss	10:30 AM 20 Apr 2021	Ongoing	Fault
Symptom of the fault	10:30 AM 20 Apr 2021	11:20 PM 20 Apr 2021	Symptom
Symptom of the fault	10:30 AM 20 Apr 2021	12:40 AM 21 Apr 2021	Symptom

- 6. The plot overlay showing when the event have been triggered is displayed.



6.6.9 Trending of Contributing Variables of Faults and Symptoms

1. Navigate to System configurations.
2. Search for setting “Honeywell.UAS.Events.IncludeVariablesToAssociatedData” and set value to "True".
This will enable attribute used in expression to be part of associated data

NOTE

By default the value is set to false.

3. Navigate to attribute tab and all the attributes with associated data will be listed.

NOTE

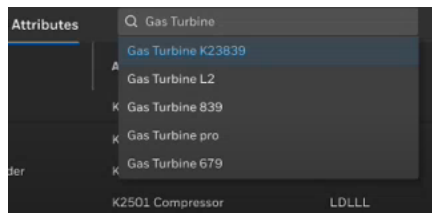
For any event which has both expression and event enrichment, this flag needs to be enabled to see the complete list of attributes used in both event expression and event enrichment as part of associated data.

6.6.10 Trend Attributes from Other Assets

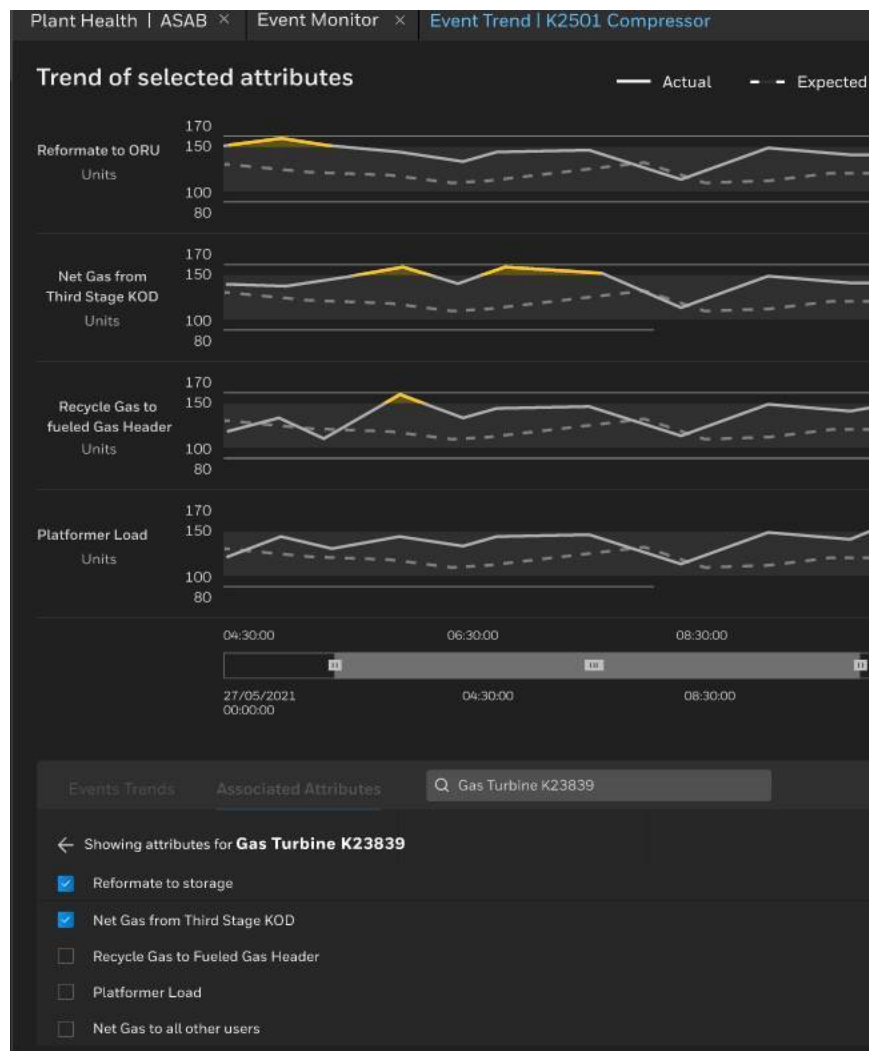
When an event is triggered for an asset the actual root cause of this fault can be caused by another asset. So to further validate the actual cause, you can pull in the attributes of different assets into the main asset.

Follow the steps given below to pull in attributes of different assets:

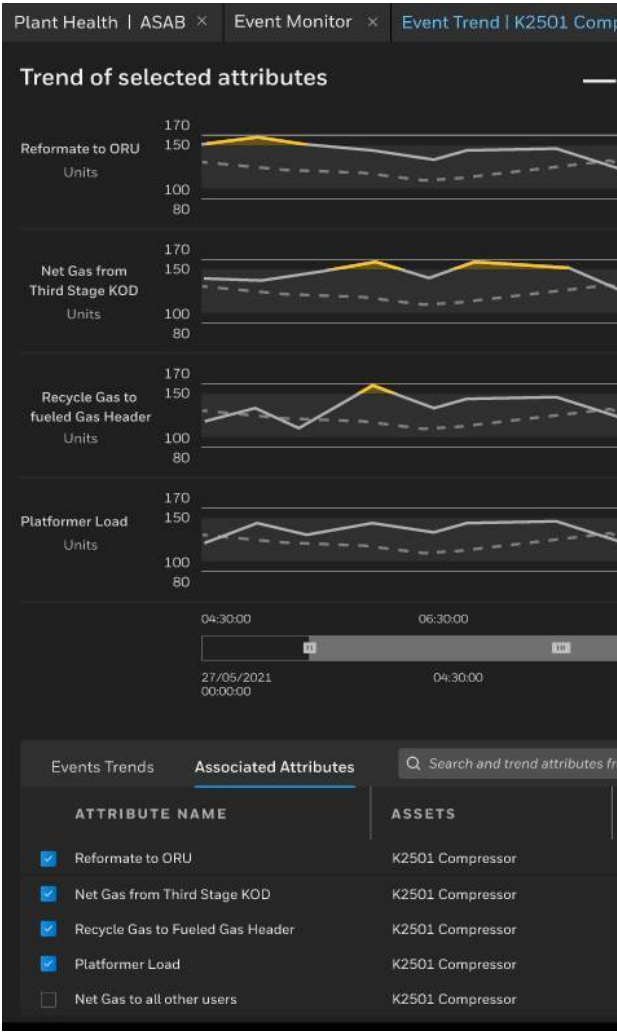
1. Right click an event from **Event Monitor** and select **Event Trend**.
2. Click **Attributes** tab appearing on the bottom of the trend.
3. Search for any asset part of asset hierarchy.



4. List of attributes that can be added to main asset appear.



5. Click the check box of attributes that needs to be added.
6. Click back arrow  icon.
7. Attribute of the searched asset is added to the main asset.

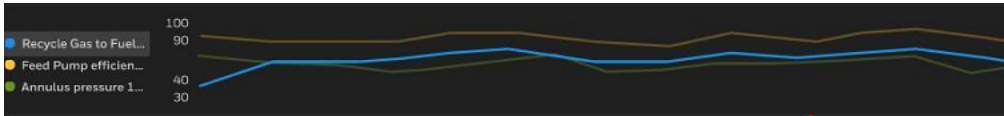


8. You can view the newly added attributes in the trend by clicking there respective check box.

6.6.11 Multiple Attributes on Trend

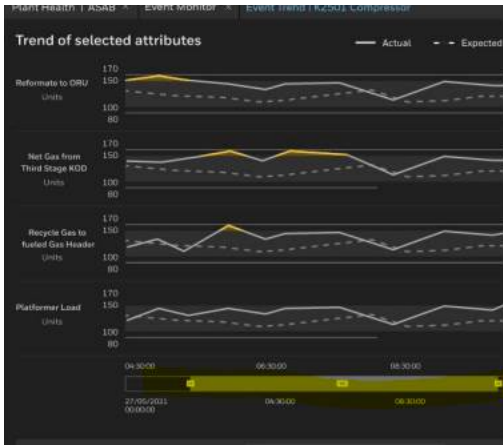
You can add multiple attributes to same plot Y1 axis by drag and drop. Y1 axis scale will be based on min and max value of all associated attributes . You can select the required attributes and accordingly axis will get reset and plots will be updated with associated limits as well

Depending on the attribute selected all the other plots associated to it will be displayed.



6.6.12 Zoom In and Zoom Out

You can select the area to zoom by left clicking on the initial and end point inside an event trend. The zoom can also be adjusted by moving the chrono view bar.



6.6.13 Multi-state Digital Data

Multi-state digital data enables the plotting of digital state enumeration numbers as stepped interpolation and show associated text in event trend.

Follow the below steps to configure digital state data:

1. Select an asset from the Asset Performance Management Landing page.
2. Click **Attribute Overview** in **Dashboard** drop-down.
3. Find **DigitalStateData** under the **ATTRIBUTE** column and double-click **[Array Data]**.

Dialog box to enter the digital state data appears.

P5/P12 DUST COLLECTOR : DIGITALSTATEDATA

VALUE	X30_MINUTE_DOWN_A...	AIR_PRESSURE_ALAR...	AIRLOCK_RUN_STA...	FAN_RUN_ST...	DELETE
35.000	3rd Reverse	Flowing			X
36.000	4th Reverse	Not Flowing			X
40.000	5th Reverse				X
15.000		Inactive			X
0.0000				Off	X

SAVE CANCEL

NOTE

If the digital state attribute is not available in the selected asset, create column header for the tabular array digital state attribute using an XML format as shown in the below example:

```
<?xml version="1.0" encoding="utf-16"?><TabularData
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xmlns:xsd="http://www.w3.org/2001/XMLSchema"><Columns><TabularDataColumn><Name>Value</Name><DataType>Integer</DataType><Values><string>1</string><string>2</string></Values></TabularDataColumn><TabularDataColumn><Name>PowerFactor
```

```
</Name><DataType>String</DataType><Values><string>ON</string><string>OFF</string></Values></TabularDataColumn><TabularDataColumn><Name>VapourPressure</Name><DataType>String</DataType><Values><string /><string /></Values></TabularDataColumn></Columns></TabularData>
```

4. Click add new  icon to add a digital state data input and click delete  icon to remove a digital state data.
5. Click **Save**.
6. The saved digital state data is displayed in event trend when **Digital State Data** is enabled in trend settings.

NOTE

Digital State Data is enabled by default, if an attribute from the configured array data is selected.

The following tools can be used to view the monitor your system status:

- Calculation Status
- Data Source Status

7.1 Calculation Status

The Calculation Status Viewer provides an overview of all configured calculations, across all assets.

To Open the Calculation Status Viewer

1. Click **Calculation Status** under **Detail Views** of Navigation pane.

Calculation Status | 0740C201A Sour Gas Compressor

Filter Criteria: --Select a Filter/ Clear applied filters--

Clear Filters

☒ Asset Type Level UOM

CALCULATION	CALCULATION INSTANCE	CATEGORY	MODE	LOCATION	STATE	LAST RUN STAT...	LAST RUN TIMESTAMP
AM_CentrComprVariableSpeed_V2	AM_CentrComprVariableSpeed_V2 (Typed)	Advanced	Both	Plant.Area02.0740C201A	Enabled	Executed (0.7 s)	8/11/2023 8:41:01 PM
CCV2_VS_Change_of_Gas_Properties_FM	CCV2_VS_Change_of_Gas_Properties_FM (Typed)	Fault	Data Driven	Plant.Area02.0740C201A	Enabled	Executed (<0.1 s)	8/11/2023 8:41:02 PM
CCV2_VS_Compressor_Inlet_Supply_Ano...	CCV2_VS_Compressor_Inlet_Supply_Ano... (Typed)	Fault	Data Driven	Plant.Area02.0740C201A	Enabled	Executed (<0.1 s)	8/11/2023 8:40:56 PM
Centrifugal_Compressor_FM1	Centrifugal_Compressor_FM1 (Typed)	Fault	Data Driven	Plant.Area02.0740C201A	Enabled	Executed (<0.1 s)	8/11/2023 8:41:02 PM
CC_PerformanceFault_Limit_CM	CC_PerformanceFault_Limit_CM (Typed)	Basic	Data Driven	Plant.Area02.0740C201A	Disabled	Not Started	
CCV2_VS_Anti_Surge_Valve_Opening_Ano...	CCV2_VS_Anti_Surge_Valve_Opening_Ano... (Typed)	Fault	Data Driven	Plant.Area02.0740C201A	Disabled	Not Started	

Last Updated Time: 8/11/2023 8:41:24 PM

11 record(s)

Expression:

Inputs/Outputs:

To View the Calculation Details

1. From the calculation status viewer, choose the calculation.
The calculation expression and input/output posts appear in the bottom viewing pane of the Status Viewer.

Filter Criteria: --Select a Filter/ Clear applied filters-- ☒ Asset Type Level UOM

CALCULATION	CALCULATION INSTANCE	CATEGORY	MODE	LOCATION	STATE	LAST RUN STAT...	LAST RUN TIMESTAMP
AM_CentrComprVariableSpeed_V2	AM_CentrComprVariableSpeed_V2 (Typed)	Advanced	Both	Plant.Area02.0740C201A	Enabled	Executed (0.7 s)	8/11/2023 8:41:01 PM
CCV2_VS_Change_of_Gas_Properties_FM	CCV2_VS_Change_of_Gas_Properties_FM (Typed)	Fault	Data Driven	Plant.Area02.0740C201A	Enabled	Executed (<0.1 s)	8/11/2023 8:41:02 PM
CCV2_VS_Compressor_Inlet_Supply_Ano...	CCV2_VS_Compressor_Inlet_Supply_Ano... (Typed)	Fault	Data Driven	Plant.Area02.0740C201A	Enabled	Executed (<0.1 s)	8/11/2023 8:40:56 PM
Centrifugal_Compressor_FM1	Centrifugal_Compressor_FM1 (Typed)	Fault	Data Driven	Plant.Area02.0740C201A	Enabled	Executed (<0.1 s)	8/11/2023 8:41:02 PM
CC_PerformanceFault_Limit_CM	CC_PerformanceFault_Limit_CM (Typed)	Basic	Data Driven	Plant.Area02.0740C201A	Disabled	Not Started	
CCV2_VS_Anti_Surge_Valve_Opening_Ano...	CCV2_VS_Anti_Surge_Valve_Opening_Ano... (Typed)	Fault	Data Driven	Plant.Area02.0740C201A	Disabled	Not Started	

Last Updated Time: 8/11/2023 8:41:24 PM 11 record(s)

CCV2_VS_Change_of_Gas_Properties_FM

Calculation Result: Calculation executed successfully in 0.003 seconds.

Expression:

```

DateTime temp
If(!isnull(DeviationPolyEfficiency) || isnull(DeviationPolyEfficiency) ||
isbad(DeviationPolyEfficiency) || isnull(DeviationPolyEfficiency) ||
isnull(DeviationPolyEfficiency.HighLimit) ||
isnull(DeviationPolyEfficiency.HighLimit) ||
isbad(DeviationPolyEfficiency.HighLimit))
then
DeviationPolyEfficiency_CGP_HIGH = (DeviationPolyEfficiency) >
(DeviationPolyEfficiency.HighLimit)
else
DeviationPolyEfficiency_CGP_HIGH = 0
endif
temp = NOW
DeviationPolyEfficiency_CGP_HIGH.Time = temp
if(!RunningStatus>0.0) then

```

Inputs/Outputs:

NAME (MAPPING)	MAPPED-TO...	I/O	TIMESTAMP	QUALITY	VALUE	UNITS
CC_Compressor_Inlet_Sup...	CC_Compressor_Inlet_Sup...	Input	8/11/2023 8:33:52 PM	Good	0	
Delta_Power	Delta_Power	Input	8/11/2023 8:40:50 PM	Bad	NaN	kW
Delta_Power_CGP_HIGH	Delta_Power_CGP_HIGH	Input	8/11/2023 8:40:50 PM	Good	0	
Delta_Power_CGP_LOW	Delta_Power_CGP_LOW	Input	8/11/2023 8:40:50 PM	Good	0	
Delta_Power_HighLimit	Delta_Power_HighLimit	Input	8/11/2023 8:40:50 PM	Bad	NaN	
Delta_Power_LowLimit	Delta_Power_LowLimit	Input	8/11/2023 8:40:50 PM	Bad	NaN	
DeviationPolyEfficiency	Deviation PolyEfficiency	Input	8/11/2023 8:40:50 PM	Good	2.0933398489631827	%

7.1.1 Status Viewer Interface

Column	Description
Name	The full name of the instance calculation.
Type	The full name of the type calculation.
Category	The category of calculation. For example, Fault Model, Calculation Model, Mechanical Model, or Advanced Model.
Mode	The calculation mode: data driven or scheduled.
Location	The location of the calculation in the asset model.
State	The current calculation state: Enabled or Disabled.
Last Run Status	The status of the most recent run. If the calculation is current running, the status reads Executing . If the last run did not succeed, the status reads Failed .
Last Run Timestamp	The time stamp of the last time the calculation ran.

7.1.2 Filtering Calculations

To filter the Calculation Status Viewer

- In the header on the column on which you want to apply the filter, click the Filter icon.
The filter options appear.
- Do all of the following:
 - From the conditions list, select one of the following filter conditions: **Is equal to**, **Is not equal to**, **Starts with**, **Contains**, **Does not contain**, **Ends with**.
 - In the text box, type the value for the filter.
- To add an additional filter condition for the column, do all of the following:

- Select the grouping condition: **And** or **Or**.
- From the second conditions list, select one of the following filter conditions: **Is equal to**, **Is not equal to**, **Starts with**, **Contains**, **Does not contain**, **Ends with**.
- In the second text box, type the value for the filter.

4. Click **Filter**.

To save the current filter settings

- Once all the filters have been added, click the New Filter button.
The **Save Applied Filters** dialog box appears.
- The **Filter Criteria Name** box, type the name for the filter criteria.
- Click **Save**.

To clear the current filters

- Click **Clear Filters**.

7.1.3 Asset Type Level UOM in Calculation Status

The **Asset Type Level UOM** check box enables you to display the selected calculation values and units in instance and type level units of measurement.

Calculation Status 0740C201A Sour Gas Compressor ⓘ			
Filter Criteria: --Select a Filter/ Clear applied filters--		Clear Filters	☒ Asset Type Level UOM
CALCULATION	CALCULATION INSTANCE	CATEGORY	MODE
AM_CentrComprVariableSpeed_V2	AM_CentrComprVariableSpeed_V2 (Typed)	Advanced	Both
CCV2_VS_Change_of_Gas_Properties_FM	CCV2_VS_Change_of_Gas_Properties_FM (Typed)	Fault	Data Driven
CCV2_VS_Compressor_Inlet_Supply_Ano...	CCV2_VS_Compressor_Inlet_Supply_Ano... (Typed)	Fault	Data Driven
Centrifugal_Compressor_FM1	Centrifugal_Compressor_FM1 (Typed)	Fault	Data Driven
CC_PerformanceFault_Limit_CM	CC_PerformanceFault_Limit_CM (Typed)	Basic	Data Driven
CCV2_VS_Anti_Surge_Valve_Opening_Ano...	CCV2_VS_Anti_Surge_Valve_Opening_Ano... (Typed)	Fault	Data Driven
Last Updated Time: 8/11/2023 8:41:24 PM			

Follow the below steps to switch between type and instance level UOM:

- Select a calculation from the **Calculation Status** viewer.
The calculation expression and input/output values for the selected calculation are displayed in the bottom pane.
- By default **Asset Type Level UOM** check box is selected and the values and units are displayed at the type level.
- Click the check box to clear the Asset Type Level UOM selection.
The values and units of the selected calculation are displayed at the instance level.

NOTE

The Asset Type Level UOM check box is reverted back to default state when you refresh the page or select a different asset.

7.1.4 Failed Calculations

If there is a problem with a calculation run, the **Last Run State** reads “Failed” and an error message appears in the calculation details.

NAME	TYPE	CATEGORY	MODE	LOCATION
601-E-19-HEX_Alarms-1334	HEX_Alarms	Fault Model	Data Driven	CDU.601-HEX.6019
601-E-18-TS_ShellTubeHeatExchanger-1324	TS_ShellTubeHeatExchanger	Embedded Model	Data Driven	MallibuRefinery.CDU.601-HEX.6018
Reboiler-135	TS_ShellTubeHeatExchanger	Embedded Model	Data Driven	MallibuRefinery.CDU.601-HEX.6015

Last Updated Time: 4/19/2018 10:58:24 AM

601-E-18-TS_ShellTubeHeatExchanger-1324

Exception Message: The required input data like Tube side fluid heat transfer coefficient, is not configured

Expression:

Input/Output Ports:

NAME	MAPPED ATTRIBUTE	I/O
EffectiveSurface...		Input
RatedShellFouli...		Input
RatedShellMass...		Input
RatedTubeFouli...		Input
RatedTubeMass...		Input
ShellFluidDens...		Input

You can use this error message to assist in debugging the calculation. For information on editing the expression, see “Configuring Calculations” in the *Asset Performance ManagementConfiguration Guide*.

7.2 Data Source Status

To view the status of the APM Services and Data Sources

1. Open Asset Performance Management, and on the Contents tree choose **System Console**.
The **System Console** page appears.
2. Scroll to the **Data Source Status** section.
The various data sources to which Asset Performance Management connects are displayed, along with the name of the server on which the service exists. The status of the service and data source connection are explained below:

a. **Running** - is indicated when the service is running and is connected to one of the servers that is configured in Asset Performance Management.

b. **Not Running** - is indicated when the service is not running and is not connected to the server.

c. **Marginal** - is indicated when the service is running but not connected to the server.
3. To update the data source status display, click **Refresh**.

You can view asset information in the following views:

- Asset Details
- Asset Attributes
- Trends
- Asset Links and Notifications

8.1 Asset Views

8.1.1 Asset Details

The asset home page provides a quick overview of the asset, including description, current state, metrics, comments, and time since the metrics were last reset.

8.1.2 Trends

The trends give you the ability to plot the asset attribute values on an x-y trend or a performance curve.

8.2 Viewing Asset Details

The asset details page lists the following information for the selected asset:

- Current Status: Monitored, Unmonitored, Out of Service, or Unlicensed.
- Current State/Date Since: The state of the asset (for example, Running) and the date/ time the asset changed to that state.
- Running hours since most recent reset: The number of hours the asset had the status Running since the last metrics reset.

NOTE

If the AssetRunning Status is set to zero and asset is not running, the output values are displayed as NAN along with timestamp. For user defined calculations, user can update the condition in the calculation expression based on Running Status.

- Cumulative running hours (total): The total number of hours the asset has had the Running status since the asset was commissioned.
- Description: Configured asset details, such as the Asset Type and Description. By default, this section shows design information from the manufacturer. See [To display an attribute on the asset details page](#) for instructions on adding attributes to the Description box.
- Metrics: All key asset metrics configured for the assets, including their current value and units. If the key metric attribute is trendable and is writing to a historical datasource, a sparkline is shown in the Last 7 Days column. See [To](#)

[display an attribute on the asset details page](#) for instructions on adding attributes to the Metrics box.

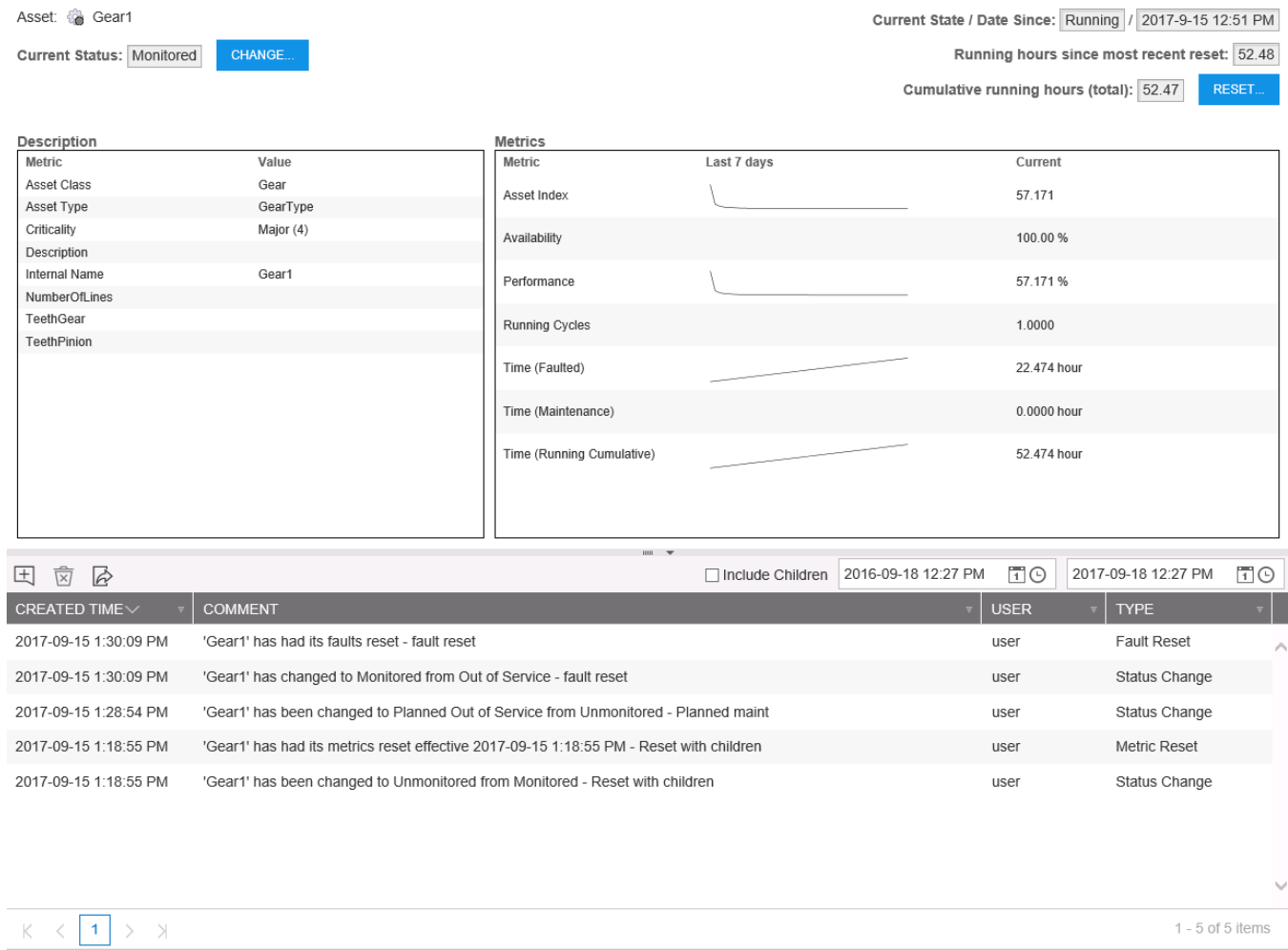
- Comments: All comments added to the asset. See “Asset Comments” on page 63 for more comment settings.

To open the asset details page

1. Do one of the following:
 - a. On the asset tree, right-click an asset, and on the context menu, click **Asset Details**.
 - b. Do all of the following:
 - a. On the asset tree, select an asset to set the context.
 - b. On the Contents tree, under **Detail Views**, select **Asset Details**.

The Asset Details page for the selected asset opens.

The page refreshes every minute to show updated values for the attributes and metrics displayed.



To open a trend from the key metrics

In the metrics area, in the Last 7 Days column, click the sparkline of the metric for which you want to view the trend.

8.2.1 Adding Asset Details and Key Metrics

Users with permissions to edit asset attributes can add them to the asset details page by including the required filter categories. (See Configure Attributes for Asset Types in the *Asset Performance Management Configuration Guide* for information on configuring other attribute settings, such as saving data and configuring trendable attributes.)

To display an attribute on the asset details page

1. On the Contents tree, click **Configuration > Templates**.
The **Asset Configuration** page appears.
2. Select the **Attribute Configuration** tab.
3. In the tree view, expand the asset class you created and then click the asset type.
The **Attribute Configuration** pane is displayed.
4. In the **Add attribute to selected asset type** group box, select the attribute in the Attribute Configuration pane, and then click the Edit  button.
5. In the **Filter Category** box, do the following:
 - a. To have the asset attribute appear in the **Description** area of the asset details page, add the filter category **Design**.
 - b. To have the asset attribute appear in the **Metrics** area of the asset details page, add the filter category **Key Metrics**.
6. Click **Save** to save the attribute.
7. Refresh the Asset Details page to see the updated attributes.

8.2.2 Changing Asset Status

To change the asset status

1. On the Asset Details page, by the Current Status, click **Change**.
The Change Asset Status dialog box appears.

New Status

☒ Unmonitored ☐ Out of Service

Comment

☐ Reset Metrics

☐ Reset Faults

☐ Include Children in the Reset(s)

2. For **New Status**, select the status to which you are changing the asset. The following statuses are possible (the current status is not listed as an option):
 - a. Monitored
 - b. Unmonitored
 - c. Out of Service
3. In the **Comment** box, type the reason for changing the status.
4. To reset the asset metrics for the asset at the same time, select **Reset Metrics**.

NOTE

By default, some metrics are automatically reset when an asset status is changed from Out of Service. In this case, selecting Reset Metrics also resets metrics that are not included in that default reset.

5. To also reset all faults for the asset at the same time, select **Reset Faults**.
6. To reset the asset metrics for all child assets of the current asset at the same time, select **Include Children in the Reset(s)**.
7. Click **OK**.

8.2.3 Asset Comments

To Add a User Comment

1. In the comment pane, click the Add Comment  button.

Add Comment

Comment



OK

Cancel

2. In the **Add Comment** dialog box, in the **Comment** box enter the comment text.
3. Click **OK**.

To Delete a User Comment

1. In the comment pane, select the user comment, and then click the Delete  button.

NOTE

Only manually added user comments can be deleted. System messages (such as an asset status change) cannot be deleted.

To Export Comments to Excel

1. Click the Export  button.
2. When prompted, save the Excel file.

To Show or Hide Comments on Child Assets

1. To show the comments from child assets, in the comments pane, select **Include Children**.
The comments list updates to show the asset column and the comments for all child assets.
2. To hide the comments from child assets, in the comments pane, clear **Include Children**.
The comments list updates to hide the asset column and the comments for all child assets.

To Change the Comments Time Range

1. In the comments pane, in the start and/or end time selectors, type or select a new date and time.
2. Click outside the time selector or press enter to update the comments list.
The comments list is filtered to comments created during the new time range.

To Page the Comments List

1. To view the next page of items, click the Page Forward  button.
2. To view the previous page of items, click the Page Backwards  button.
3. To jump to the last page, click the Last Page  button.
4. To jump to the first page, click the First Page  button.
5. To view a specific page, in the Page box, type the page number, and then press Enter.

8.2.4 Reseting Asset Metrics

From the asset details page, you can reset the saved metrics for the current asset, including reseting the metrics for all child assets if selected. You can also choose to reset the faults for the asset at the same time.

The metrics reset is configured in the metrics calculation (for example, the Equipment Metrics Template). By default, all metrics are reset.

To Reset the Asset Metrics

1. On the Asset Details page, click **Reset**.
2. To reset the asset metrics starting at a time other than the current time, do all of the following:
 - a. Click the Calendar icon, and then select the day.
 - b. Click the Clock icon, and then select the time.
3. In the **Comment** box, type the reason for resetting the asset metrics.
4. To reset the asset metrics for all child assets of the current asset, select **Include Children in the Reset(s)**.
5. To also reset all faults for the asset, select **Reset Faults**.
6. Click **OK**.

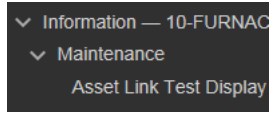
The metrics are reset effective at the selected time. For example, if they were reset for the current time, they begin calculating from the time of the reset. If the selected time was set to a week ago, then they are recalculated from a week earlier.

8.3 Asset Links and Notifications

Some additional tools for asset analysis can be configured on your system as required.

8.3.1 Asset Links

In addition to the default Contents tree listings, your system can also have custom links configured for individual assets or asset types.



These asset links contain more information about the asset, which can be presented in a variety of formats. Some options for asset link content includes:

- Graphics, including schematics.
- Custom Workspaces, which can include a single layout for asset-related graphics, trends, and additional features, including available widgets from Uniformance Insight.
- PDF Documents with additional information.

8.3.2 Notifications

When configuring the system, e-mail notifications can be setup for users. These e-mails will contain template information related to the asset, as required.

In addition, if your system is configured to send notifications to a SharePoint list, it is possible to configure a SharePoint or Ninetex workflow using the notifications as a trigger.

VIEWING REPORTS

This chapter explains the procedure to view reports.

9.1 Reports Overview

Reports are generated and can be viewed in Report Manager tool. Report Manager is a web based tool used to manage all the report activities, which provides a secure and centralized location to produce and manage timely reports. In addition, you can use your templates to integrate reports displayed by Report Manager using your organization's individual style. Report Manager offers a complete solution for report management.

To access the Reports view, your user account must belong to a group that has been granted access permissions or access must be granted to the user account. (We recommend adding user groups during post-installation. See “Add group or user access to view SQL Reports” in the *Installation Guide*.)

Features of Report Manager include:

- Provides a centralized report distribution system for reports.
- Enables information producers to submit and publish standard set of reports to a central repository.
- Provides the ability to export the report data to Excel, CSV and PDF formats.
- Permits report availability in a future timeframe.
- In addition, to viewing spreadsheets or PDF files for further analysis, it also allows you to sort retrieved data.
- Filters that limits the scope of data being displayed in the reports are provided. You can also set default values for the filters.
- Provides an ability to bind data with standard charts like pie chart, bar chart, and trends.
- Enables the publishing of reports that have been submitted as part of request sets or report sets.
- Enables you to upload reports with respect to your permission limits.

There are three types of standard reports that you can access through the AM Reports home page:

- Assets: reports that provide a list of assets and its properties.
- Faults: contains reports related to faults.
- KPI Reports: reports that provide accurate data information and feedback on certain areas.

There are two roles available to access AMReports.

- Content Manager: is the administrator who has permission to perform all the tasks.
- Browser: is an AMUser who can only view the information.

ATTENTION

The data displayed in reports is only related to the collected Asset Performance Management data.

9.1.1 Accessing the Asset Performance Management Reports

To Access the Asset Performance Management Reports

1. The Contents tree, click Reports.
2. If required, click Open this content in a new window.
3. Click AMReports.
4. Browse the following report sections:
 - a. Assets (see [Asset Reports](#)).
 - b. Faults ([Fault Reports](#)).
 - c. KPI Reports ([Key Performance Indicator](#) (KPI)).

Toolbar options available for reports

1. First Page: To view the first page of the document, click  icon.
2. Last Page: To view the last page of the document, click  icon.
3. Previous page: Click the Previous page icon  to view contents of the previous page
4. Next Page: Click the Previous page icon  to view contents of the next page
5. Click the 100% drop-down list and select the percentage of zoom required.
6. Find: In the box provided next to Find, type the word which you want to search in the document and click **Find**. Click **Next** to search the word in different pages.
7. Export: Click the **Export** button and select the required option to export reports.
8. Refresh: Click the Refresh icon  to refresh the report.
9. Print: Click the Print icon  to print the report.

NOTE

To view the properties of the report, click the Properties tab.

9.2 Asset Reports

Assets are components or subsystems present in a plant.

Asset reports categorizes assets based on priority number and criticality of the fault in a machine/component. This displays the mode of the asset. Asset modes like monitored or unmonitored or out of service can be viewed.

Assets reports page displays a list of assets and its properties. The following reports are available in the **Assets Report** page.

- AssetsStatus
- UrgentPriorityAssets
- MonitoredAssets

- Normal Priority Assets
- OutofService
- UnMonitoredAssets
- ImportantPriorityAssets

Here, we have considered the asset 'Controller' and explained all the asset reports based on the Controller. The child report of the controller is displayed when you select the option 'True' for IncludeChildItems. The report of the controller displays the details of priority number defined, mode (monitored or unmonitored or outofservice), and the performance number.

9.2.1 Asset Status Report

Asset Status reports are generated to provide information about the asset's failure condition during operation and maintenance.

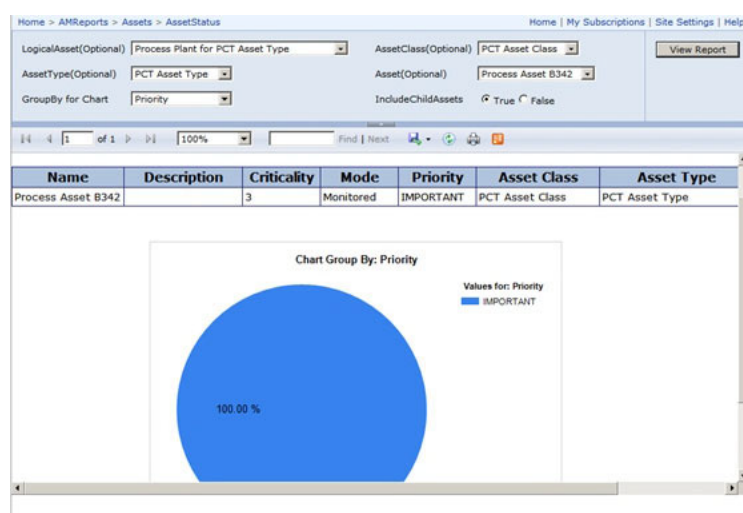
Viewing the Asset Status Reports

You have provided all possible options to view a report. These reports are published to a report server and are viewed in an HTML viewer, which includes a report toolbar and various tool bar options. The report toolbar also appears in a browser window when you access reports through a URL.

To view assets reports

1. On the AM Reports page, click Assets.
2. From the list of reports, select a report you want to view. For example **Asset Status** report.

The **Asset Status** page appears.



3. Provide the following parameters to view the report.
 - a. From the **LogicalAsset (Optional)** drop-down list and select the asset.
 - b. From the **AssetClass (Optional)** drop-down list and select the asset class.
 - c. From the **AssetType (Optional)** drop-down list and select the asset type.
 - d. From the **Asset (Optional)** drop-down list and select the asset.
 - e. From the **GroupBy for chart** drop-down list and select the required option.
 - f. Select the required option for IncludeChildAssets. By default, "False" is selected.

True: Select True to include the child assets.

False: Select False to exclude the child assets.
4. Click the **View Report** button.

The report is generated in the HTML viewer.

9.2.2 Important Priority Assets

ImportantPriority Assets reports are generated for providing information about the assets, which are critical and prone to faults in the operation, and whose priority value ranges between 11 to 20.

Home > AMReports > Assets > ImportantPriorityAssets

LogicalAsset(Optional) (Null) AssetClass(Optional) (Null) View Report

AssetType(Optional) (Null) Asset(Optional) AEDA Check

IncludeChildAssets ☐ True ☒ False

1 of 1 100% Find | Next

ImportantPriority Report

Name	Description	Criticality	Mode	Priority	Asset Class	Asset Type
AEDA Check	AEDA Check	4	Monitored	IMPORTANT	Engine	Diesel Engine (piston)

Done

9.2.3 Monitored Assets

Monitored Assets reports are reports generated when the asset is monitored in the operation..

Home > AMReports > Assets > MonitoredAssets

LogicalAsset(Optional) Process Plant for PCT Asset Type AssetClass(Optional) (Null) View Report

AssetType(Optional) (Null) Asset(Optional) Process Asset B342

IncludeChildAssets (Optional) ☐ True ☒ False

1 of 1 100% Find | Next

Monitored Assets

Name	Description	Criticality	Mode	Priority	Asset Class	Asset Type
Process Asset B342		3	Monitored	URGENT	PCT Asset Class	PCT Asset Type

9.2.4 Normal Priority Assets

NormalPriority Assets reports are reports generated to provide information about the assets which are in the normal operation condition without the presence of any severe/critical fault. These reports are for assets whose priority value ranges between 1 to 10.

Home > AMRports > Assets > NormalPriorityAssets

LogicalAsset(Optional) (Null) AssetClass(Optional) (Null) View Report

AssetType(Optional) (Null) Asset(Optional) Bearing_ManualData

IncludeChildAssets ☐ True ☒ False

1 of 1 100% Find Next

NormalPriority Report

Name	Description	Criticality	Mode	Priority	Asset Class	Asset Type
Bearing_ManualData	Bearing_ManualData	5	Monitored	NORMAL	Bearing	Brg_Bearing

Done

9.2.5 Out of Service

OutOfService reports are reports generated to provide information about the assets which are damaged and are not in working condition.

Home > AMRports > Assets > OutofService

LogicalAsset(Optional) (Null) AssetClass(Optional) (Null) View Report

AssetType(Optional) (Null) Asset(Optional) Time Interval Fault

IncludeChildAssets ☐ True ☒ False

1 of 1 100% Find Next

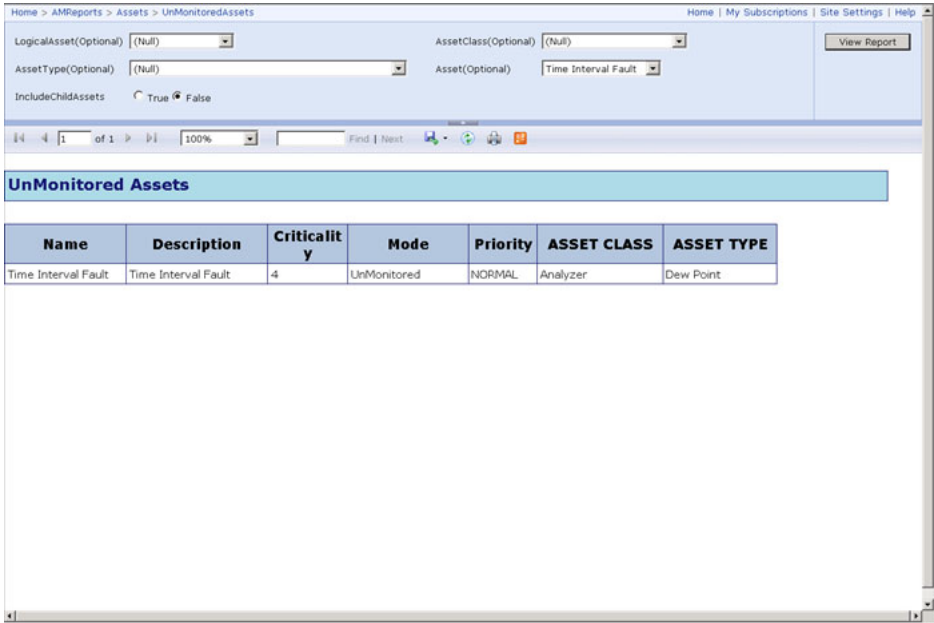
OutOfService Report

Name	Description	Criticality	Mode	Priority	Asset Class	Asset Type
Time Interval Fault	Time Interval Fault	4	OutOfService	NORMAL	Analyzer	Dew Point

Done

9.2.6 UnMonitored Assets

UnMonitored Assets reports are reports generated to provide information about the assets that have stopped their functioning.



9.2.7 Urgent Priority Assets

Urgent Priority Assets reports are reports generated to provide information about assets that require urgent attention. If the value of priority is above 20, then the asset is categorized as “UrgentPriorityAsset” and requires immediate attention.

9.2.8 Viewing KPI Reports

This section explains the procedure to view all KPI Reports.

NOTE

The procedures for viewing AssetFailureRate, BadActors-Fault, MTBF and MTTR are same.

To View a AssetFailureRate Report

1. On the **AMReports** page, click **KPI Reports**.
2. From the list of reports, click **AssetFailureRate**.
The **New Subscription** page is displayed.
3. Enter the following information to view the report.
 - a. From the **LogicalAsset (Optional)** drop-down list and select the required asset.
 - b. From the **AssetClass (Optional)** drop-down list and select the class the asset.
 - c. From the **AssetType (Optional)** drop-down list and select the type of the asset.
 - d. From the **Asset** drop-down list and select the asset.
 - e. Click the calender icon next to **STARTTIME** box and choose the required start time.
 - f. Click the calender icon next to **ENDTIME** box and choose the required end time.
4. Click **View Report**. The report is generated in the HTML viewer.

Asset Failure Rate Report				
Asset Name	Description	Number of Faults	Total hours Running	Failure Rate
Electrical Motor	To drive the Shaft i.e. Act as the prime mover	4	1080	0.0037

ATTENTION

- The procedures for viewing the properties of a KPI reports, modifying their parameters, specifying datasources, specifying execution, specifying report history, specifying security, and creating report subscriptions are same for all KPI reports.
- For more information on the toolbar options refer to, [Toolbar options available for reports](#).

To View a AssetHealthIndex Report

1. On the **AM Reports** page, click **KPI Reports**.
2. From the list of reports, click **AssetHealthIndex**.
The **New Subscription** page is displayed.
3. Enter the following information to view the report.
 - a. From the **LogicalAsset (Optional)** drop-down list and select the required asset.
 - b. From the **AssetClass (Optional)** drop-down list and select the class the asset.
 - c. From the **AssetType (Optional)** drop-down list and select the type of the asset.
 - d. From the **Asset** drop-down list and select the asset.
4. Click **View Report**. The report is generated in the HTML viewer.

ATTENTION

- The procedures for viewing the properties of a KPI reports, modifying their parameters, specifying datasources, specifying execution, specifying report history, specifying security, and creating report subscriptions are same for all KPI reports.
- For more information on the toolbar options refer to, [Toolbar options available for reports](#).

To View a AssetRunningHours Report

1. On the **AM Reports** page, click **KPI Reports**.
2. From the list of reports, click **AssetRunningHours**.
The **New Subscription** page is displayed.
3. Enter the following information to view the report.
 - a. From the **LogicalAsset (Optional)** drop-down list and select the required asset.
 - b. From the **AssetClass (Optional)** drop-down list and select the class the asset.
 - c. From the **AssetType (Optional)** drop-down list and select the type of the asset.
 - d. From the **Asset** drop-down list and select the asset.
 - e. Clear the **NULL** check box next to the **FROMDATE** box. The box is enabled.
 - f. Click the calendar icon next to **FROMDATE** box and choose the required start time.
4. Click **View Report**. The report is generated in the HTML viewer.

Asset Running Hours Report

Asset Name	Description	Total Running Hours
Electrical Motor	To drive the Shaft i.e. Act as the prime mover	1077:24

ATTENTION

- The procedures for viewing the properties of a KPI reports, modifying their parameters, specifying datasources, specifying execution, specifying report history, specifying security, and creating report subscriptions are same for all KPI reports.
- For more information on the toolbar options, refer [Toolbar options available for reports](#).

NOTE

The procedures for viewing AssetRunningHours, AssetIdleTime and AssetDownTime reports are same.

To View a BadActors-Asset Report

1. On the **AM Reports** page, click **KPI Reports**.
2. From the list of reports, click **BadActorsAsset**.
The **New Subscription** page is displayed.
3. Enter the following information to view the report.
 - a. From the **LogicalAsset(Optional)** drop-down list and select the required asset.
 - b. From the **AssetClass (Optional)** drop-down list and select the class the asset.
 - c. From the **AssetType (Optional)** drop-down list and select the type of the asset.
 - d. From the **Asset** drop-down list and select the asset.
 - e. From the calender icon next to **FROMDATE** box and choose the required start time.
 - f. From the calender icon next to **TODATE** box and choose the required end time.
4. Click **View Report**. The report is generated in the HTML viewer.

Bad Asset Type Report	
Asset Name	Number of Faults
Electrical Motor	4

ATTENTION

- The procedures for viewing the properties of a KPI reports, modifying their parameters, specifying datasources, specifying execution, specifying report history, specifying security, and creating report subscriptions are same for all KPI reports.
- For more information on the toolbar options, refer [Toolbar options available for reports](#).

9.3 Fault Reports

Fault Reports are reports related to faults. The following reports are available in the faults page.

- AMFaultReports: reports that display the fault, fault description, status of the fault, event category, start time, and end time for the fault on the selected asset.
- EliminatedFaults: reports that display the details of the fault eliminated faults.
- CloseOutFaults: reports that indicate to the user that the fault has been repaired and closed out.
- ContextHistory: reports that display the details of the variable for a selected asset status like monitored,

unmonitored, out of service, and so on including their time stamp.

- **LatchedFaults:** reports that display the details of the fault which are manually locked (latched) and auto locked (latched).
- **ManuallyInitiatedFaults:** reports that display the details of manually initiated faults, status of the faults, start time, end time and event category.

9.3.1 Viewing Fault reports

You are provided with all the possible options to view a report related to faults. These reports are published to a report server and are viewed in HTML viewer, which includes a report toolbar and other various tool bar options.

ATTENTION

Reports can be viewed by the browser and content manager roles.

To View a Fault Report

1. On the **AM Reports** page, click **Faults**.
2. From the list of reports, select a report you want to view. For example **AMFaultReports**.

The **AM Fault Reports** page is displayed.

Asset Name	Fault Name	Description	Status	Start Time	End Time	Event Category
pmp	Plant1.Unit1.Pmp.TestSym	testin	Active	3/2/2011 7:44:52 PM		SYMPTOM

3. Specify the following parameters required to view the report.
 - a. From the **LogicalAsset (Optional)** drop-down list and select the asset.
 - b. From the **AssetClass (Optional)** drop-down list and select the asset class.
 - c. From the **AssetType (Optional)** drop-down list and select the asset type
 - d. From the **Asset (Optional)** drop-down list and select the asset
 - e. Select the required option for **IncludeChildAssets**. By default, "False" is selected.
True: Select True to include the child assets.
False: Select False to exclude the child assets.
 - f. Click the **STATUS** drop-down list and select the status of the fault.
 - g. Clear the **NULL** check box next to the **STARTTIME** box. The box is enabled.
 - h. Click the calendar icon next to **STARTTIME** box and choose the required start time from when the faults are generated.
 - i. Clear the **NULL** check box next to the **ENDTIME** box. The box is enabled.
 - j. Click the calendar icon next to **ENDTIME** box and choose the required end time till when the faults are generated.

ATTENTION

You can add time along with date. To add time for both StartTime and EndTime fields, select the date and type the time in the format “HH:MM:SS” (24 hour format). By default, the time specified is 12:00:00 AM.

4. Click **View Report**.

The report is generated in the HTML viewer.

ATTENTION

- All associated child reports are also displayed.
- The procedure for viewing the properties of a fault, modifying their parameters, specifying datasources, specifying execution, specifying report history, specifying security, and creating report subscriptions are same.
- For more information on the toolbar options, refer “Toolbar options available for reports” on page 81.

9.4 Key Performance Indicator (KPI) Reports

KPI reports provide accurate data information and feedback on certain areas. KPI reports are generated to measure the performance of an activity. The following reports are available in the KPI reports page.

- **AssetFailureRate**: reports generated to display the failure rate of the component. Failure rate is the frequency at which the engineered system or component fails. AssetFailureRate is calculated using the formula:

$$\text{AssetFailureRate} = \text{NOF} / \text{Total hours}, \text{ where NOF is the Number of Faults.}$$
- **AssetHealthIndex**: report generated to display the health of rotating equipment. AssetHealthIndex is a statistical measure that quantifies one or more aspects of a rotating equipment’s health. It is calculated from the vibration data HealthIndex.
- **AssetDownTime**: is calculated as:
 - When then RunningStatus attribute changes from non-zero to zero, the asset status is changed to “UnMonitored”.
 - For every one hour, Asset Performance Management checks the status of all the assets. If the status of the asset is “UnMonitored”, the down time of the asset is increased by one hour.
- **AssetRunningHours**: is calculated as:
 - When then RunningStatus attribute changes from zero to non-zero, the asset status is changed to “Monitored”.
 - For every one hour, Asset Performance Management checks the status of all the assets. If the status of the asset is “Monitored”, the RunningHours of the asset is increased by one hour.

NOTE

If RunningStatus attribute is not tag mapped, the asset status is considered as “Monitored” by default, and hence RunningHours can be increased by one. However, the status of the asset can be changed manually to “UnMonitored” and hence the DownTime can be calculated.

- **MTBF**: reports that display the predicted elapsed time between inherent failures of the system during its operation. It is calculated using the following formula.

$$\text{MTBF} = (\text{Total hours} / \text{NOF}) \text{ where Total hours is the time given by the user and NOF=Number of faults considered.}$$

Fromdate: is a mandatory field.

Todate: if Todate is null, then **CurrentDateTime** is considered.

- **MTTR**: reports that display the average time taken to repair a failed component. MTTR is calculated using the following formula:

$MTTR = \text{TotalHours} / \text{No. of times asset is OutofService}$.

Fromdate: is a mandatory field.

Todate: if ToDate is null, the **CurrentDateTime** is considered.

OutofService is calculated as the number of times the status of the asset is changed to OutofService within the specified time by the user.

NOTE

You must consider the difference in minutes and not in seconds.

- **BadActors-Asset**: reports that display the number of faults that occurred for selected assets only. The filter for type-Asset is not available in this report.
- **BadActors-Faults and Symptoms**: reports that display the number of times the fault appeared.

9.5 Changing Report Properties

Certain properties of the reports such as, the date of the report when it was generated, the date when the report was modified, and the size of the report can be viewed in the **Properties** tab.

ATTENTION

A Browser can only view the properties of the report. A Content Manager can modify the parameters of the properties page.

To Modify the Properties of the Report

ATTENTION

Ensure that you contact the Honeywell Support Team before modifying the properties of the report.

1. On the **AM Reports** page, click the type of report: **Assets, Faults, KPI Reports**.
2. From the list of reports, click the ellipses button for which you want to modify its properties, and then select **Manage**. For example Asset Status report.
3. Type the name of the report in the **Name** box.
4. In the **Description** box, type the description of the report.
5. Select the **Hide in list** view check box to hide the description of the report when it is displayed in the properties page.
6. To edit the definition of the report, do all of the following:

- a. Click **Download** and save the .rdl file and make the necessary modifications in the file.
- b. To upload this file, click **Replace**.
- c. Click **Browse** and navigate to the .rdl file which you modified.
- d. Click **OK** to upload the modified file.

TIP

You can also modify the report by clicking **Edit** in Report Builder.

7. In the **Properties** page, click **Apply** to apply the changes.
8. If you want to create a new linked report, click **Create Linked Report**, type the name, description and click **Change Location** if you want to change the location of the report.

To Specify the Parameters of a Report

1. Click **Properties > Parameters**.
2. Select the appropriate parameters that all the users can change.
3. Select the default values and click **OK**.

To Specify the Datasource Properties of a Report

1. Click **Properties > Datasource**.
2. Select the appropriate option and click **Apply**.

To Specify the Properties that Determine How the Report Server Runs the Report

1. Select the appropriate option and click **Apply**.
2. Click **Properties > Execution**.

To Specify the Properties that Determines How Report History is Retained

1. Click **Properties > History snapshots**.
2. Click **Schedule and settings**.
3. Select the appropriate option and click **Apply**.

To Specify Security Settings

1. Click **Properties > Security**.
2. Click **Customize security**.
3. Read the message and click **OK** if you want to change the security settings.

NOTE

By default, security settings are inherited from the parent item. If you customize the security settings for an individual report, changes to the parent folder will no longer be applied to that report.

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